

1. HEAT

- Heat is a form of energy
- The transfer of heat is always from hotter to colder body.
- Its S.I. unit - joule (J)
- Its C.G.S. unit - calorie
1 calorie = 4.18 J
- Calorimetry deals with the measurement of heat and the quantities involved in it.

Temperature :

- The degree of hotness or coldness of a body known as temperature of that body.

Measurement of Temperature :

- The device used to measure the temperature of a body is called thermometer.
- S.I. unit of temperature is “Kelvin”.
- Other units of temperature is $^{\circ}\text{C}$, $^{\circ}\text{F}$, degree Reaumur.

Scale of Temperature :

- To measure temperature of a body following temperature scales are used.

Celsius Scale :

- Temperature of ice point = 0°C
- Boiling Point of Water = 100°C
- No. of divisions between LFP and UFP = 100

Fahrenheit scale :

- Temperature of ice point = 32°F
- Boiling Point of Water = 212°F
- No. of equal divisions between LFP & UFP = 180

Kelvin or Absolute Scale :

- Temperature of ice point = 273 k
- Boiling point of water = 373 k
- No. of equal divisions between LFP & UFP = 100

Reaumur Scale :

- Temperature of Ice point = 0°R
- Boiling point of water = 80°R
- No. of equal divisions between LFP & UFP = 80
- Where lower fixed point (LFP) = Ice point and upper fixed point (UFP) = Boiling point of water.

Relation between Different Scales of Temperature.

$$\frac{C}{100} = \frac{F - 32}{180} = \frac{K - 273}{100} = \frac{R}{80}$$

- $K = ^{\circ}\text{C} + 273$
- At temperature: $-40^{\circ}\text{C} = -40^{\circ}\text{F}$

- At temperature : $574.58^{\circ}\text{F} = 574.58\text{ K}$
- The temperature at which the three phases of water remains at equilibrium is called triple point of water (273.15K) (pressure is 610 Pa)

Thermometers :

- The device is used to measure the temperature of a body is called thermometer.

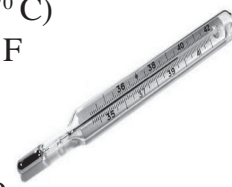
Types of thermometers :

1) Clinical thermometer :

- It is used to measure human body temperature
- In celsius scale normal temperature of human body = 36.9°C (37°C)
- In Fahrenheit scale normal temperature of human body = 98.4°F
- In Kelvin Scale normal temperature of human body = 310 K

2) Mercury thermometer :

- This thermometer works in the temperature range of -39°C to 357°C . But in colder place where temperature goes below -39°C mercury thermometer does not work because mercury freezes at -39°C .



Note: Alcohol freezes at -115°C and boils at 78°C so alcohol thermometer is used to measure temperature below -39°C

3) Maximum and Minimum thermometer :

- This thermometer was invented by James Six.
- This is used to measure maximum and minimum temperature during the last 24 hours.

4) Radiation thermometer (Pyrometer):

- To measure very high temperature radiation thermometer or pyrometers are used.
- Pyrometers are used to measure temperature of sun.

5) Electrical resistance thermometer:

- The type of thermometer is based on the phenomenon of increase in resistance of a conductor with rise in temperature. A platinum - resistance thermometer is such a thermometer, which can measure temperature between 260°C to 1200°C

6) Thermo Electric Thermometers (Thermocouple thermometers):

- It is based on the thermo electric effect. It can measure temperature upto 3000°C

Thermal Expansion :

- The expansion of a body caused by heat is known as thermal expansion.

Thermal Expansion of Solids :

- Thermal Expansion of solids is of three types.
- Expansion in length on heating is called linear expansion.
- The increase in length of a rod of unit length of a substance due to increase in its temperature by 1°C is called the coefficient of linear expansion of the substance of that rod. It is represented by α

$$\alpha = \frac{\text{Increase in length}}{\text{Initial length} \times \text{rise in temperature}}$$

$$\alpha = \frac{\Delta L}{L \times \Delta t}$$

- Its unit is $(^{\circ}\text{C})^{-1}$
- Expansion in area on heating is called superficial expansion.
- coefficient of superficial expansion is given as

$$\beta = \frac{\Delta A}{A \times \Delta t}$$

- Its unit is $(^{\circ}\text{C})^{-1}$
- Expansion in volume on heating is called volume expansion or cubical expansion.
- Coefficient of cubical expansion is given as

$$\gamma = \frac{\Delta V}{V \times \Delta t}$$

- Its unit is $(^{\circ}\text{C})^{-1}$

Relation between coefficients of expansions:

$$\beta = 2\alpha \text{ and } \gamma = 3\alpha \text{ and } \alpha : \beta : \gamma = 1 : 2 : 3$$

- Thermal expansion follows the order

Increase in volume
Gases > Liquids > Solids

Initial volume rises as temperature increases

Applications of Thermal expansion :

- 1) In laying railway line, a small gap is always left between the rail due to for allowing expansion on summer.
- 2) Since metals expand much more than glass, metal caps of glass bottles can be loosened by heating them under hot water.
- 3) Telephone wires are connected between the poles to allow for their contraction in winter, otherwise in winter wires may be broken.
- 4) Thermostats are used for temperature regulator in electrically heated rooms, ovens, refrigerators, Air conditioners and in fire alarms.

Thermal expansion of liquids :

- In liquids, only expansion in volume takes place on heating.
- Expansion of liquid is of two types.
 - (i) The observed change in volume of a liquid related to the container is called the apparent expansion of the liquid.
 - (ii) If the real change in volume of the liquid is called the real expansion.
- Real expansion > Apparent expansion

Anomalous Expansion of Water :-

- When temperature of water is increased from 0°C , then its volume decreases upto 4°C , becomes minimum at 4°C and then increases. This peculiar behaviour of water is called anomalous expansion of water.

- water has maximum density at 4°C and minimum volume at 4°C

Applications :

- 1) Radiators in motor vehicles break in winter in colder regions due to this phenomenon.
- For the minimising of breakage of radiators ethyleneglycol or alcohol mixed with water.
- 2) Lumps of wet clay break into pieces during winter due to this phenomenon.
- 3) The aquatic animals (fish, frogs) can survive at bottom of the lakes in colder regions.
- 4) Breakage of water pipes in colder regions. For minimising this breakage of water pipes painted with black colour.

Thermal expansion of gases :-

- There are two types of coefficient of expansion of gases.
- (i) At constant pressure, the change in volume per unit volume per 1°C is called volume coefficient of a gas.
- (ii) At constant volume, the change in pressure per unit pressure per 1°C is called pressure coefficient of a gas.

Specific Heat:

- The quantity of heat required to raise the temperature of unit mass of a body through 1°C is called specific heat.
- Where Q - quantity of heat
 m - mass of substance
- rise of temperature
- Its S.I. unit is
- Its C.G.S unit is
- Specific heat of some common substances in $\text{cal/gm } ^{\circ}\text{C}$
Water = 1
Iron = 0.12
Mercury = 0.03
Copper = 0.09
- The specific heat of water is $4200 \text{ Jkg}^{-1} \text{ K}^{-1}$ (S.I) which is high compared with most other substances therefore water is used as coolant in radiator in vehicle and in nuclear reactors.
- Specific heat of solids and liquids determined by using “Principle of calorimetry”

Principle of calorimetry can be stated as follows:

- Heat lost by a body at high temperature = Heat gained by a body at lower temperature.

Latent Heat (L):

- It is the amount of heat taken in or given out by unit mass of the substance when it changes from one state to other at constant temperature.
- Its S.I. Unit = J/kg
- Its C.G.S. Unit = calorie / gm

Latent heat of Fusion:

Definition: The amount of heat is required to change the state from solid to liquid at constant temperature is called latent heat of fusion.

$$L_{\text{ice}} = 80 \text{ cal/gm.}$$

Latent heat of fusion of some solids in cal/gm

Aluminium - 93
Copper - 50.6
Naphthalene - 35.6

Applications:

- Bottled drinks are cooled more effectively when surrounded with lumps of ice than with iced water because ice absorbs more heat than does the same amount of iced water. Each gram of ice that melts absorbs 80 calorie of heat.

Latent heat of Vaporization:

- The amount of heat is required to change the state from liquid to vapour at constant temperature is called latent heat of vaporization.

$$L_{\text{water vapour}} = 540 \text{ cal/gm.}$$

Applications:

- (Scalding) Burning with steam (at 100° C) is more severe and painful than that caused by boiling water (at 100° C) because steam contains more energy than Boiling water at the same temperature.

Latent heat vaporization of some liquids in cal/gm

Mercury - 70
Alcohol (Ethyl) - 204

Evaporation :

- The phenomenon of change of a liquid into vapour at any temperature below its boiling point is called evaporation.
- Evaporation is a surface phenomenon.
- The rate of evaporation increases with surface area.
- The rate of evaporation increases with temperature.
- The rate of evaporation increases with speed of wind.
- The rate of evaporation decreases with increase in the humidity of air.

Applications:

- 1) Earthen pot (matka) becomes cool during summers particularly hot day because water evaporation from the pot become easier.
- 2) Dog keeps their tongue usually out in summer water evaporate through the tongue produces cooling effect.
- 3) When a sweat evaporates from the skin it draws much heat from the body and produces a cooling sensation.

Change of State :-

- The process due to which a substance changes from one physical state to another physical state with the absorption or release of heat energy is known as change of state.

(i) Fusion or melting:

- When a solid changes into liquid at some fixed temperature is known as melting.
- Ex:- Ice is converted into water.

(ii) Freezing or Solidification :

- When a liquid changes into solid state at some fixed temperature is known as freezing.
- Ex:- Water is converted into Ice.

(iii) Boiling or Vaporisation:

- The fixed temperature at which a liquid changes into gaseous state, due to the process of vaporisation is known as Boiling point.
- Ex:- Boiling point of water is 100°C

(iv) Condensation:

- The process due to which a gas changes into liquid state at some fixed temperature with the release of heat energy (or by cooling) is known as condensation.
- Ex:- Formation of water droplets on the outer surfaces of a glass containing ice-cold water.

Sublimation:

- The change of state from solid to gas without changing into liquid state is called sublimation.

Ex:- 1) Camphor on heating changes directly into vapours.

2) Naphthalene

3) Ammonium Chloride

Deposition:

- The Change of state directly from gas to solid state is called deposition.

Ex:- 1) Water vapour to ice. This process occurs on windows during the winter months

2) CO_2 (gas) $\longrightarrow$ Dryice (Solid CO_2)

Transmission of Heat:

- Heat can be transmitted from one place to other by three methods.
 - 1) Conduction
 - 2) Convection
 - 3) Radiation

1) CONDUCTION:-

- Material medium is required its transmission of heat. Transfer of heat from one place to another without the actual movement of particles of medium is called conduction.
- Conduction takes place mainly in solids, Mercury is the only liquid where conduction takes place.
- Ex:- The heat energy transmitted through the metallic spoon when it is in a gas flame.

Good Conductors :

- Those substances which allow the heat energy to flow through them easily are called good conductors of heat.
- 'Silver' is the best conductor of heat.
- The next in order are Copper, Aluminium, Gold, Iron.

Practical Applications of good Conductors of Heat :-

- 1) cooking vessels are made of metals so that they can readily absorb heat energy and transfer it to the food.
- 2) Mercury is used as a thermometric liquid as it is a good conductor of heat.
- 3) Copper tubing is used in automobile radiators as it readily takes heat from the hot water coming from the side of the engine.

Bad conductors of Heat or Heat Insulators:

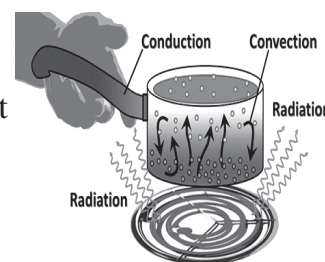
- Those substances which do not allow the heat energy to flow through them easily, are called bad conductors of heat.
- Ex:- Wood, Plastic, Cotton, Gases (air), Ice.

Practical Applications of Insulators of Heat :

- 1) Two thin blankets are warmer than a one thick blanket because there is an extra layer of air trapped between thin blankets which acts as an insulator of heat.
- 2) Eskimos make igloo out of snow because snow contains a fairly large amount of "trapped air" which acts as insulator of heat.
- 3) Appliances like kettles, electric irons, electric ovens, soldering rods, cooking pan are provided with wooden or plastic handles because they are insulators of heat.
- 4) Slabs of ice are covered with saw dust or gunny bags to prevent them from melting. The saw dust or gunny bags contain a "trapped air", which acts as insulator of heat.

CONVECTION :

- Transfer of heat from one place to another place by the actual movement of particles of medium is called convection.
- Convection takes place in liquids and gases only.
- Material medium is required for this transmission of heat

**Applications :**

- 1) Formation of land-breeze, sea-breeze.
- 2) Heating of water in Geyser and electric kettle

- 3) Cooling unit (freezer) in a refrigerator is fitted near the top to cool the whole interior. The air near the freezer cools and descends due to increased density and warm air (less density) goes up near the freezer and keeps the cool the whole interior.
- 4) In houses, the warm air (CO_2) passes out through the ventilators and cold and fresh air from doors and windows flows in. This circulation of air maintained within the room by convection.

RADIATION :

- The transfer of heat from one place to another without any medium is called radiation.
- Medium is not required for this method of transmission of heat.

Applications :

- 1) The heat from sun reaches the earth by the process of radiation
- 2) On standing near fire, we feel hot as heat comes to us through radiation.

Note : Radiation is the fastest mode of transmission of heat.

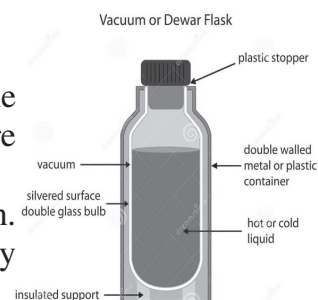
- It takes place at a speed of $3 \times 10^8 \text{ ms}^{-1}$

Absorption of radiant heat by a body :

- 1) Black bodies are good absorbers and good radiators of heat.
- 2) White or shining bodies poor absorbers of heat.

Thermos Flask:-

- It consists double walled glasses and vacuum between the walls. The two inner glass surfaces facing each other are silvered.
- Vacuum prevents heat loss by conduction and convection. Silvered surfaces of the walls prevents heat loss by radiation.
- Thus in a thermos flask, the transmission of heat by conduction, convection and radiation is minimised and therefore, its content (Tea, Milk) remains at nearly the same temperature for a long time.



The principle of pressure cooker :

- The boiling point of liquid depends on external pressure.
- When pressure is increased the boiling point is raised.
- Water boils at 120°C in a pressure cooker.
- In a pressure cooker, water boils at temperature higher than 100°C due to increased pressure.
- At higher altitudes, (hill stations), atmospheric pressure is reduced. This lowers the boiling point of water and food takes much longer to cook.

Principle of Skating :

- Skating on ice is made possible due to Regelation.

Regelation :

- When pressure applied (increased) to block of ice, its melting point decreases. When pressure is released (decreases) it regains its original melting point. This is the phenomenon of regelation.



HEAT - Practice Bits

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| <p>1. The degree of hotness or coldness of a body is called.
 1) heat energy 2) temperature
 3) radiant heat 4) none of the above</p> <p>2. The device used for measuring temperature of a body is called.
 1) barometer 2) odometer
 3) thermometer 4) calorimeter</p> <p>3. The most common liquid used in a laboratory thermometer.
 1) alcohol 2) ether
 3) mercury 4) glycerine</p> <p>4. The marking between the melting point of pure water and boiling point of pure water on celsius thermometer are.
 1) 100 2) 180
 3) 273 4) none of the above</p> <p>5. On Kelvin scale the temperature which corresponds to 0° C
 1) 0 K 2) 273 K
 3) 373 K 4) None of the above</p> <p>6. 20° C corresponds to.
 1) 20 K 2) 393 K
 3) 293 K 4) 193 K</p> <p>7. 273 K corresponds to:
 1) 273 °F 2) 173 °F
 3) 73 °F 4) 32 °F</p> <p>8. 212°F corresponds to.
 1) 373 K 2) 100 °C
 3) 0 °C 4) both (1) and (2)</p> <p>9. The number of divisions between the lower and upper standard points on Fahrenheit thermometer are.
 1) 100 2) 180
 3) 80 4) 253</p> | <p>10. 1 °C rise in temperature corresponds to
 1) 1.8 °F 2) 1.8 K
 3) 1 K 4) both (1) and (3)</p> <p>11. A thermometer used for recording temperature of human beings is called.
 1) doctor's thermometer
 2) fever thermometer
 3) clinical thermometer
 4) all the above</p> <p>12. The temperature of a healthy person Celsius scale is.
 1) 35 °C 2) 38 °C
 3) 37 °C 4) 34 °C</p> <p>13. The temperature of a healthy person on Fahrenheit scale is.
 1) 98.4 °F 2) 98.8 °F
 3) 98.2 °F 4) 98.6 °F</p> <p>14. Heat energy can bring about change in.
 1) Temperature of a body
 2) Dimension of a body
 3) State of a body
 4) Any of the above</p> <p>15. The process due to which a given state of matter expands on the absorption of heat energy is called.
 1) molecular expansion
 2) matter expansion
 3) thermal expansion
 4) thermal stretching</p> <p>16. The process due to which a solid changes into a liquid state at some fixed temperature by the absorption of heat energy is called.
 1) melting 2) fusion
 3) liquation 4) either (1) or (2)</p> |
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17. The fixed temperature at which a solid changes into liquid state on the absorption of heat energy is called.
1) melting point 2) fusion point
3) either 1 or 2 4) none of the above
18. The process due to which a liquid changes into solid state, at some fixed temperature, by the release of heat energy is called.
1) solidification 2) freezing
3) either 1 or 2 4) none of the above
19. The process due to which a liquid changes into gaseous state at some fixed temperature, with the absorption of heat energy, is called.
1) vaporisation 2) boiling
3) either 1 or 2 4) none of the above
20. If the melting point of ice is 273 K, the freezing point of water is.
1) less than 273 K
2) more than 273 K
3) equal to 273 K
4) none of the above
21. The change of liquid into vapour form at any temperature is called.
1) vaporisation 2) evaporation
3) gasification 4) none of the above
22. The rate of evaporation of a liquid increases with the.
1) increase in temperature of liquid
2) increase in temperature of the surroundings
3) increase in exposed surface area of the liquid
4) all the above
23. Combustion is said to take place when substance burns in____with the release of large amount of heat energy.
1) air 2) oxygen
3) any of 1 or 2 4) none of the above
24. The substances which easily catch fire in air or oxygen are called.
1) flammable substances
2) combustible substances
3) either (1) or (2)
4) heat sensitive substances
25. The increase in length of a solid heating is called.
1) length expansion
2) linear expansion
3) superficial expansion
4) cubical expansion
26. Iron rims of slightly smaller diameter mounted on the wooden wheels of because.
1) Iron does not allow the wooden wheel to wear off
2) Iron expands on heating and slips over the wooden wheel
3) Iron strengthens the wheel
4) None of the above
27. Gaps are left between the rails to allow for.
1) contraction
2) expansion
3) both (1) and (2)
4) for easy transportation of rails
28. When boiling hot water is poured into a thick glass tumbler, it cracks because.
1) its outer surface expands more than the inner surface
2) its outer surface expands less than the inner surface
3) its outer and inner surfaces expand in opposite directions
4) none of the above
29. A bimetallic strip is used in.
1) automatic fire alarm
2) automatic electric iron
3) automatic electric oven
4) all the above

- 30. The process of transmission of heat energy in solids, without actual visible movement of particles, from their mean position is called.**
1) conduction 2) convection
3) radiation 4) both (1) and (2)
- 31. Which of the following is not an insulator of heat?**
1) Air 2) Wood
3) Aluminium 4) Rubber
- 32. The metal used in the cooling coils of air conditioners and refrigeration is made of.**
1) Silver 2) Iron
3) Copper 4) Zinc
- 33. We wear woolen clothes in winter because.**
1) woolen clothes are warm clothes
2) they contain trapped air, which acts as an insulator
3) both (1) and (2)
4) none of the above
- 34. The phenomenon due to which particles of a medium actually move to the source of heat is called.**
1) radiation 2) conduction
3) convection 4) none of the above
- 35. The materials in which convection takes place are.**
1) solids 2) liquids
3) gases 4) both 2 and 3
- 36. The process by which warm air goes out of a room from the top and cold air moves in from doors and windows naturally is called.**
1) air circulation 2) convection
3) air rotation 4) none of the above
- 37. The heat energy from the sun comes to us by the process of.**
1) radiation 2) conduction
3) convection 4) none of the above
- 38. Black bodies are.**
1) good absorbers and poor radiators of heat energy
2) poor absorbers and good radiators
3) good absorbers and good radiators of heat energy
4) poor absorbers and poor radiators of heat energy
- 39. Cooking vessels are blackened from the base and kept shining at the sides, because.**
1) black bodies are good absorbers of heat and shining bodies are poor radiators of heat
2) black bodies are poor absorbers of heat and shining bodies are good radiators of heat
3) black bodies, as well as shining bodies are good absorbers of heat
4) black bodies, as well as shining bodies are poor radiators of heat.
- 40. A device which can detect the presence of heat radiations is called.**
1) thermometer 2) calorimeter
3) radiation meter 4) thermoscope
- 41. Vacuum is created in between the two glass bottles of thermos flask, to prevent heat losses by.**
1) conduction 2) convection
3) radiation 4) both 1 and 2
- 42. In summer, people wear white clothes because white clothes are.**
1) good reflectors of heat
2) good absorbers of heat
3) poor absorbers of heat
4) both (1) and (3)
- 43. Wooden handles are provided with cooking vessels, because.**
1) wood is a heat insulator
2) wood is a heat conductor
3) either (1) or (2)
4) none of (1) and (2)

- 44. The mode of transmission of heat which requires no medium is called.**
1) conduction 2) radiation
3) convection 4) none of the above
- 45. The intermolecular spaces in case of liquids is.**
1) less than solids
2) more than gases
3) more than solids
4) more than solids and gases
- 46. The intermolecular spaces are maximum in case of.**
1) solids 2) gases
3) liquids 4) none of the above
- 47. The intermolecular forces of attraction is least in case of**
1) gases 2) liquids
3) solids 4) liquids and solids
- 48. With the increase in pressure on the enclosed gas, the volume of the gas.**
1) increase 2) remains same
3) decreases 4) none of the above
- 49. Calorie is the old unit of heat energy. The substance whose 1 g mass on gaining 1°C rise in temperature is called calorie. The substance is.**
1) alcohol 2) mercury
3) water 4) none of the above
- 50. One kilocalorie is equal to.**
1) 4.2 J 2) 42 J
3) 420 J 4) 4200 J
- 51. The amount of heat energy contained in a body is.**
1) directly proportional to the mass of a substance
2) inversely proportional to the mass of substance
3) equal to the mass of the substance
4) none of the above
- 52. Calorie is the.**
1) amount of heat energy required to raise the temperature of any amount of water through 1°C
2) amount of heat energy required to raise the temperature of 1g of water through 1°C
3) amount of heat energy required to raise the temperature of 1 kg of water through 100°C
4) none of the above
- 53. The substance with the highest specific heat capacity is.**
1) mercury 2) gold
3) tungsten 4) water
- 54. The specific heat capacity of water in SI system is.**
1) $4.2 \text{ Jg}^{-1} ^{\circ}\text{C}^{-1}$ 2) $4.2 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$
3) $420 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$ 4) $4200 \text{ J kg}^{-1} \text{ K}^{-1}$
- 55. During daytime, water bodies are cooler than the land, because specific heat capacity of land is.**
1) less than water
2) more than water
3) not playing any role
4) none of the above
- 56. The land breeze blows from land to sea in coastal regions, because, the land.**
1) cools slowly as compared to seawater
2) cools rapidly as compared to seawater
3) cools at the same rate as seawater
4) none of the above
- 57. Near coastal regions, the sea breeze blows.**
1) during day time only
2) during night time only
3) throughout the day and night
4) none of the above

- 58. The liquid used as a coolant in automobile engines is.**
1) engine oil 2) alcohol
3) benzene 4) water
- 59. The heat energy required to change a solid at its melting point into liquid state without any rise in temperature is called.**
1) latent heat of vaporisation
2) specific heat capacity
3) latent heat of fusion
4) thermal capacity
- 60. The heat energy required to change a liquid at its boiling point, into gaseous state, without any rise in temperature is called.**
1) latent heat of vaporisation
2) specific heat capacity
3) thermal capacity
4) latent heat of fusion
- 61. The wind cycles are formed due.**
1) high specific heat capacity of water
2) low specific heat capacity of the earth
3) both (1) and (2)
4) rotation of earth about its own axis
- 62. During the fusion of a solid its temperature.**
1) increases 2) decreases
3) remain same 4) none of the above
- 63. The temperature at which a solid changes into its liquid state is called.**
1) fusion point
2) vaporisation point
3) liquefaction point
4) boiling point
- 64. More is exposed area of the surface of a liquid.**
1) more is the rate of evaporation
2) less is the rate of evaporation
3) there is no change in the rate of a vaporisation
4) none of the above
- 65. More is the amount of water vapour present in the air.**
1) more is the rate of evaporation
2) lesser the rate of evaporation
3) there is no change in the rate of evaporation
4) none of the above
- 66. Consider the following statements.**
a) Aquatic life is preserved under frozen water in a lake
b) Water has minimum density at 4 °C
Which of the following statements is /are true?
1) Both a and b 2) Neither a nor b
3) Only a 4) Only b
- 67. In cold countries alcohol is preferred to mercury as a thermometric liquid because.**
1) Alcohol is a better conductor of heat
2) Alcohol can be coloured and its level seen easily
3) Alcohol has a very low freezing point
4) Alcohol is cheaper than mercury
- 68. On heating a circular metallic disc with a circular hole at the centre, the diameter of the hole will**
1) Decrease
2) Remain the same
3) Increase
4) First increase, then decrease

69. Fish can survive inside a frozen lake because.

- 1) Fish are warm blooded animals
- 2) Fish hibernate in ice
- 3) Water near the bottom does not freeze
- 4) Ice is a good conductor of heat

70. What happens when water is cooled from 8°C to 0°C .

- a) Its volume decreases uniformly, becoming minimum at 0°C
- b) Its density increases uniformly, attaining maximum value at 0°C
- c) Its volume decreases up to 4°C and then increases.
- d) Its density increases up to 4°C and then decreases

Which of the following is correct

- 1) 'a' only
- 2) 'b' only
- 3) 'c' and 'd' only
- 4) 'c' only

71. A thermostat is device for.

- 1) Switching off an electrical appliance
- 2) Measuring temperature
- 3) Regulating temperature
- 4) Producing heat

72. Woolens keep us warm in winter because they.

- 1) Produce more warmth
- 2) Are good absorbers of heat
- 3) Do not allow our body heat to escape
- 4) Do not allow cool air to enter

73. The freezer in a refrigerator is fitted near the top.

- 1) To keep it away from the hot compresor which is near the bottom
- 2) Because of convenience
- 3) So that it can cool the whole interior by setting up convection currents
- 4) Without any specific purpose

74. Light coloured clothes are preferred in summer because.

- 1) Light colours are good radiators of heat
- 2) Light colours are poor absorbers of heat
- 3) They soak sweat faster
- 4) They allow body heat to escape faster.

75. By which of the following methods could a cup of hot coffee lose heat if placed on a metal table in a room?

- 1) Conduction and radiation
- 2) Conduction and convection
- 3) Conduction and radiation
- 4) Conduction, convection, radiation and evaporation

76. Drinks remain hot or cold for a long time in a thermos flask because of the

- 1) Presence of air, which is a bad conductor of heat, in between the double walls
- 2) Silver coating on the inside of the double walls
- 3) Vaccum between the double walls
- 4) Combined effects of (2) and (3)

77. For a given liquid, the rate of evaporation depends on.

- a) The temperature of the liquid
- b) The temperature of air
- c) The surface area of the liquid

Which of the above factors is / are correct?

- 1) a only
- 2) a and b only
- 3) b and c only
- 4) a, b and c

78. Scalding (Burning) with steam is more severe than scalding by boiling water because.

- 1) Steam can penetrate the skin
- 2) Steam is at a higher temperature
- 3) Steam contains more energy than boiling water
- 4) Steam is at a higher pressure

79. Which one of the following thermometers is known as pyrometers.

- 1) Thermoelectric thermometer
- 2) Radiation thermometer
- 3) Gas thermometer
- 4) Liquid thermometer

80. Which one of the following is the temperature at which the Celsius and Fahrenheit readings are numerically equal but opposite in signs.

- 1) 40
- 2) 80/7
- 3) 56
- 4) 68

Answers

1) 2	2) 3	3) 3	4) 1	5) 2	6) 3	7) 4	8) 4	9) 2	10) 4
11) 3	12) 3	13) 1	14) 4	15) 3	16) 4	17) 3	18) 3	19) 3	20) 3
21) 2	22) 4	23) 3	24) 3	25) 2	26) 2	27) 3	28) 2	29) 4	30) 1
31) 3	32) 3	33) 2	34) 3	35) 4	36) 2	37) 1	38) 3	39) 1	40) 4
41) 4	42) 3	43) 1	44) 2	45) 3	46) 2	47) 1	48) 3	49) 3	50) 4
51) 1	52) 2	53) 4	54) 4	55) 1	56) 2	57) 1	58) 4	59) 3	60) 1
61) 3	62) 3	63) 1	64) 1	65) 2	66) 3	67) 3	68) 3	69) 3	70) 3
71) 3	72) 3	73) 3	74) 2	75) 4	76) 4	77) 4	78) 3	79) 2	80) 1

2. SOUND

- “Acoustics” is branch of physics that deals with study of sound.
- Sound is a form of energy.
- All sounds are produced by the vibration of material objects.

Ex:- 1) In a sitar the sound is produced by the vibrating string.
2) In a tabla or a drum by the vibrating stretched skin
3) The voice results from the vibration of vocal chords in the larynx.

- Sound waves are longitudinal mechanical wave and cannot travel in vacuum.

Wave:

- The disturbance produced by a vibrating body in the surrounding medium is called wave.
- waves transport energy only and there is no transportation of medium.
- The waves can be classified into two types on the basis of material medium.
 - i) Mechanical waves (Elastic wave)
 - ii) Non-mechanical waves (Electromagnetic wave)

i) Mechanical Waves:-

- The waves which can be produced or propagated only in a material medium are called mechanical wave.

Ex:- Sound waves.

ii) Non-Mechanical Wave (E.M.Waves):-

- The waves which do not require any material medium for its transmission and can travel even in vacuum are called E.M.waves.

Ex:- Light waves, Radio waves, X-rays.

Mechanical waves are of two types:-

- a) Progressive waves
- b) Stationary waves

a) Progressive waves:-

- A wave that transfers the energy from a point into an infinite medium and never returns to the origin is called a progressive wave.
- All the particles in the medium execute simple harmonic motion about their mean position.
- Further progressive waves are classified into two types:
 - i) Longitudinal waves
 - ii) Transverse waves

i) Longitudinal waves:-

- If the particles of the medium vibrate parallel to the direction of propagation of the wave, then the wave motion is said to be longitudinal

Ex:- 1) sound waves in gases (air)

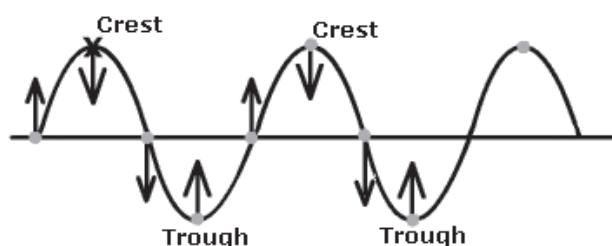
- 2) vibrations of air column in organ pipe.
- 3) vibrations of air column above the surface of water in the tube of a resonance apparatus are longitudinal.
- Longitudinal waves travel through the medium in the form of compressions (C) and rarefactions (R).
- The distance between any two successive compressions or rarefactions is equal to wavelength of the wave (λ).



- Longitudinal waves can be travels in all the media i.e, solids, liquids and gases.

Transverse waves:-

- If particles of the medium vibrate perpendicular to the direction of propagation of the wave, then the wave motion is said to be transverse.



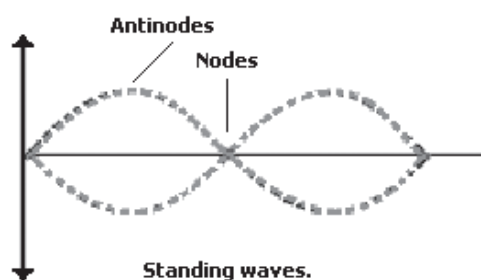
- Transverse waves travel through the medium in the form of crests (C) and troughs (T).
- The distance between two successive crests or troughs is called wavelength of the wave (λ).

Examples of Transverse waves:-

- 1) Transverse waves can be produced in solids and on the surface of a liquid but cannot be produced in the interior of a liquid (or) in a gas.
- 2) Movement of membrane of a Tabla or dholak.
- 3) Movement of string of a sitar or Violin.
- 4) Movement of a kink on a rope.

Stationary Wave (Standing Wave):-

- When two waves having same frequency and amplitude travelling along the same straight line with same speed but in opposite direction are supreposed the result is a stationary wave.



- There is no flow or transfer of energy along the stationary wave.
- Stationary waves travel through the medium in the form Nodes (N) and Antinodes (A).
- There are certain points in the medium in a standing wave, which are permanently at rest. These are called nodes.
- The distance between two successive nodes is .
- There are certain points in the medium in a standing wave, displacement is maximum are called Antinodes.
- The distance between two successive antinodes is .
- The distance between a node and adjoining antinode is .

Examples of stationary waves:-

- 1) Stationary waves produced in organ pipes and in air column of resonance tube apparatus are longitudinal stationary wave.
- 2) Stationary waves produced on the vibrating string of a sonometer are transverse stationary waves.

Characteristics of a wave:-

1. Amplitude
2. Wavelength
3. Frequency
4. Time period
5. Wave velocity

1) Amplitude:-

- The maximum displacement of particles in the medium on either side of mean position is called amplitude of wave.
- It is represented by letter 'A'.
- The units of amplitude is 'cm' or m.
- Loudness of sound is determined by on the basis of its 'amplitude'.

2) Wavelength:-

- The distance between two nearest points in a wave which are in the same phase of vibration is called the wavelength.
- It is represented by Greek letter ' λ '. (or)
- Wavelength of a wave is the length of one wave. (or)
- The distance between two successive compressions or two successive rarefactions is called the wavelength of a sound wave.
- The distance between two successive crests or troughs is called the wavelength of sound wave.
- S.I. unit of wavelength is 'metre' (m).

3) Frequency:-

- The number of vibrations completed by a particle in one second is called frequency.
- It is represented by a letter ‘ f ’ (or) ‘ n ’
- $f = \frac{1}{T}$; where ‘ T ’ is time period.

Units of frequency:-

- Hertz (or) cycles/sec (or) vibrations/sec (or) oscillations/sec (or) second^{-1}
- Its SI unit is:- Hertz.

4) Time period:-

- The time taken to complete one oscillation is called the time period of the sound wave.
- Its S.I. unit is ‘second’.
- It is represented by letter ‘ T ’.

5) Wave velocity:-

- It is the distance travelled by a wave in one second.
- If a sound wave propagates (travels) from one medium to another, then its speed and its wavelength changes but its frequency remains constant.

On the basis of frequency sound waves are classified into three types:-

- 1) Infrasonics (Infrasonic waves)
- 2) Audible range (Audible waves)
- 3) Ultrasonics (Ultrasonic waves)

1) Infrasonics:-

- Sound of frequency less than 20Hz is known as infrasonic sound.
- Animals such as elephants and whales produce sounds of frequencies less than 20 HZ.
- Some fishes can hear sounds of frequencies as low as 1 to 25 Hz.
- Rhinoceroses communicate using infrasonic sounds of frequency around 5 Hz.
- Earthquake waves, volcanic eruptions produce sounds of frequency less than 20Hz.
- We cannot hear these sounds.

Audible range:-

- The sound waves of frequency lies between 20 Hz to 20,000Hz are called audible range.
- Audible range is 20 Hz to 20,000 Hz (or) audible range is 20 Hz to 20 KHz.

Ultrasonic Waves:-

- The sound waves of frequency greater than 20,000 Hz are called ultrasonic waves.
- Bats, dogs, dolphins, cats, insects can hear these sounds.
- A dog can hear sounds of frequency upto 50,000 Hz.
- Bats can hear the sounds of frequency upto 100000 Hz.
- We cannot hear these sounds.

Applications of ultrasonic sounds:-**1) Echocardiography:-**

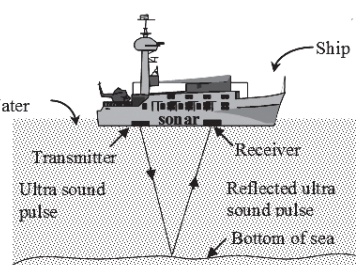
- It is a technique in which ultrasonic waves, reflected from various parts of heart, form an image of the heart.

2) Ultrasonography:-

- Ultrasonography is routinely used to show doctors, the image of a patient's organs such as the liver, gall bladder, uterus etc.
 - It helps doctors to detect abnormalities such as stones in the gall bladder tumors etc.
 - These are used to break small stones formed in the kidneys.
 - It is also used to monitor the growth of a foetus inside the mother's womb.
- 3) Ultrasonic waves can be used to detect flaws in metal.
 - 4) Ultrasonic waves can be used for cleaning utensils, washing clothes removing dust and soot from the chimney of factories.
 - 5) Ultrasonic waves can be used for drilling and cutting processes in metals.
 - 6) These waves can also be used for soldering for example aluminium cannot be soldered by normal methods. For solder aluminium ultrasonic wave along with electrical soldering iron are used.
 - 7) The animals like rats, frogs, fishes can be killed or injured by high intensity ultrasonics.
 - Ultrasonic waves are also used in cataract removal.
- 8) Treatment of neuralgic pain:-**
- The body parts affected due to neuralgic or rheumatic pains on being exposed to ultrasonics get great relief from pain.
- 9) In sterilizing of a liquid.
 - 10) Ultrasonic waves are used in SONAR.

SONAR (Sound Navigation and Ranging):-

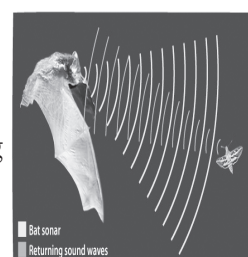
- This is a method for detecting and finding the distance of objects under water by means of reflected ultrasonic waves. The device used in this method is also called SONAR.

**Uses of SONAR:-**

- 1) Determining the depth of the sea and to locate underwater hills, valleys and sunken ship and submarine.
- The following formula is used in SONAR for finding the depth of the sea.

$$d = \frac{V \times t}{2}$$

- Where V - speed of sound in water
t - Time between sending ultrasonic wave and receiving its echo from the object
d - distance between SONAR and underwater object.



- “Bats” can produce and receive ultrasonic waves. During flight, a bat emits ultrasonic waves. The bat receives back these waves after being reflected by the obstacle in its path. From the time interval between transmission and reception of ultrasonic waves the bat gets information about the distance, nature of obstacle and its direction of location.

Characteristics of sound:-

- 1) Pitch
- 2) Loudness
- 3) Quality (or) Timbre

1) Pitch:-

- Pitch is the subjective sensation of hearing while “frequency” is physically measurable quantity.
 - It depends on frequency of sound wave.
 - If frequency of sound wave increases pitch also increases.
 - If frequency of sound wave decreases pitch also decreases.
- order of pitch.**
- Insects > childrens > women > men > Lion.

Examples:-

- 1) Pitch of women voice is higher than men voice.
- 2) Pitch of children voice higher than elders.
- 3) The pitch of the buzzing of bee is higher than the roar of lion.
- 4) The sound produced by a violin has a higher pitch than the bass of a drum.

2) Loudness:-

- Loudness is the subjective sensation of hearing while intensity is physically measurable quantity.
- It depends both on the intensity of sound and response of the ear. (sensitivity of ear)
- The greater the intensity the greater the loudness.
- The intensity of the wave is directly proportional to the square of the amplitude of the wave.
- $\therefore I \propto A^2$
- Intensity of sound is the rate of transfer of energy per unit area of a surface perpendicular to the direction of propagation.
- Loudness of sound is proportional to the square of the amplitude of the vibration producing the sound.

Decending order of Loudness:

- Lion > men > women > children > Insects.
- The loudness is expressed in a unit called **decibel** (dB).

“bel” is the bigger unit of loudness of sound.

$$1dB = \frac{1}{10} \text{ bel}$$

S.No.	Source	Loudness
1.	Normal breathing	10 dB
2.	Soft whisper	30 dB
3.	Normal conversation	60 dB
4.	Busy traffic	70 dB
5.	Average factory	80 dB
6.	Explosion of bomb	90-110 dB
7.	Jetplanes	120-140 dB

- Above 80 dB the noise becomes physically painful.

Examples:-

- 1) The roar of lion is louder than the buzzing of bee.
- 2) Loudness of men voice is higher than women voice.

3) Quality (or) Timbre:-

- The property by which the two sounds of the same pitch and loudness can be distinguished is called quality.

Examples:-

- 1) We can distinctly recognise the voice of a person due to the quality of the characteristic notes from his vocal chords.
- 2) The sound from a piano and guitar can be distinguished on the basis of quality.
- 3) It is due to the quality of sound that one can recognise the voice of one's friend without seeing him.

Some basic definitions about musical note:

Music: The sound which has a pleasing sensation to the ears is called music.

Noise: The sound which has a non-pleasing sensation to the ears is called noise.

Tone: A sound of single frequency is called a tone.

Note: The sound which is a mixture of several frequencies is called a note.

Natural frequency:

- The number of vibrations executed per second by freely vibrating body is called natural frequency

Natural Vibrations:

- The vibrations produced in a body, on being slightly disturbed from its mean position are called natural vibrations.

Examples:-

1. A freely suspended pendulum vibrating about its mean position.
2. Strings of various musical instruments on being plucked.
3. A metal blade clamped at one end it gently disturbed.
4. A tuning fork on being struck on a rubber pad.

Forced Vibrations:-

- The vibrations which take place under the influence of an external periodic force are called forced vibrations.

Examples:-

- 1) When the handle of a vibrating tuning fork is pressed against table top, a loud sound is heard.
 - It is because, the vibrating tuning fork forces the table top, to vibrate with its own frequency. As the area of table top is very large, therefore, forced vibrations produce louder sound.
- 2) When the wire of sitar or guitar is plucked, its board and wind box make forced vibrations.

Resonance:-

- Resonance is the phenomenon in which if one of the two bodies of the same natural frequency is set into vibration the other body also vibrates with larger amplitude under the influence of the first body.

Examples:-

- 1) When soldiers cross a bridge, they are asked to break their steps. This is because if the frequency of vibration of their marching becomes equal to the natural frequency of the bridge, the bridge would vibrate with a large amplitude due to resonance and the bridge could collapse.



(or)

- Soldiers break-step while crossing a bridge to avoid setting up resonance in the bridge.
- 2) When we tune a transistor radio, we are actually adjusting its natural frequency to that of the in-coming carrier electromagnetic waves from a radio transmitter when the two frequencies match, maximum energy is absorbed from the in-coming wave and the sound is clearly heard with appreciable amplitude. (Tuning of radio and TV)
 - 3) Consider a swing on which a child sits, when the swing is given a series of pushes by the person of child with a frequency equal to the natural frequency of the swing, the swing oscillates with larger amplitude.
 - In this case, resonance occurs between the oscillatory force applied by the person and the swing.
 - 4) A pronounced rattling sound may occur in a car when it is travelling at a particular speed, but if it travels faster or slower than this speed no rattle occurs. This is due to the resonance between the car engine and the body of the car.
 - 5) The working of flute and whistle also based on resonance.

**The Speed of Sound:**

- For the propagation of sound material medium is required.

Ex: Solid, Liquid or gas.

- Bell-jar experiment proves for the propagation of sound medium is necessary.
- The speed of sound depends on the medium.
- Speed of sound is more in solids, less in liquids, and least in gases.

$$V_{\text{Solid}} > V_{\text{Liquid}} > V_{\text{Gas}}$$

- The material medium which transmits sound should possess two important properties
 - (i) Elasticity
 - (ii) Inertia
- The speed of sound is less than the speed of light, that's why thunder is heard much after the flash of lightning is seen.
- Speed of sound in vacuum is zero.

Factors effecting on speed of sound in a gas (air):

1) Temperature:-

- The speed of sound in a gas is directly proportional to the square root of its absolute temperature.
- $V \propto \sqrt{T}$
- So, the speed of sound increases with the increase in temperature of the gas.
- For 1°C rise of temperature of a gas, the speed of sound in a gas increases by 0.61 ms⁻¹.
- At 0°C, speed of sound in air is 331 ms⁻¹
- At 22°C, speed of sound in air is 344 ms⁻¹

2) Effect of density:-

- The speed of sound in a gas is inversely proportional to the square root of its density.
- $V \propto \frac{1}{\sqrt{d}}$ Where 'd' is density of gas.
- So, the speed of sound increases with decrease in density of gas.
- density of Hydrogen is less than all other gases. Therefore speed of sound in Hydrogen is more.

Effect of Humidity:-

- The percentage of water vapour present in the air called humidity.
- The speed of sound is more in humid air (wet air) than in dry air because the density of humid air less than the density of dry air.
- If humidity increases, the speed of sound is also increases.

Effect of pressure:-

- Pressure has no effect on the speed of sound in a gas.

Effect of wind:-

- As the sound is carried by air, so the speed of sound increases when the wind blows in the direction of sound and speed of sound decreases when the wind blows in the opposite direction of sound.

Note: The speed of sound in air, water, steel at 0°C is given in the following table.

Medium	Speed (m/s)
Air	331
Water	1450
Steel	5000

Subsonic Speed:

- When a body moves in air with a speed less than the speed of sound, it is said to have a subsonic speed.

Ex: car, bus are travelling with this speed.

Sonic Speed:

- The object which moves with the speed of sound is called sonic speed.

Ex: Speed of sound in air is 331 ms^{-1} .

Supersonic speed:

- When a body moves in air with a speed greater than the speed of sound, it is said to have a supersonic speed.

Ex: Bullets, supersonic air craft are travelling with this speed.

- The speed of supersonic air craft is 1200 km/hr (or) $768 \text{ miles per hour}$.

Shock wave:-

- When a supersonic air craft moves through air, it leaves behind itself a conical region of disturbance which spreads continuously. Such a disturbance is called a shock wave.
- Shock waves travel at supersonic speeds and carry a huge amount of energy.

Sonic boom:-

- When a shock wave reaches a person, he hears a sharp, loud sound called a sonic boom.

Bow waves:-

- When a motor boat in a sea travels faster than sound, then waves are produced on the surface of water. These waves are called bow waves.

Hypersonic speed:-

- When a body moves in air with a speed 5 times greater than the speed of sound, it is said to have a hypersonic speed.

Ex: Missiles are travelling with this speed.

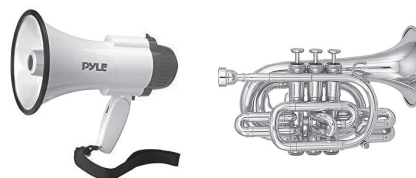
Mach Number (M.N.):

- It is defined as the ratio of speed of source of sound to the speed of sound in the same medium.
- If $M.N. > 1$, body is called supersonic
- If $M.N. < 1$, body is called subsonic
- If $M.N. > 5$, the body is said to be moving with hypersonic speed.

ECHO:

- It is the phenomenon of repetition of sound due to its reflection from the surface of a large obstacle.
- To hear echo, the minimum distance between the observer and the reflector should be 17.2 m . (17m)

- Human ear can hear two sounds separately only, if they reach the ear after an interval of $\frac{1}{10}$ second. This is also called as persistence of hearing.
- (or)
- The minimum time gap to hear the echo is 0.1 second.
- Stethoscope, Megaphone, Trumpet works on the principle of multiple reflection or reflection of sound.
- The stethoscope was invented by “Laennec”

**Uses of ECHO:**

- 1) This principle is used in SONAR for finding depth of sea.
- 2) Bats can fly skillfully at night in the utter darkness of caves.

Reverberation:

- The persistence of sound in a closed room as a result of multiple reflections of sound even after the source of sound is turned off is known as Reverberation.
- Reverberation time depends on the size of enclosure, nature of the reflecting materials and the area of the reflecting surfaces.

Methods of reducing reverberation:-

- 1) By covering the walls and roof of the auditorium with sound absorbent materials like compressed fibreboard, rough plaster.
- 2) Providing heavy curtains.
- 3) By using good sound absorbing materials for the seats.

Doppler's effect in sound:-

- Whenever there is a relative motion between a source of sound and listener, the apparent frequency of sound heard by the listener is different from the actual frequency of sound emitted by the source.

Doppler effect in light:

- Whenever there is a relative motion between a source of light and observer, the apparent frequency of light received by observer is different from the actual frequency of light emitted from the source of light known as Doppler's effect.
- i) When source and observer approach each other, apparent frequency increases (wavelength decreases) this is called Blue shift.
- ii) When source and observer are moving away from each other apparent frequency decreases (wavelength increases). This is called Red shift.

Applications of Doppler's effect

- 1) By police to check over speeding of vehicles.
- 2) at airports to guide the aircraft.
- 3) In the military to detect enemy aircraft.
- 4) By astrophysicists to measure the velocities of planets and stars.
- 5) To study heart beats and blood flow in different parts of the body.

RADAR:

- It works on the principle of Doppler's effect.
- Radio waves are used in RADAR.
- A radar transmitter sends out pulses of radio waves from a rotating aerial. A solid metal object in the path of the beam would reflect some of the waves back to the aerial.
- The time gap between transmitting the pulse and receiving the reflection allows a computer to calculate how far away the object is.
- This type of system is useful for "navigating ships" and "aircraft" especially in the fog or at night.

**Speed of sound in different media at 25°C.**

State	Substance	Speed in ms ⁻¹
Solids	Aluminium	6420
	Nickel	6040
	Steel	5960
	Iron	5950
	Brass	4700
	Glass (Flint)	3980
Liquids	Water (sea)	1531
	Water (distilled)	1498
	Ethanol	1207
	Methanol	1103
Gases	Hydrogen	1284
	Helium	965
	Air	346
	Oxygen	316
	Sulphur dioxide	213

SOUND - Practice Bits

- | | |
|--|---|
| <p>1. A device used in laboratory to produce pure sound of some particular frequency is called.
 1) sonometer
 2) tuning fork
 3) sonar
 4) none of the above</p> <p>2. When sound energy propagates through a material medium, the particles of medium move.
 1) in forward direction
 2) in backward direction only
 3) forward and backward about their mean position
 4) none of the above</p> <p>3. When a sound wave travels through air, such that at some particular instant, the particles of air come closer to one another, then such a region of air is called.
 1) pressurised air
 2) compression
 3) rarefaction
 4) none of the above</p> <p>4. When a sound wave travels through air, at some places, the pressure of air decreases. These places of low pressure are called.
 1) rarefactions
 2) depressurisation
 3) compressions
 4) none of these</p> <p>5. The sound will not travel through.
 1) gases 2) solids
 3) liquids 4) vacuum</p> <p>6. The sound travels fastest in case of.
 1) vacuum 2) solids
 3) liquids 4) gases</p> | <p>7. A sound wave is made to travel in air as well as in water. It will cover in same time.
 1) more distance in air
 2) more distance in water
 3) same distance in air and water
 4) none of the above</p> <p>8. A boy 2 km from a train puts his ear on the rails. He finds that the sound of train coming through the tracks is decreasing. He conclude that.
 1) the train is moving towards him
 2) the train is moving away from him
 3) the train is stationary
 4) none of the above</p> <p>9. During thunderstorm, lightning is seen at once, but the crack of thunder is heard after sometime. It is because sound travels.
 1) slower than light
 2) faster than light
 3) at the same speed as light
 4) none of the above</p> <p>10. The speed of sound in air is 330ms^{-1}. It the crack of thunder is heard 6.5 sec after the lightning is seen, the approximate height of the cloud is.
 1) 2054 m 2) 2045 m
 3) 2154 m 4) 2145 m</p> <p>11. The number of vibrations produced by a vibrating body in one second is called.
 1) oscillation 2) amplitude
 3) time period 4) frequency</p> <p>12. The SI unit of frequency is.
 1) hertz 2) metre
 3) metre / second 4) metre / second²</p> |
|--|---|

- 13. One mega hertz is equal to.**
 1) 10^{-6} Hz 2) 10^{-3} Hz
 3) 10^3 Hz 4) 10^6 Hz
- 14. The time taken by a body to complete one vibration is called.**
 1) frequency time 2) vibration time
 3) time period 4) none of the above
- 15. The repetition of sound on reflection from a high building or hills is called.**
 1) echo 2) reverberations
 3) resonance 4) none of the above
- 16. Minimum distance required for producing an echo is.**
 1) 15 m 2) 17 m
 3) 16 m 4) 14 m
- 17. The minimum time required for producing an echo is.**
 1) $1/20$ of second 2) $1/15$ of second
 3) $1/40$ of second 4) $1/10$ of second
- 18. A device used to find the depth of sea is called.**
 1) ocean meter 2) sonometer
 3) depth meter 4) sonar
- 19. The loudness of sound is measured in.**
 1) decibel 2) newton / metre
 3) dyne / cm^2 4) none of the above
- 20. Just audible sound has loudness level between.**
 1) 15 dB - 45 dB 2) 30 dB - 45 dB
 3) 20 dB - 50 dB 4) 10 dB - 25 dB
- 21. Transference of energy in a medium due to the oscillatory motion of the particles of a medium about their mean position is called.**
 1) wave front 2) wave length
 3) wave motion 4) wave velocity
- 22. When a wave is formed on the surface of water, its particles vibrate in the.**
 1) direction of wave propagation
 2) direction of right angles to the wave propagation
 3) in all directions
 4) none of the above
- 23. The maximum displacement of a vibrating particle from its mean position is called.**
 1) frequency 2) amplitude
 3) time period 4) none of the above
- 24. The SI of amplitude is.**
 1) Centimetre 2) millimetre
 3) metre 4) none of the above
- 25. The number of vibrations produced by a body in one second is called.**
 1) amplitude 2) time period
 3) frequency 4) none of the above
- 26. Consider the following statements and select the correct code.**
 A) Ultrasonic sound waves are in audible to human ears.
 B) Ultrasonic waves are used to destroy bacteria in water and to drill holes in hard and brittle materials
 C) Volcanic eruptions emit ultrasonic waves
Code:
 1) A, B and C are correct
 2) A and B are correct
 3) Only B is correct
 4) None is correct
- 27. Time taken to complete one vibration is called.**
 1) vibration time 2) vibration period
 3) time period 4) none of the above

- 28. A vibrating tuning fork is making 100 vibrations in 20 seconds. The period of tuning fork is.**
1) 0.4 s 2) 0.2 s
3) 0.5 s 4) 0.1 s
- 29. The frequency of tuning fork, having a time period 0.05 s is.**
1) 50 Hz 2) 70 Hz
3) 20 Hz 4) 200 Hz
- 30. The unit of frequency is named after.**
1) Issac Newton 2) H.R. Hertz
3) Maxwill Clark 4) Prescott Joule
- 31. Which one of the following frequencies are audible.**
1) 15 Hz 2) 30,000 Hz
3) 25,000 Hz 4) 2500 Hz
- 32. The vibrations with less than 20 Hz frequency are called.**
1) sonic vibrations
2) infrasonic vibrations
3) both (a) and (b)
4) none of the above
- 33. The vibrations with more than 20,000 Hz frequency are called.**
1) supersonic vibrations
2) hypersonic vibrations
3) ultrasonic vibrations
4) none of the above
- 34. The animal that cannot hear ultrasonic vibration is.**
1) deer 2) dog
3) bat 4) dolphin
- 35. The bats and dolphins locate their prey by emitting.**
1) infrasonic wave 2) sonic waves
3) ultrasonic waves 4) all the above
- 36. Watch dogs respond to special device called.**
1) Galton's whistle
2) Geraldand's whistle
3) Greame's whistle
4) None of the above
- 37. Ultrasonic waves are used in.**
1) sonar to find depth of sea
2) locating fish shoals
3) ultrasound scanning
4) all the above
- 38. Sound energy travels through.**
1) solids only 2) liquids only
3) gases only 4) all the above
- 39. The sound will travel fastest in.**
1) liquids 2) vacuum
3) solids 4) none of the above
- 40. You can predict the approaching train that is not in the line of sight by.**
1) listening the sound of engine through air.
2) listening the sound of engine through steel rails
3) both (1) and (2)
4) none of the above
- 41. A person fires a gun at night. An observer half a kilometre away.**
1) first sees the flash and then hears the sound of gun
2) first hears the sound of gun and then sees the flash
3) sees the flash and sound of gun at the same time
4) all the above
- 42. The astronauts communicate with each by radiotelephony on the surface of moon because.**
1) there is no air for the propagation of sound
2) the gravitational pull of moon is too small
3) it is too hot or too cold on the surface of moon
4) none of the above
- 43. An explosion made at a distance of 1km is heard by an observer in 3 s. The speed of sound in air is.**
1) 330 m/s 2) 333.3 m/s
3) 350 m/s 4) 332.3 m/s

44. An electric bell is fitted in a vessel, connected to a vacuum pump. The bell is switched on. The air is gradually withdrawn from the vessel, when the bell is working. The loudness of sound of the bell will.
- 1) increase when air is withdrawn
 - 2) decreases when air is withdrawn
 - 3) does not change when air is withdrawn
 - 4) none of the above
45. The product of frequency and time period of wave is always.
- 1) 1
 - 2) more than 1
 - 3) less than 1
 - 4) none of the above
46. Which is not a mechanical wave.
- 1) seismic wave
 - 2) electromagnetic wave
 - 3) sea wave
 - 4) sound wave
47. Which of the following needs a material medium for propagation?
- 1) light wave
 - 2) infrared wave
 - 3) radio wave
 - 4) sound wave
48. The mechanical wave produced deep inside the earth are called.
- 1) transverse wave
 - 2) longitudinal wave
 - 3) seismic wave
 - 4) gravitational wave
49. When the particles of a medium oscillate at right angles to the direction of wave propagation, the kind of wave produced is.
- 1) oscillatory wave
 - 2) transverse wave
 - 3) longitudinal wave
 - 4) electromagnetic wave
50. The transverse sound wave are possible only in.
- 1) solids
 - 2) liquids
 - 3) gases
 - 4) (1) and (2)
51. The light waves are.
- 1) transverse in nature
 - 2) electromagnetic in nature
 - 3) longitudinal in nature
 - 4) both (1) and (2)
52. The highest point on the hump of a transverse wave is called.
- 1) trough
 - 2) crest
 - 3) compression
 - 4) amplitude
53. The lowest point on the depression (below) of a transverse wave is called.
- 1) trough
 - 2) crest
 - 3) rarefaction
 - 4) compression
54. The linear distance between a crest and the nearest trough is equal to.
- 1)
 - 2)
 - 3)
 - 4)
55. Wavelength of transverse wave is.
- 1) Linear distance between two consecutive crests
 - 2) Linear distance between two consecutive troughs
 - 3) Linear distance between two consecutive points in the same phase
 - 4) all the above
56. When the particles of a medium oscillate in the direction of wave propagation the wave is.
- 1) transverse wave
 - 2) electromagnetic wave
 - 3) longitudinal wave
 - 4) seismic wave

57. A part of a longitudinal waves in air where particles are closer to one another than normal for a moment is called.
- 1) region of rarefaction
 - 2) region of depression
 - 3) region of compression
 - 4) none of the above
58. A part of a longitudinal wave in air where particles are farther away from one another than normal for a moment is called.
- 1) region of rarefaction
 - 2) region of depression
 - 3) region of compression
 - 4) none of the above
59. The linear distance between two consecutive compressions or rarefactions is.
- 1) 2) 3) 4) none of these
60. Noise level above which of the following limits can cause death in human beings.
- 1) 120 dB 2) 180 dB
 - 3) 80 dB 4) 110 dB
61. Which one of the following is incorrect about ultrasound?
- 1) 'Ultrasound' literally means sound beyond the range of human beings
 - 2) Ultrasound waves are produced by the effect of piezoelectric effect.
 - 3) Like electromagnetic radiation, sound waves can pass through vacuum
 - 4) When the ultrasound wave is introduced through a homogenous tissue, they pass unimpeded till they meet another tissue or organ.
62. Select the correct statements from the codes given below.
- A) The loudness of sound is determined by its Amplitude
B) The speed of sound is maximum in vacuum
- 1) A only 2) B only
 - 3) Both A and B 4) Neither A or B
63. A clearly separate echo of a sound is heard when the minimum distance of the reflecting surface from the source of sound is.
- 1) 10 m 2) 17 m
 - 3) 34 m 4) 100 m
64. Bats can fly in the dark because.
- A) They have better vision in the dark
B) The pupils of their eyes are very big
C) They are guided by ultrasonic waves produced by them
- Which of the above is/are correct?
- 1) Only A 2) Only B
 - 3) Only C 4) A and B
65. A police van sounding its siren is moving away from an observer. The pitch (shrillness) of the sound from the siren relative to its normal pitch is.
- 1) Lower 2) Same
 - 3) Higher
 - 4) Lower or higher depending on the speed of the van
66. Sound waves in air are which one or the following waves?
- 1) Transverse waves
 - 2) Longitudinal waves
 - 3) May be of both types
 - 4) None of the above

67. Which one of the following is incorrect with respect to the factors which affect the velocity of sound in air or gas?

- 1) Velocity of sound is independent of change in pressure
- 2) Velocity of sound increases with the rise in temperature and is proportional to the square root of the temperature.
- 3) The velocity of sound decreases with the increase of humidity in the atmosphere
- 4) The velocity of sound waves is more in the direction of the wind and less in the opposite direction.

68. Consider the following statements and select the correct code.

- A) The ear is able to distinguish two sounds as separate sounds only if they reach the ear at least $1/10^{\text{th}}$ second apart
- B) In order that an echo may occur, the following surface must be at least 33 metres away

Code:

- 1) Both A and B are correct
- 2) Both A and B are incorrect
- 3) Only A is correct
- 4) Only B is correct

Answers

1) 2	2) 3	3) 2	4) 1	5) 4	6) 2	7) 2	8) 2	9) 1	10) 4
11) 4	12) 1	13) 4	14) 3	15) 1	16) 2	17) 4	18) 4	19) 1	20) 4
21) 3	22) 2	23) 2	24) 3	25) 3	26) 3	27) 3	28) 2	29) 3	30) 2
31) 4	32) 2	33) 3	34) 1	35) 3	36) 1	37) 4	38) 4	39) 3	40) 2
41) 1	42) 1	43) 2	44) 2	45) 1	46) 2	47) 4	48) 3	49) 2	50) 4
51) 4	52) 2	53) 1	54) 2	55) 4	56) 3	57) 3	58) 1	59) 4	60) 2
61) 3	62) 1	63) 2	64) 3	65) 1	66) 2	67) 3	68) 3		

3. LIGHT

- Light is form of energy.

OPTICS: Optics is the branch of physics which deals behaviour and properties of light.

- According de-Broglie light has dual nature. Clearly light behaves as wave and particle both.
- Speed of light was first measured by Roemer (1678 A.D.)
- Speed of light is maximum in vacuum and air $3 \times 10^8 \text{ ms}^{-1}$

Medium	Speed of light
Vacuume	$3 \times 10^8 \text{ ms}^{-1}$
Water	$2.25 \times 10^8 \text{ ms}^{-1}$
Glass	$2 \times 10^8 \text{ ms}^{-1}$

Luminous bodies:-

- A body which emits light on its own is called luminous bodies.

Ex:- Sun, Stars, Lamp etc.

Non-Luminous bodies:-

- A body which emits light does not on its own is called non-luminous bodies.

Ex:- Earth, Moon, Wall

Transparent bodies:-

- The bodies which allows the light energy passing through them are called transparent bodies.

Ex:- glass and pure water.

Translucent bodies:-

- The bodies which allows the light partially is called translucent bodies.

Ex:- Butter paper, ground glass, oiled paper.

Opaque bodies:-

- The bodies which does not allows the light energy passing through them are called opaque bodies.

Ex:- Table, chair, wall, earth etc.

Velocity of light in decreasing order is:

- Vacuum > Air > Water > Glass > Diamond
- Light takes 8 minute 19 second (499 sec) to reach from sun to earth.
- The light refelected from moon takes 1.28 second to reach earth.

Theories of Light:

1) Newton's Corpuscular Theory of Light:

- The first understanding of nature of light was put forth by Newton (1642-1727) and is called Newton's corpsuscular theory.

Sailent Points of Newton's corpuscular theory of light:-

- It assumes that light consists of extremely small and perfectly elastic particles called corpuscles.
- Rectilinear propagation of light is explained by the straight line motion of corpuscles.
- Colour vision is explained on the basis of different size of corpuscles.
- Reflection and refraction are on the basis of repulsion and attraction of corpuscles.
- Polarisation, diffraction and interference of light cannot be explained.
- It predicts velocity of light is more in denser medium than in rarer medium, which is incorrect.

Huygen's wave theory of light:-

- The wave theory of light was put forward by Dutch physicist **Christian Huygens**.
- The basic assumption of this theory is that light travels in the form of waves from the source of light.

Assumptions of Huygen's wave theory:-

- It assumes the light travels in the form of waves.
- The rectilinear propagation of light explained by the advancement of wave-front along the direction of normals drawn to it.
- Colour vision is explained on the basis of difference in wavelengths.
- Reflection and refraction are explained on the basis of secondary wavefronts.
- Polarisation, diffraction and interference of light cannot be explained.
- It predicts velocity of light is more in denser medium than in rarer medium which is incorrect.

Electromagnetic theory of light:-

- This theory was put forward by **James Clark Maxwell** in 1865 which predicted theoretically that heat, light and electric energy are propagated in free space at the speed of $3 \times 10^8 \text{ms}^{-1}$ (speed of light) as electromagnetic waves.
- Twenty years later in 1885 Hertz confirmed this theory by producing and detecting electromagnetic waves experimentally.
- A varying (changing) electric field can generate magnetic field conversely, a changing magnetic field can generate electric field. So, an electromagnetic wave propagates continuously due to the mutual generation of electric and magnetic fields.
- In electromagnetic wave, the electric field and magnetic field always mutually perpendicular to each other and also to the direction of propagation of wave. From this it implies that electromagnetic waves are transverse in nature.
- Electromagnetic waves carry energy. They can transfer this energy to the objects placed in their path. Infact every electromagnetic wave is transportation form of energy.
- The energy carried by an electromagnetic wave is distributed equally between the electric and magnetic fields.

- Originally, the transmission of energy occurs in the form of electromagnetic waves through space. So no material medium is required for their propagation. They are also called non-mechanical waves.

Quantum theory of Light:-

- Quantum theory of light was proposed by “**Max Plank**” in 1900.
- This theory stated that light or any other emission is stream of tiny particles (packets) were called quanta (or) photon.
- So, in the quantum theory we have a dual nature, i.e. it behaves like an electromagnetic wave as well as particle.

Sailent features of Quantum theory:-

- Every electromagnetic wave is associated with a photon. The photon is a massless and chargeless particle.
- The photon is particle form of electromagnetic energy.
- Every electromagnetic wave is the form of certain quanta of energy. Quanta means a discrete packet of energy. A quanta of energy is also called photon.
- The magnitude of energy (E) in a packet or quanta or photon of an electromagnetic wave of frequency (ν) is given by the expression.

$$E = h\nu \quad \text{----- (1)}$$

$$\text{But } \nu = \frac{c}{\lambda}$$

$$E = \frac{hc}{\lambda} \quad \text{----- (2)}$$

Where, E - energy of photon

h - Plank's constant = 6.62×10^{-34} J.S.

ν - Frequency of electromagnetic wave

λ - Wavelength of E.M. wave

c - velocity of light.

From above equation (1)

$$E \propto \nu \quad (\text{h - is constant})$$

- Energy of photon is directly proportional to the frequency of electromagnetic wave. i.e more the frequency more is the energy possessed by electromagnetic wave.

From above equation (2)

$$E \propto \frac{1}{\lambda} \quad (\text{C is constant})$$

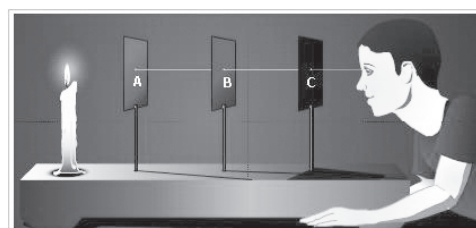
- Energy of photon is inversely proportional to the wavelength of electromagnetic wave. i.e. less the wavelength more is the energy possessed by electromagnetic wave.
- The Nobel prize in physics 1918 was awarded to Max Plank for this theory.

Properties of Light:

1. Rectilinear propagation of Light
2. Reflection of Light
3. Refraction of Light
4. Scattering of Light
5. Total internal reflection of Light
6. Interference of Light
7. Diffraction of Light
8. Polarisation of Light
9. Dispersion of Light

1) Rectilinear Propagation of Light:

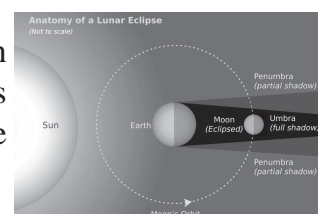
- The light travels always in straight line path is called Rectilinear propagation of light.

**Applications:**

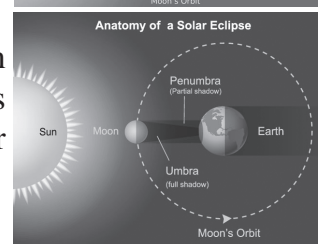
- 1) Formation of images
 - 2) Formation Eclipses.
- The light travels straight lines in a homogeneous transparent medium is called rectilinear propagation of light.
 - When an opaque object is placed between a source of light and a screen, a shadow of the object is formed on the screen due to rectilinear propagation of light.
 - The formation of Lunar and Solar eclipse is based on rectilinear propagation of light.

Lunar eclipse:

- When the earth comes between the sun and the moon, then shadow of the earth falls on the moon and the sun light is not incident on the moon. Therefore, moon is not visible at the earth. This phenomenon is called lunar eclipse.

**Solar eclipse:**

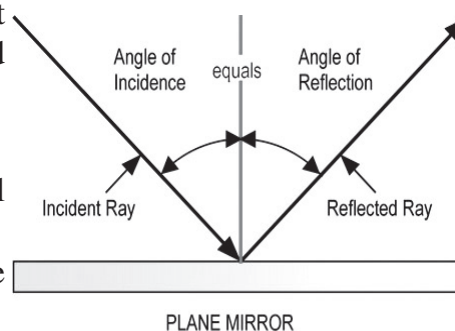
- When the moon comes between the sun and the earth, then the shadow of the moon falls on the earth and the sun is not visible at earth. This phenomenon is known as solar eclipse.

**2) Reflection of Light:-**

- A part of light on striking the surface is sent back in the original medium. It is called reflection of light.

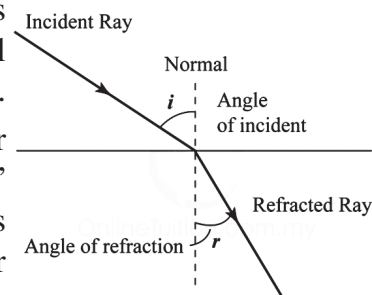
Laws of Reflection:

- 1) The incident ray, the reflected ray and the normal lie in the same plane, at the point of incidence.
- 2) The angle of incidence is always equal to the angle of reflection.



3) Refraction of Light:-

- The phenomenon due to which a ray of light deviates from its original path, while travelling from one optical medium to another optical medium is called refraction.
- When light travels from one medium to another medium, its wavelength changes but its “frequency” remains the same. (wavelength of light wave decreases when it travels from a rarer medium (air) to denser medium (glass))

**The direction in which a ray of light bends depends upon:-**

1. The medium through which the ray of light is initially travelling
2. The optical density of the two media which gives rise to the phenomenon of refraction.

Cases in Refraction of Light:**Case (i):**

- When ray of light travels from an optically rarer medium (air or vacuum) to an optically denser medium (glass or water) it always bends towards the normal, drawn at the point of incidence.
- In this case, the angle of incidence is always greater than the angle of refraction.
()

Case (ii):

- When a ray of light travels from an optically denser medium (glass or water) to an optically rarer medium (air or vacuum) it always “bends away” from the normal, drawn at the point of incidence.
- In this case ()

Case (iii):

- When an incident ray strikes normally at the point of incidence it does not deviate from its path i.e. it suffers no refraction.
- In this case, the angle of incidence is equal to the angle of refraction and their magnitudes are zero.

Laws of Refraction:-**1 Law: Snell's law of Refraction:-**

- This law was discovered by willebrord snell in 1621.

Statement:-

- The ratio between the values of sine of angle of incidence and the sine of angle of refraction is called refractive index of material.
- It is denoted by (μ) .

- Thus, if the angle of incidence is situated in air and the angle of refraction is situated in glass then the refractive index of glass is $\mu_g = \frac{\sin i}{\sin r}$
- The superscript denotes the medium in which light is initially travelling and the subscript the medium in which light travels after refraction.
- The ratio gives the refractive index of the second medium w.r.to the first medium.
- If the superscript and the subscript are not given, it is assumed that light is initially through air.

2nd Law:

- The incident ray, the refracted ray and the normal lie in the same plane at the point of incidence.

Another definition for Refractive index is:

Refractive Index:

- It is the ratio between the velocity of light in vacuum and the velocity of light in the given optical medium.

$$\mu = \frac{\text{Velocity of light in vacuum}}{\text{Velocity of light in given medium}}$$

- Refractive index of Some common substances.

Substance	Refractive Index
Vacuum	1
Air (STP)	1.0003
Water (20°C)	1.33
Glass	1.5
Crown glass	1.52
Flint glass	1.65
Diamond	2.42
Cyclohexane	1.46
Chloro benzene	1.52

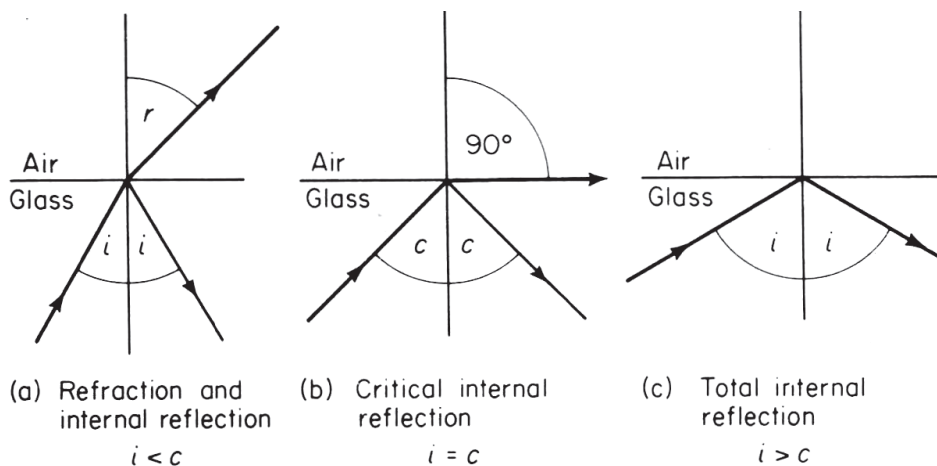
Applications of Refraction:

1. Twinkling of stars
 2. Sun appears bigger during sun-set and sun-rise.
 3. A stick appears bent short when immersed in water.
 4. The bottom of beaker filled with water appears raised
 5. When we sit around a camp fire the face of the person sitting opposite to you, appears to shimmer. This happens due to refraction of light.
 6. An object in a denser medium when seen from a rarer medium appears to be at a smaller distance, that's why
 - i) A fish in a pond when viewed from air appears to be at a smaller depth than actual depth
 - ii) A coin at the base of a vessel filled with water appears raised.
- Early sunrise and late sun-set due to atmospheric refraction.



4) Total Internal Reflection (TIR):-

- The phenomenon due to which a ray of light, while traveling from an optically denser medium to an optically rarer medium, gets reflected into the optically denser medium, at the surface of separation is called T.I.R.



Critical Angle:

- Angle of incidence in a denser medium, for which the angle of refraction in the rarer medium is 90° is called critical angle.

Conditions for T.I.R.:-

1. Rays of light must travel from an optically denser medium to an optically rarer medium.
2. The angle of incidence in the optically denser medium, must be greater than the critical angle.

Applications:-

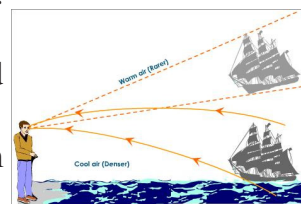
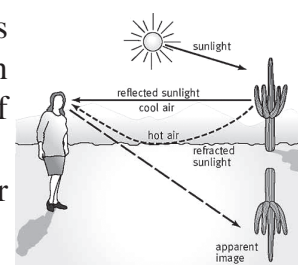
1. Sparkling of diamond (or) the brilliance of diamond due to TIR.
2. Formation of Mirage and looming.

Looming:-

- Mirages of Arctic region where air closer to the ground is much colder than the air above it, known as looming. In very cold regions, some times an erect virtual image of distant object is seen above the object.

Ex:- A ship moving in the sea is observed hanging in mid-air above the ship itself.

3. Shining of air bubble in water (or) Glittering of air bubble.
4. Cracked glass sparkles.
5. The upper surface of water contained in a beaker and held above the eye level appear silvery.
6. An empty test tube held obliquely in water and viewed from the top appear silvery.
7. A crack in a glass window pane appears silvery on account of the presence of air in the crack.



8. Bubbles of air rising up in a fish tank (aquarium) appear silvery (If the rays of light strike them at an angle which is greater than 48° i.e. critical angle of water).

9. Optical Fibre:-

- The working of optical fibre is based on TIR.
- Its inner part is core of higher refractive index surrounded by another layer of glass (cladding) of lower refractive index.
- When light enters from the one end of the core and moves towards cladding, then TIR takes place again and again and light propagates through it.

Uses:-

- i) Doctors use optical fibres to examine patients stomach which is called Endoscopy.
- ii) Optical fibre can be used to send telephone signals.
- iii) Optical fibres are used in decorative table lamps.

5) Scattering of Light:-

- When light is incident on small particles it is absorbed by them. This absorbed light is then send into all directions. This phenomenon is called scattering.

Applications of scattering of light:-

- According to Rayleigh scattering formula
- $\text{scattering} \propto \frac{1}{\lambda^4}$; Where λ is wavelength
- According to above formula, Red colour of light is scattered least (wavelength is max) and violet colour of light is scattered most (wavelength is least).
- Danger signals are of red colour because red colour of light scatters least, therefore these signals can be seen from far away.
- The sky appears blue because in sunlight the blue colour of light is scattered maximum and this scattered blue light enters our eyes and therefore sky appears blue.
- Clouds appear white because clouds are consists of large size of droplets of water and dust particles which scatter all colours of light equally. The mixture of all seven colours of sun light is white.
- Sky appears black to astronauts from space because there is no atmosphere in space and therefore no scattering of light takes place.

The red sunset (or) sunrise:

- At sunset or sunrise, the sun's rays have to pass through a larger distance in the atmosphere. Most of the blue and other shorter wavelengths are removed by scattering. The least scattered light reaching our eye, the sun looks reddish. This explains the reddish appearance of the sun near the horizon.
- Distant hills appear blue because of the scattering of blue light by the water vapour present in air.
- The blue colour of sea water or deep water is due to scattering of light.

6) Interference of Light:

- The physical effect of superposition of waves from two sources vibrating with same frequency and amplitude is called the interference of waves.
- The formation of anti-nodal and nodal lines results in relatively bright and dark regions called interference pattern of waves.
- Interference is not limited to water waves but is a characteristic phenomenon of all waves including light waves.
- The phenomenon of interference was first explained by Young.

Applications:-**1) Formation of Newton's rings:**

- Newton's rings are formed due to interference between the waves reflected from the top and bottom surfaces of the air film enclosed between the lens and the plate.
- 2) The beautiful colour of thin films such as soap bubble or a thin layer of oil on water are seen due to interference of light reflected from the upper and lower surfaces of the this film.

Types of Interference:-

- i) Constructive
- ii) Destructive

i) Constructive Interference:-

- At some points in the medium, intensity of light is maximum this interference is said to be constructive.
- In constructive interference there is overlap of crest of one with crest of the other and resultant amplitude is max.

ii) Destructive:

- At certain other points, intensity of light is minimum, this interference is said to destructive interference. In destructive interference there is overlap of crest of one wave on trough of the other and the resultant amplitude is min (zero)

7) Diffraction:-

- The bending of waves around the edges of an obstacle (aperture) is called diffraction.
- This phenomenon was discovered by Grimaldi.
- The theoretical explanation was first given by Fresnel.

Applications:

- 1) The silver lining surrounding the profile of a colud when the sun rises behind the clouds is due to the diffraction.
- 2) The silver lining surrounding the profile of a mountain just before sun rise in north is due to the diffraction.
- 3) The colourful rings formed around the street light when denser frost in winter is due to the effect of diffraction.
- 4) The formation of colours pattern when light fall on the gramophone record (CD or DVD) due to diffraction.
- 5) Hold your two fingers close sothat these is a narrow slit formed between the fingers, dark lines are seen due to diffraction.

8) Polarisation of Light:-

- The phenomenon due to which the vibrations of light (electric vector) are restricted in a particular plane (in particular direction) is called polarisation of light.
- The materials such as tourmaline crystal or Quartz crystal (or) nicol prism which can cause polarisation are called polariser.
- Polarisation proves the transverse character of light waves.

**Polaroids:-**

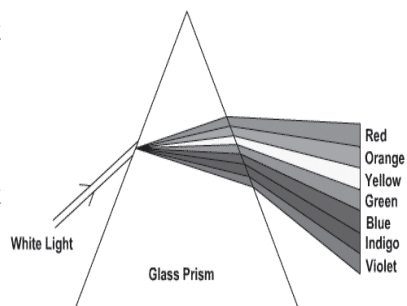
- The materials which polarise the light are called polaroids.
- Quartz, tourmaline calcite crystals are natural polarising materials.

Uses of Polaroids:-

- i) Wind-shield of automobiles are also made of polaroid sheets.
- ii) Polaroids are used to improve colour contrast in old oil paints.
- iii) The polaroid sheets also reduced head-light glare of motor car driving at night.
- iv) Sunglasses fitted with polaroid sheets protect the eyes from glare.
- v) Polaroids are useful in three dimensional picture i.e. Holography.

9) Dispersion of Light:-

- The splitting of white light into its constituent colours is called dispersion of light.
- The band of colours formed on a screen due to dispersion is called “SPECTRUM”.
- In 1665 sir Isaac Newton showed that natural light actually consists of seven colours. All the colours of light mixed together appears white.
- When white light is incident on a glass prism, it splits into its seven colour components in the sequence of VIBGYOR. These colours are violet, indigo, Blue, Green, Yello, Orange and Red. This phenomenon is called dispersion of white light.
- The refractive index of glass is maximum for violet colour and minimum for red colour of light. Therefore, violet colour of light deviated maximum and red colour of light deviated least.

**Angular dispersion:-**

- The angular dispersion is defined as the difference in the angles of deviation of two extreme colours.
- The mean deviation of all the colours is noticed yellow colour.

Application:-

- 1) Prism can split light into constituent colours due to dispersion.

2) Formation of RAINBOW:-

- Rainbow is produced due to the dispersion of light by small raindrops floating in the air after rain.
- The rainbow is seen when the sun is behind the observer.
(or)
- Rainbow formed always opposite to the sun.



Primary rainbow:-

- The primary rainbow is formed when sun light suffers “one reflection” and “two refractions” before emerging from water droplets. In this case violet colour is on the lower side and red colour is on upper side of the rainbow. It has been observed that violet colour makes an angle 40° and red makes angle 42° at eye with the axis of rainbow.

Secondary Rainbow:-

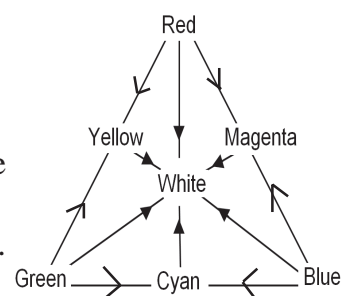
- This is formed by the sunrays which suffer “two internal reflections” and “two refractions” from the water droplets and giving rise violet colour on upperside and red on lower side of the rainbow.

Colours:-**i) Primary Colours:-**

- Red, Blue and Green are primary colours.

ii) Secondary Colours:-

- The colours obtained by mixing two primary colours are called secondary colours.
- Magenta, Cyan and Yellow are called secondary colours.

**iii) Complementary pair of Colours:-**

- A pair of colours, one primary colour and other secondary colour, which on mixing form white light is called complementary pair of colours.

Ex:- 1. Yellow + Blue = White

- Yellow and Blue colours are called complementary pair colours.
- 2. Red + Cyan = White
- Red, Cyan colours are complementary pair.
- 3. Green + Magenta = White
- Green, Magenta colours are complementary pair.

Colours of Objects:-

- When light is on an object, it reflects only a part of it. The reflected light gives the objects of their colour.
- A red rose appears red when white light is incident on it because it reflects only red colour of light and absorbs other all colours of light. When same rose is viewed in green light, it appears black because it absorbs green light and reflects no colour of light.

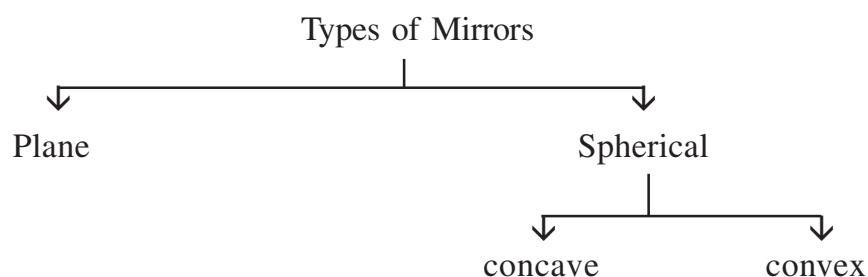
Colour Television:-

- Light from the scene to be televised is split up in red, green and blue components with the help of dichroic mirrors in the TV camera.
- The three colours are then incident on three separate camera tubes, which produce corresponding electrical signals. These signals are transmitted to distant colour TV receivers, which reproduce the scene by adding the three colours.

Substance	White light	Red light	Blue light	Green light
Red petals	Red	Red	Black	Black
Green leaves	Green	Black	Black	Black
Yellow stamens	Yellow	Black	Black	Black
Blue insect	Blue	Black	Black	Black

MIRROR:

- A highly polished surface which is smooth enough to reflect a good fraction of light incident on it is called a mirror.

**1) Plane Mirror:-**

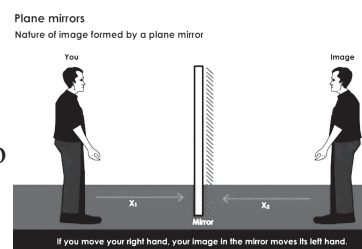
- If the image formed is of same size as our face but laterally inverted (i.e. left face looks right and vice-versa) for all positions, then it is a plane mirror.
- It is the cut part of a plain glass with highly polished one side.

Image:

- When the rays of light, diverging from a point, after reflection or refraction either actually meet at some other point or appear to meet at some other point, then that point is called image of the object.

Characteristics of Image formed by plane mirror:

- The image is virtual.
 - virtual image means cannot be taken on screen.
 - The size of image is equal to that object.
 - The distance of image from the mirror is equal to distance of object from the mirror.
 - The image is erect.
 - Image is laterally inverted. i.e. If a person is wearing a watch on his left hand, the watch appears to be on right hand of the image.
 - If an object moves towards (or away from) a plane mirror with speed ' V ', the image moves towards (or away) with a speed ' $2V$ ' relative to the object.
 - If a plane mirror is rotated by an angle ' θ ' keeping the incident ray fixed, the reflected ray is rotated by an angle ' 2θ '
- Real image means that images can be taken on a screen.
 - To see his full image in a plane mirror, a person requires a mirror of at least half of his height (size).



Images formed by inclined mirrors:-

- When an object is placed between two inclined mirrors, several images of the object are formed. The number of images formed depends on the angle between the mirrors and the total number of images (n) formed is given by

where θ is angle between two plane mirrors.

- If value is even no then “-1” can be taken into the calculation part.

Examples:

- 1) When the angle between two plane mirrors is 60° , and object is in between the two mirrors, then number of images formed.

$$= 6$$

Here 6 is even number

- $\therefore n = 6 - 1 = 5$

Total number of images = 5

- If value is odd number then “-1” cannot take in calculation part.

- 2) When the angle between two plane mirrors is 72° , and object is in between the two mirrors, then number of images formed.

$$= 5$$

- Here 5 is odd number

$\therefore$ Total number of images formed $n = 5$

Ex: If $\theta = 0^\circ$, (Two plane mirrors parallel to each other)

(infinity)

Ex:- If $\theta = 90^\circ$ (Two plane mirrors perpendicular to each other)

$$= 4$$

- Here ‘4’ is even no.

- $n = 4 - 1 = 3$

Applications of plane mirrors:-

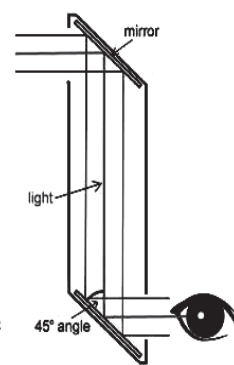
- 1) They are used as looking glass.

2) Kaleidoscope:

- This device based on the principle of multiple reflections in inclined mirrors.
- In Kaleidoscope the idea of formation of a number of images with the help of plain mirrors placed at angle to one another and it is used to make number of beautiful patterns.
- Various types of coloured designs may be seen by a Kaleidoscope.

3) Periscope:

- It is used in construction of reflecting periscope.
- It consists two plane mirrors arranged at an angle 45° each other.
 1. It is used to see above the head of crowds
 2. It is used by soldiers in trench warfare.
- Periscope works on the principle of light reflect on plane mirror.

**Uses of Periscope:**

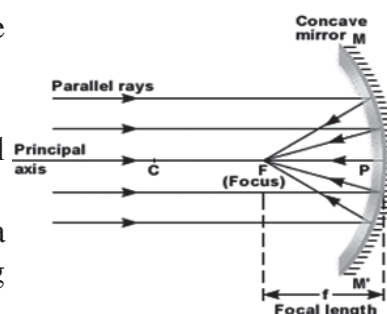
- It is a device which is used to see objects which are not in the direct line of light.
- This is used in submarines tanks and also things outside by soldiers in bunkers to see.

2) Spherical Mirrors:-**Types of Spherical Mirrors:-**

- A spherical mirror is a part of hollow sphere, whose one side is reflecting and other is opaque.

Ex:- Concave mirror and Convex mirror.

- The following terms in connection with spherical mirrors.
- In the Fig., let MPM^1 be a principal section of a spherical mirror i.e. the section cut by a plane passing through pole and centre of curvature of the mirror.



Pole (P):- It is the middle point of the spherical mirror.

Centre of Curvature (C):-

- It is the centre C of the sphere of which the mirror forms a part.

Radius of Curvature (R):-

- It is the radius (R) of the sphere of which the mirror forms a part.

Principal axis:-

- The line passing through the pole and the centre of curvature of mirror is called its principal axis.

Principle focus (F):

- It is point F on the principal axis where a beam of light parallel to the principal axis either actually converges to or appears to diverge from after reflection from a mirror.

Focal Length(f):-

- It is the distance $(f) = PF$ between the focus and the pole of the mirror.

Focal Plane:

- The vertical plane passing through the principal focus and perpendicular to the principal axis is called focal plane.
- Where a parallel beam of light is incident on a concave mirror at a small angle to the principal axis, it is converged to a point in the focal plane of the mirror.

Spherical Mirrors are two types

- i) Concave mirror
- ii) Convex mirror

i) Concave Mirror (Converging mirror):-

➤ A spherical mirror is if it is silvered on the outer bulged surface and reflects from the inner hollow surface.

➤ Relation between f and R :-

➤ $PC = R$

➤ $PF = f$ and $FC = f$

➤ From fig:

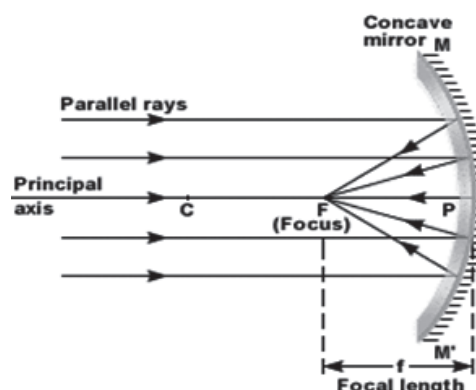
➤ $PC = R$

➤ $PF + FC = R$

➤ $f + f = R$

➤ $2f = R$

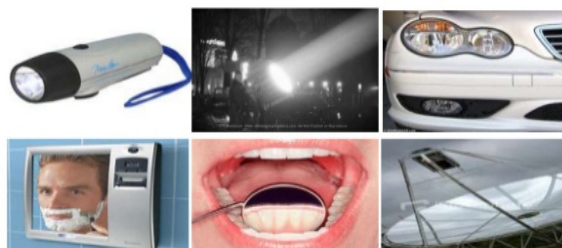
➤ Focal length = $\frac{\text{Radius of curvature}}{2}$

**Characteristics of image formed by concave mirror.**

Position of object	Position of the image	Size of the image	Nature of the image
1. At infinity	at the focus 'F'	Highly diminished point sized	Real and Inverted
2. beyond C	between F and C	Diminished	Real and Inverted
3. at C	at C	same size	Real and Inverted
4. between C and F	beyond C	enlarged	Real and Inverted
5. at 'F'	at infinity	highly enlarged	Real and Inverted
6. between P and F	behind the mirror	enlarged	Virtual and erect.

Uses of Concave Mirror:-**1) Shaving Mirror:-**

➤ A concave mirror is used as a shaving mirror or make-up mirror because it forms erect and enlarged image of the face when it is held closer to the face.

**2) As Head Mirror:-**

➤ E.N.T. specialists use a concave mirror on their forehead. The light from a lamp after reflection from the mirror is focussed into the throat, ear or nose of the patient making the affected part more visible.

3) In Headlights:-

- Concave mirrors are used as reflectors in headlights of motor vehicles, Railway engines, torchlights.
- The source is placed at the focus of concave mirror. The light rays after reflection travel over a large distance as a parallel beam of high intensity.

4) In Solar Furnaces:-

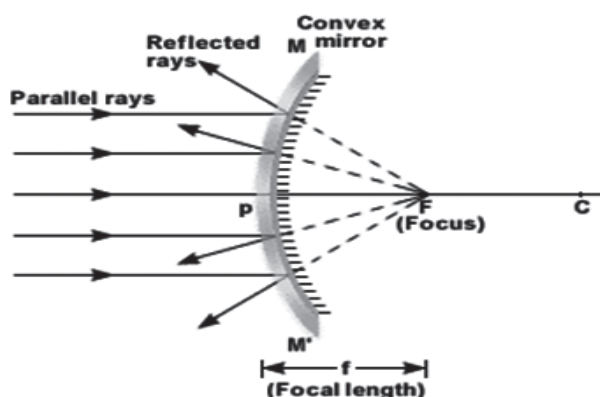
- Large concave mirrors are used to concentrate sunlight to produce heat in solar furnace.

5) In Ophthalmoscope:-

- It consists of a concave mirror with a small hole at its centre. The doctor looks through the hole from behind the mirror while a beam of light from a lamp reflected from it is directed into the pupil of patient's eye which makes the retina visible.

6) In Astronomical Telescope:-

- A concave mirror of large diameter (5m or more) is used as objective in an astronomical telescope. It collects light from the sky and makes visible even those faint stars which cannot be seen with naked eye.
- 7) Concave mirrors used by dentists to see the large images of the teeth of patients. This is because when tooth is within the focus of concave mirror, then an enlarged image of the tooth is seen in the concave mirror. Due to this, it becomes easier to locate the defect in the tooth. The concave mirrors used by dentists are very small in size. They are fitted in a frame with a long handle.

ii) Convex Mirrors (Diverging mirror):-**Characteristics of image formed by Convex Mirror**

Position of the Object	Position of Image	Nature of Image	Size of Image
1. between pole and infinity	between P and F. behind the mirror	Virtual and erect	diminished
2. at infinity	at the focus (F) behind the mirror	Virtual and erect	Highly diminished point sized

Uses of Convex Mirror:-

1. It is used as a rear view mirror in automobiles, because it can cover a very wide field behind the driver and hence enables to see him the traffic behind, without turning his head.
2. It is used as a reflector for street light bulbs, because it diverges rays of light over large area.

**Mirror Formula:-**

- The mirror formula gives the relation between the distance (u) of an object from a spherical mirror, the distance of its image (V) and the focal length (f) of the mirror.

where u - object distance
 v - image distance

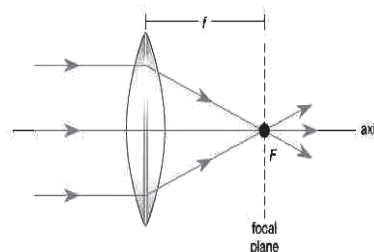
LENSES

- A lense is a piece of transparent glass bounded by two spherical surfaces.
 Lenses are mainly classified into two types.

- i) Convex lens (Converging lens)
- ii) Concave lens (diverging lens)

i) Convex Lens (Converging Lens):

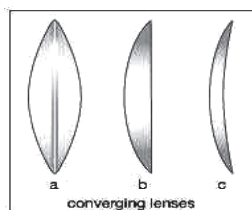
- A convex lens is thick at the centre but thinner at the edges. It has a real focus. (or)
- A piece of transparent, optical material, having one or two spherical surfaces, such that it is thicker in the middle and thinner (tapering) at the edges is called a convex lens.
- Images formed by convex lens
- Except one case, remaining all cases it forms Real and Inverted.

**Exceptional Case:-**

- If the object is placed between focus (F) and optical centre (o). It forms Erect and Virtual and size of the image is Enlarged.

Types of Convex Lens:

- a) Bi-convex lens
- b) Plano-convex lens
- c) Concavo-convex lens

**Uses of Convex Lens:-**

- 1) Convex lenses are used in spectacles to correct the defect of vision called hypermetropia (long sightedness)
- 2) Convex lens is used for making a simple camera.

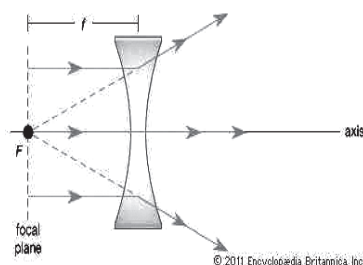
- 3) It is used as a magnifying glass (by palmists, watchmakers)
- 4) Convex lenses are used in making microscopes, telescopes and slide projectors (film projectors) or (cinema projectors) used as burning glass.
- 5) It is used as eye lens for astronomical telescope.
- 6) It is used as an erecting lens in terrestrial telescope.
- 7) It is used in searchlights, spotlights.
- 8) It is used as a simple microscope for
 - a) observing weave patterns of cloth
 - b) observing lines of the hand by palmists.
 - c) For reading fine print
 - d) Observing biological specimens.
 - e) It is used for correcting long sight.

Images formed by convex lenses

Position of object	Position of image	Size of image	Nature of image
1) At infinity	At the focus	Highly diminished	Real & inverted
2) At focus	At infinity	Highly enlarged	Real & inverted
3) Between F_1 & $2F_1$	Beyond $2F_2$	Enlarged	Real & inverted
4) At $2F_1$	At $2F_2$	Same size of object	Real & inverted
5) Beyond $2F_1$ and	Between F_2 & $2F_2$		Real & inverted
6) Between focus (F) Optical centre (o)	On the same side	Enlarged	Virtual & erect

Concave Lens (Diverging lens):

- It is thinner at the centre than at the edges. It diverges a beam of light on refraction through it. It has a virtual focus.

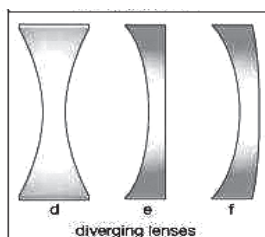


Nature, position and relative size of the image formed by a concave lens for various positions of the object:-

Position of object	Position of image	Size of image	Nature of image
1) At infinity	At focus F_1	Highly diminished point - sized	Virtual and erect
2) Between infinity & optical centre O of the lens.	Between focus F_1	Diminished	Virtual and erect

Types of Concave Lens:

- d) Bi-concave lens
- e) Plano-concave lens
- f) Convexo-concave lens



Uses of Concave Lens:-

1. It is used as an eye lens in Galilean telescope.
2. It is used for correcting short sightedness.

Lens formula:

- A formula which gives the relationship between image distance (v), object distance (u) and focal length (f) of a lens is known as lens formula.

Lens Maker's Formula:-

- It is used while making a lens and is given by
- where μ = refractive index of the material of the lens.
 R_1 and R_2 = radii of curvature of the surfaces of the lens.

Equivalent focal length of a lens combination:

- When two lenses are placed in contact
- When two lenses are placed at a distance 'd'.

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

(P) Power of Lens:

- Power of lens is defined as the reciprocal of its focal length expressed in metres.

$$P = \frac{1}{f(\text{in meters})}$$

- S.I. unit of power of lens is dioptre (D)
- $P = +ve$ for convex lens
- $P = -ve$ for concave lens

Power of a combination of lenses:

- If a no. of lenses are placed in close contact then power of the combination of lenses is equal to the algebraic sum of the powers individual lenses.

$$P = P_1 + P_2 + P_3 + \dots$$

Linear magnification (m):-

- The linear magnification produced by a lens is defined as the ratio of the size of the image formed by the lens to the size of the object.
- $m = \frac{v}{u}$
- where V - image distance
u - object distance

- $m = +ve$ for a virtual image formed by a lens.
- $m = -ve$ for a Real image formed by a lens.
- If $|m| > 1$, the image is magnified
- $|m| < 1$, the image is diminished.
- $|m| = 1$, the image is of the same size as the object.

Human Eye-Camera:-

- The human eye works on the refraction of light through a natural convex lens made of transparent living material and enables us to see things around us.
- The eye lens is a convex lens made of a transparent, soft and flexible material like a jelly made of proteins.
- The focal length of the convex lens used in a camera is fixed and cannot be changed but the focal length of the convex lens present inside the eye can be changed by the action of ciliary muscles.

Retina:-

- The screen on which the image is formed in the eye is called retina.
- It is a delicate inner membrane on the backwall of the eye ball. It contains light sensitive cells called 'rods' and 'cones'.
- Rods are sensitive to intensity of light while cones are sensitive to colours.
- These cells change light energy into electrical signals which send message to the brain via the optic nerves.
- The **retina** of an eye is just like the **film** in a camera.
- The human eye is like a camera. In the eye convex lens (called eye-lens) forms a Real and Inverted image of an object on the light sensitive screen called retina whereas in a camera the convex lens (called camera - lens) forms a Real and Inverted image of an object on the light sensitive photographic film.
- It should be noted that there is an **Eye-lid** in front of the **eye** which is just like the **shutter** in a **camera**.
- In the eye, a convex lens (called eye lens) forms a Real and Inverted image but our mind (brain) automatically translates the image as that of an erect object.

Differences between Human eye and Camera:

Human Eye	Camera
1. Image is formed by the eyelens (convex lens) made of fibrous jelly like material.	1. Image is formed by a convex lens made of glass.
2. A real and inverted image is formed on the retina. The brain automatically interprets the erect image.	2. A real and inverted image is formed on the photographic film.
3. Pupil in the iris controls the amount of light entering the eye.	3. Diaphragm controls the amount of light entering the camera.
4. Time of exposure is controlled by eye-lids.	4. Time of exposure is controlled by the shutter.

Accommodation:-

- It is the ability of the eye lens due to which it can change its focal length so that images of objects at various distances can be formed on the same retina.

Power of accommodation:

- The power of accommodation of the eye is the maximum variation of its power for focussing on near and far objects.
- For normal eye, the power of accommodation is about 4 dioptries.

Persistence of Vision:-

- The phenomenon of the continuation of the impression of an image on the retina for some time even after the light from the object is cut off is called persistence of vision.
- The impression of the image remains on the retina for about $\frac{1}{16}$ of a second.
- Cinematography works on the principle of persistence of vision.
- Modern movies run at a speed of 24 frames per second.

Range of Normal Vision:-

- The distance between infinity and 25 cm point is called the range of normal vision.

Least distance of distinct vision (D):

- The minimum distance from the eye, at which the eye can see the objects clearly and distinctly without any strain is called the least distance of distinct vision.
- For normal eye its value is 25 cm.

Near Point:-

- The nearest point from the eye, at which an object can be seen clearly by the eye is called its near point.
- The near point of a normal eye is at a distance of 25 cm.

Far point:-

- The farthest from the eye at which an object can be seen clearly by the eye is called far point of the eye. For a normal eye, the far point is at “infinity”.

Defects of Vision:**1) Myopia (or) Short-sightedness (or) Near sightedness:**

- It is a vision defect in which a person can see nearby objects clearly but cannot see the distant objects clearly.

Causes of Myopia:-

- The eye-ball being too long, so that the distance between the eye lens and the retina becomes larger.
 - The focal length of the eye lens becomes too short due to the excessive curvature of cornea.
- The far point of an eye suffering from myopia is less than infinity.
 - In a myopic eye, image of distant object is formed in front of the retina.

Correction of myopia:-

- This defect is corrected by Concave lens.

2) Hypermetropia (Long-sightedness) (or) Far-sightedness:-

- It is a vision defect in which a person can see the distant objects clearly but cannot see the nearby objects clearly.

Causes of Hypermetropia:-

- i) The eye ball becomes too small, so that the distance between the eye lens and Retina is reduced.
- ii) The focal length of eye lens becomes too large resulting in the low converging power of the eye lens.

Correction of Hypermetropia:-

- A hypermetropic eye is corrected by using a convex lens of suitable focal length.

3) Presbyopia:-

- This defect is similar to hypermetropia, i.e. a person having this defect cannot see nearby objects distinctly, but can see distant objects without any difficulty. This defect differs from hypermetropia in the cause by which it is produced. It usually occurs in elderly persons. Due to the stiffening (weakening) of the ciliary muscles, the eye lens loses flexibility and hence the accommodating power of the eye lens decreases.
- Presbyopia defect is corrected in the same way as hypermetropia by using spectacles having convex lenses.
- It is also possible that the same person has both the defects of vision myopia as well as hypermetropia. Such a person uses spectacles having “**Bifocal lenses**” in which upper part consists of a concave lens (to correct myopia) used for distant vision and lower part consists of a convex lens (to correct hypermetropia) used for reading purposes.

4) Astigmatism:-

- It is a defect of vision in which a person cannot simultaneously see both the horizontal and vertical views of an object with the same clarity.

Cause of Astigmatism:

- This defect occurs when the cornea of the eye is not perfectly spherical. This results in objects in one direction being well focussed while those in perpendicular direction are not well focussed.

Correction of Astigmatism:-

- It can be corrected by using cylindrical lenses.

5) Colour Blindness:-

- It is a defect of vision in which a person cannot distinguish between various colours.
- It is due to the absence of some cones in the retina of eye.
- This defect cannot be removed.

6) Night Blindness:

- It is most common eye problem and it is caused by the deficiency of Vitamin 'A' in our diet.
- Raw carrots, broccoli are rich in vitamin 'A'.

7) Blind Spot:-

- At the junction of optic nerve and the retina, there are no sensory cells, so no vision is possible on that spot. This is called the blind spot.

8) Cataract:-

- In this defect, an opaque, white membrane is developed on cornea due to which a person loses power of vision partially or completely. This defect can be removed by surgery.

Optical Instruments:**Lens Cameras:**

- It consists of light proof box with a lens system in front and photographic film at the back.
- The lens system, which converges light onto the film, consists of a number of lenses.
- The purpose of using more than one lens is to minimise defects of the image.
- Objects at different distances are focussed on the film by moving the lens system.
- There is a shutter placed between the lens system and the film. When a photograph is taken the shutter opens and closes rapidly.

Simple Microscope:

- It consists of convex lens of very short focal length held close to the eye.
- It forms an erect virtual and enlarged image.

- magnification of simple microscope (m) =

where, D - Least distance of distinct of vision
 f - focal length of lens.

Uses:-

- This is used by watch makers, for reading fine print.

Compound Microscope:-

- A microscope is an instrument used for magnifying minute objects.
- A compound microscope consists of two convex lenses.
 - i) objective (shorter focal length)
 - ii) eye-lens (larger focal length)

- The objective, which is placed close to a well-illuminate minute object, produces a real and enlarged image.
- This image acts as an object for the eye-lens which produces a further enlarged virtual image.
- The image seen in a microscope is inverted.
- Magnifying power (m) =
- Where L = Distance between objective and eye lens.
 f_o = focal length of the objective.
 f_e = focal length of the eye lens.
 D = Least distance of distinct vision

Uses:-

- For observing biological specimens like living cells, Bacteria animal cells, blood cells.

Telescope:

- A telescope provides angular magnification of distant object and therefore produces an effect as if the object is closer to the eye.
- magnifying power of telescope (m) = f_o/f_e
- Where, f_o - focal length of objective lens.
 f_e - focal length of eye lens.

Astronomical Telescope:-

- It consists of an objective of large focal length and an eye-lens of short focal length. The objective forms a real and inverted image of a distant object (a planet or a star) on its focus. This image can be brought within the focus of the eye-lens by adjusting its position. The eye-lens produces an enlarged virtual secondary image. An astronomical telescope produces an inverted image.

Uses:-

- This telescope is used for observing celestial bodies like sun, stars and planets.

Terrestrial Telescope:-

- In a terrestrial telescope an extra lens is placed between the objective and the eye-lens to produce an erect image of the object.

Uses:-

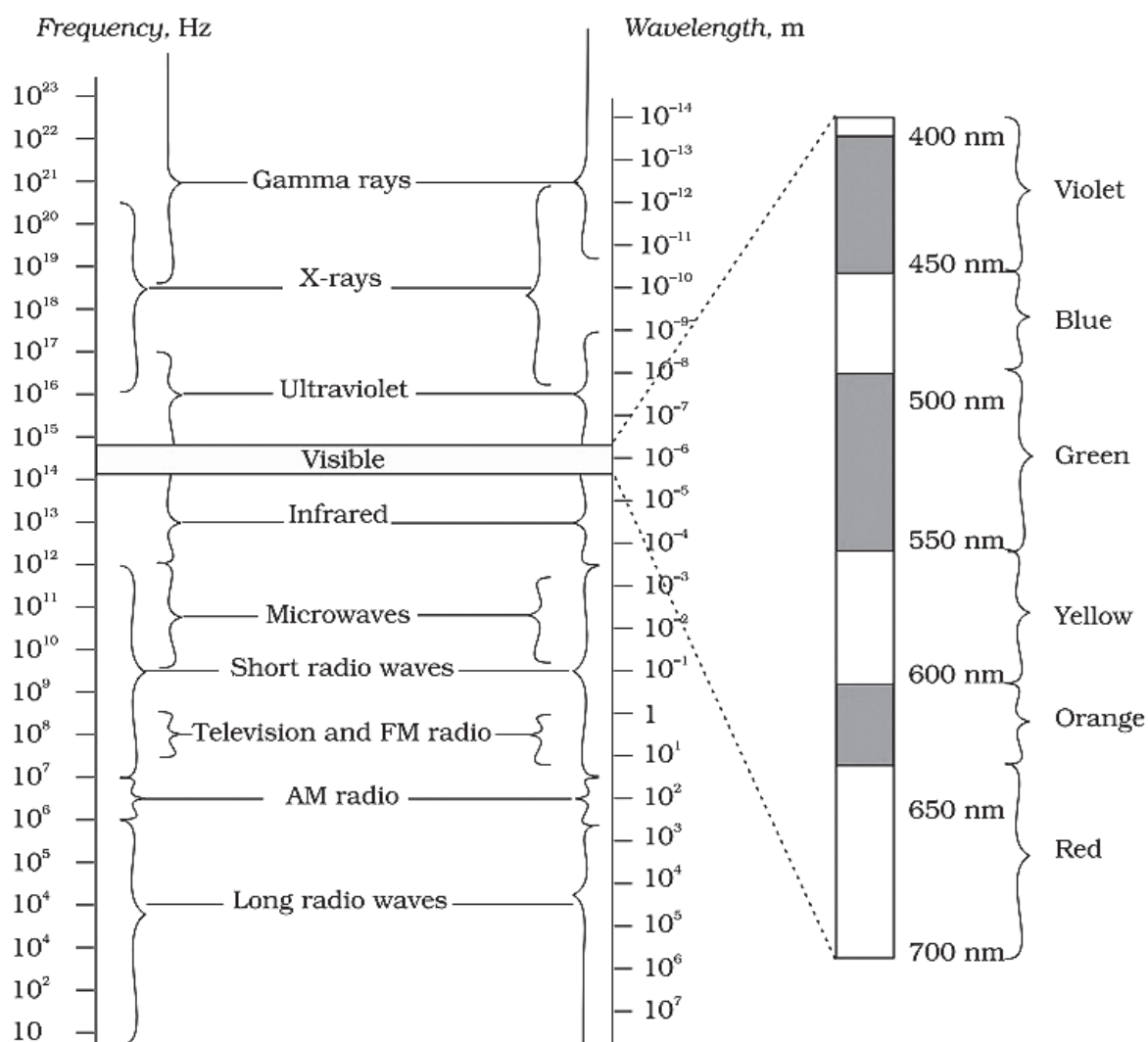
- This is used for observe object on the earth surface

Electromagnetic Spectrum:-

- The electromagnetic spectrum is spread over a vast range in terms of wavelength. It can have wavelength as large as $6 \times 10^6 \text{ m}$ to as small as $6 \times 10^{-15} \text{ m}$.
- There are various regions of electromagnetic spectrum, such as electron waves, radio waves, microwaves, infrared radiations, visible light, ultraviolet light, X-rays, γ -rays, cosmic rays etc. However these above stated regions do not have sharp boundaries, but differ in wavelength and frequency and the effects they produce in matter.
- The table given below gives basic information of various regions of electromagnetic spectrum.

No.	Electromag	Wavelength	Frequency	Source	Applications
1.	Electron waves	$5 \times 10^6 \text{ m}$ to $6 \times 10^6 \text{ m}$	60 Hz - 50 Hz	Weak radiations from A.C. circuits	Lightning
2.	Radiowaves	$1 \times 10^{-1} \text{ m}$ to 10^4 m	$3 \times 10^9 \text{ Hz}$ to $3 \times 10^4 \text{ Hz}$	Oscillating circuits	Radio telephony and broadcasting
3.	Microwaves	$3 \times 10^{-1} \text{ m}$ to $1 \times 10^{-3} \text{ m}$	$1 \times 10^{11} \text{ Hz}$ - $1 \times 10^8 \text{ Hz}$	Oscillating currents in special vacuum tubes	Radar, T.V., Cell phone, micro wave ovens
4.	Heat Radiations	$1 \times 10^{-1} \text{ m}$ to $1 \times 10^{-5} \text{ m}$	$3 \times 10^9 \text{ Hz}$ to $3 \times 10^{13} \text{ Hz}$	All hot bodies	Source of heat for various home and industrial heat applications
5.	Infra-red radiation	$1 \times 10^{-5} \text{ m}$ to $8 \times 10^{-7} \text{ m}$	$1 \times 10^{13} \text{ Hz}$ - $8 \times 10^{14} \text{ Hz}$	Excitation of electrons around atoms or molecules	In night vision devices infra-red cameras, reveal secret documents on ancient walls and treatment of muscular strains
6.	Visible light	$8 \times 10^{-7} \text{ m}$ to $4 \times 10^{-7} \text{ m}$	$4 \times 10^{14} \text{ Hz}$ to $8 \times 10^{14} \text{ Hz}$	Excitation of electrons around atoms during sparking or burning	Gives sensation of vision and colour
7.	Ultra-violet rays	$4 \times 10^{-7} \text{ m}$ to $3 \times 10^{-8} \text{ m}$	$3 \times 10^{14} \text{ Hz}$ to $1 \times 10^{16} \text{ Hz}$	Excitation of electrons around atoms	1. To kill germs and bacteria 2. To detect finger prints 3. To detect fake currency 4. To sterilising surgical instruments 5. Purification of water

8.	X-rays	1×10^{-8} m to 1×10^{-10} m	3×10^{16} Hz 3×10^{18} Hz	Due to knocking of electrons from the inner orbits of atoms	1. Medical radiography 2. Industrial radiography 3. Crystallography 4. Medical X-ray therapy
9.	Gamma (γ) rays	1×10^{-10} m to 6×10^{-15} m	3×10^{18} Hz - 5×10^{22} Hz	Decay of radio-active elements naturally or during fission of fusion	1. Used to study nuclear structure 2. Used in the cure of cancerous tissues
10.	Cosmic rays	Less than 10^{-15} m	More than 10^{23} Hz	Coming from cosmos in all direction	Unknown



Invisible Spectrum:

- The regions of the spectrum which do not excite the retina, and hence, are not visible are collectively called the invisible spectrum.

Infrared Spectrum:-

- The electromagnetic radiations beyond red end of the visible spectrum are called infrared radiations and the spectrum is called infrared spectrum.
- Their wavelength extends from 8000×10^{-10} m to $4,00,000 \times 10^{-10}$ m.
- This part of the spectrum was discovered by William Herschel in about 1800.

Sources of Infrared Radiations:-

- As a matter of fact any substance which radiates out heat energy is a source of infrared radiation.
Ex: Human body, all living beings, animals or plants are infrared emitters.
- However, for the infrared radiations of higher intensity white hot sources of light, such as the sun, arc lamp, burning gases or any other material are the most suitable.

Properties of Infrared Radiations:-

1. Being electromagnetic waves, they are not affected by electric or magnetic fields.
2. They travel with the velocity of light, i.e., $3 \times 10^8 \text{ ms}^{-1}$.
3. They obey the laws of reflection and refraction. It is on account of this reason that parabolic mirrors or lenses can focus these rays. If the focusing is done on a piece of paper or cotton wool, it catches fire.
4. Their wavelength is between 8000×10^{-10} m to $4,00,000 \times 10^{-10}$ m. They can be easily absorbed by glass if the wavelength increases beyond $20,000 \times 10^{-10}$ m. They can easily pass through rock salt.

Uses of Infrared Radiations:-

1. Since they produce heating effect, therefore they are used by the doctors for therapeutic purposes patients with swollen joints, muscles etc, feel relieved when exposed to these radiations.
2. As these radiations have a very long wavelength, they do not easily get scattered in fog or smoke. Thus, they can be used for taking infrared **photographs** with suitable photographic plates and filters in **foggy weather**.
3. Since all hot objects are emitters of infrared radiations, therefore specially prepared devices can see objects in darkness, by receiving infrared radiation. These devices are generally called night vision devices.
4. They are used in surface to **air missiles** for tracking and **destroying enemy planes**. The missile is fired in the direction of the enemy plane. The on-board sensor of infrared radiation in the nose of the missile detects the heat trail left by the enemy plane and feeds it to a computer. The computer then guides the missile towards its target.
5. These waves are used in TV remote sensors.

Ultraviolet Spectrum:-

- The region of spectrum which extends beyond the violet colour of the visible spectrum is called ultraviolet spectrum.
- The portion of the spectrum was discovered by prof.J.Ritter in 1801.

Properties of Ultraviolet Radiations:-

1. They are electromagnetic waves and as such are not affected by electric or magnetic fields.
2. They travel with the velocity of light, i.e., $3 \times 10^8 \text{ ms}^{-1}$
3. The range of these radiations is between $4000 \text{ \AA}$ to $1000 \text{ \AA}$
4. They obey the laws of reflection and refraction.

Uses of Ultraviolet Radiations:-

1. Ultraviolet radiations when absorbed by the skin, stimulate the body to produce vitamin D, which is very good for the growth of strong bones and teeth.
➤ However, they are harmful to the eyes and can cause permanent blindness. It is for this reason, that one should not see the sun directly, as ultraviolet radiation produces a blinding effect. It is for the same reason that welders use thick a glass screen to protect their eyes.
2. Ultraviolet radiations have lethal effect on micro organisms. Thus they are used for sterilising surgical instruments and drinking water.
3. Ultraviolet radiations are used to detect adulteration in ghee, because of its property of fluorescence.
4. Ultraviolet radiations are used to detect forgery of cheques or important documents, prepared by erasing original writing when such a document is placed under ultraviolet lamp, the alteration clearly shows up.
5. Ultraviolet radiations are used for detecting original diamonds from fake diamonds.

Spectrum of Visible Light:

Colour	Wavelengthy (approximately)
Red	$6200 \text{ \AA}$ to $8000 \text{ \AA}$
Orange	$5920 \text{ \AA}$ to $6200 \text{ \AA}$
Yellow	$5780 \text{ \AA}$ to $5920 \text{ \AA}$
Green	$5000 \text{ \AA}$ to $5780 \text{ \AA}$
Blue	$4640 \text{ \AA}$ to $5000 \text{ \AA}$
Indigo	$4460 \text{ \AA}$ to $4640 \text{ \AA}$
Violet	$4000 \text{ \AA}$ to $4460 \text{ \AA}$

LASER

- The word LASER is an acronym for “Light Amplification by stimulated Emission of Radiation”.
- The principle of LASER proposed by Dr.Charles H.Towenes in 1954.
- The efforts of large number of scientists led to the development of first laser called pulsed laser in 1960.
- Basically, laser is a source of light which has extra-ordinary properties. These properties are not present in the light from common sources of light such a sun, in candescent lamp (electric bulb) or Sodium vapour lamp.

Characteristics of Laser Light:-

- 1) Coherence of Light
- 2) Directionality of Light
- 3) Monochromaticity of Light
- 4) Intensity of Light

1) Coherence of Light:-

- In a laser source, electronic transitions of the excited electrons take place in orderly manner, such that light waves emitted have a consistent phase relation, which does not change with time. This kind of emission is called coherence of light. It is the most important characteristic of laser light.

2) Directionality of Light:-

- Let us consider a beam of laser light. It travels 1km, it spreads by about 1 cm. This is called directionality of laser light. From directionality it implies that laser beam continuously moving in same direction and does not diverge appreciably. Which helps us to focus it on an object at a very large distance.

Monochromaticity:-

- Monochromaticity of light means of single colour or of single wavelength.
- The spread of wavelength (or frequency) about the wavelength of maximum intensity is called band width (or range).
- The band width of this monochromatic light is $1000 \text{ \AA}$.

Intensity:-

- Energy of wave flowing per unit time through a unit normal area is called intensity of light.
- There are highly advanced lasers available whose intensity is so high that an enormous power of 10^{15} watt can be concentrated over a surface 1 cm^2 .

Definition of LASER:-

- Laser is an extraordinary light emitted under stimulated and amplified conditions, such that the beam is characterised by high degree of coherence, specific directionality, high monochromaticity and high intensity.

Types of Lasers:-

- The lasers can be broadly classified as
 - i) solid lasers
 - ii) liquid lasers
 - iii) gaseous lasers

i) Solid Laser:

- Ruby laser is example of solid laser.
- In this laser a ruby (Al_2O_3 ; Cr^{3+}) crystal is used as an active system.
- The pumping in this laser is achieved by Xenon-discharge tube, which provides flash light.
- The wavelength of ruby laser is $6943 \text{ \AA}$.

Liquid Laser:-

- Example of liquid laser is Europium benzoyl acetate dissolved in alcohol.

Gaseous Lasers:-

- Helium-Neon (He-Ne) is an example of a gaseous laser.
- 'Neon' acts as active system.
- The pumping in this laser is achieved by a radio frequency generator (R.F.generator).
- The wavelength of He-Ne laser is $6328 \text{ \AA}$.

Applications of Lasers:-

- The lasers are employed in field of medicine, defence and space science, industry and science and technology.

1) Lasers in medicine:

1. Laser beams are used in bloodless surgery.
2. Laser beams are used in killing tumours in liver and lungs.
3. Laser beams are used in fibre optics to detect ulcers in the alimentary canal.
4. Laser beam is extensively used in correcting the curvature of cornea and reattachment of detached retina.

2) Lasers in defence and space science:

1. Lasers are used in guided missiles and spiner rifles.
2. Lasers are used in (i) space communication, radars and satellites.

3) Lasers in industry:

1. Laser beams are employed in cutting, drilling or welding of steel and other metals.
2. Laser beams are used in reading prefixed price on the articles from the bar code printed on article.

4) Laser in science and technology:

1. Development of laser has helped in establishing a new method of fast communication known as "fibre optic communication".
2. Development of laser has helped in establishing a new branch of science called, "Atmospheric optics" for measuring polluting gases in air.
3. Laser beam helps in studying brownian motion in particles of fluids.
4. He-Ne laser helps in determining the rate of rotation of earth accurately.
5. Laser beam is used in computers to retrieve information from the compact disc (CD).
6. Laser helps in extending the observable distance and is used in the study of stellar objects in astronomy.
7. With the help of laser it is possible to count number of atoms in a given substance.
8. Lasers with high intensity and directionality are used in micro Raman-spectroscopy for the analysis of biological or biomedical specimens.
9. Lasers are employed in 'holography' which is three dimensional photography.
10. Lasers are used to separate isotopes from a given element.
11. Lasers are used to study the nature of plasma state of matter and to achieve high temperatures for causing thermonuclear fusion.

1. **Light energy travels at a speed of.**
 - 1) $3 \times 10^8 \text{ kms}^{-1}$
 - 2) $3 \times 10^6 \text{ kms}^{-1}$
 - 3) $3 \times 10^5 \text{ kms}^{-1}$
 - 4) $3 \times 10^7 \text{ kms}^{-1}$
2. **The primary source of light for the earth is.**
 - 1) stars
 - 2) moon
 - 3) sun
 - 4) both (2) and (3)
3. **The bodies which emit light of their own are called.**
 - 1) light giving bodies
 - 2) luminous bodies
 - 3) bright bodies
 - 4) all the above
4. **Which is not a luminous body.**
 - 1) Burning kerosene oil lamp
 - 2) Stars
 - 3) Sun
 - 4) Moon
5. **The bodies which do not give light energy of their own, but reflect the light falling on them are called.**
 - 1) opaque bodies
 - 2) reflecting bodies
 - 3) non-luminous bodies
 - 4) all the above
6. **The old unit of luminosity of light is.**
 - 1) watt
 - 2) candle power
 - 3) lumen
 - 4) both (1) and (3)
7. **One lumen of light is equivalent to.**
 - 1) 12.56 candle power
 - 2) 21.56 candle power
 - 3) 11.56 candle power
 - 4) 13.56 candle power
8. **Any thing material or non-material through which light energy can pass called.**
 - 1) optical medium
 - 2) light medium
 - 3) homogenous medium
 - 4) none of the above
9. **Which is not an optically homogeneous medium.**
 - 1) Vacuum
 - 2) Smoke
 - 3) Glass
 - 4) Distilled water
10. **A medium which allows most of the light energy to pass through it is called.**
 - 1) opaque
 - 2) transparent
 - 3) translucent
 - 4) none of the above
11. **A medium which partially allows the light energy to pass through it is called.**
 - 1) translucent medium
 - 2) transparent medium
 - 3) heterogeneous medium
 - 4) homogeneous medium
12. **Which is not a translucent medium?**
 - 1) Clouds
 - 2) Fog
 - 3) Oiled paper
 - 4) Vacuum
13. **The bodies which do not allow light energy to pass through them are called.**
 - 1) reflecting bodies
 - 2) opaque bodies
 - 3) solid bodies
 - 4) non-transparent bodies
14. **Rectilinear propagation of light means that light travels in.**
 - 1) rectanles
 - 2) straight lines
 - 3) both (1) and (2)
 - 4) none of the above

- 15. A shadow is formed only if.**
1) there is a source of light
2) there is a source of light and an opaque body to obstruct light
3) there is a source of light, an opaque body to obstruct light and a screen
4) any of the above
- 16. Solar eclipse takes place on.**
1) full moon 2) half moon
3) crescent moon 4) new moon
- 17. Lunar eclipse takes place on.**
1) full moon 2) half moon
3) crescent moon 4) new moon
- 18. When the sun, the moon and the earth happen to be in the same straight line and the same plane, such that the moon is in the middle, the eclipse caused is known as.**
1) solar eclipse
2) lunar eclipse
3) can be either (1) or (2)
4) none of the above
- 19. When a ray of light travelling through some optical medium, on striking some surface, is returned to the same medium, the phenomenon is called.**
1) rectilinear propagation of light
2) reflection of light
3) refraction of light
4) none of the above
- 20. The phenomenon due to which left hand side of an object appears as right hand side in a plane mirror is called.**
1) inversion
2) lateral inversion
3) upside down inversion
4) none of the above
- 21. Which is not the characteristic of the image formed in a plane mirror?**
1) Image is virtual
2) Image is erect
3) Image is of different size than object
4) Image is formed as far behind the mirror as the object is in front of it.
- 22. A reflecting periscope uses.**
1) single plane mirror inclined at an angle of 45°
2) two plane mirrors facing each other and inclined at an angle of 45°
3) two plane mirrors facing each other and inclined at an angle of 60°
4) none of the above
- 23. Two plane mirrors are inclined at an angle of 30° . The number of images of an object in between them will be.**
1) 11 2) 7
3) 5 4) infinite
- 24. When two plane mirrors are facing each other, such that an object is placed in between them, then the number of images formed in each mirror will be.**
1) 2 2) 20
3) 10 4) infinite
- 25. The phenomenon due to which a ray of light deviates from its path, which travelling from one optical medium to another optical medium is known as.**
1) scattering of light
2) defraction of light
3) reflection of light
4) refraction of light

26. A stick held obliquely and held in water appears bent and short. It is on account of the phenomenon of.
- 1) reflection
 - 2) refraction
 - 3) scattering
 - 4) dispersion
27. The bottom of tank filled with water appears.
- 1) raised
 - 2) depressed
 - 3) normal
 - 4) none of the above
28. In case of photographic camera, the image formed on the film is.
- 1) real, erect and diminished
 - 2) virtual, erect and diminished
 - 3) virtual, inverted and diminished
 - 4) real, inverted and diminished
29. The phenomenon due to which white light splits into component colours is called.
- 1) refraction of light
 - 2) diffusion of light
 - 3) dispersion of light
 - 4) none of the above
30. The band of seven colours formed on a screen when the white light suffers dispersion is called.
- 1) solar coloured band
 - 2) spectrum
 - 3) sonar
 - 4) none of the above
31. Red, orange, _____, _____, _____, indigo and violet is the order of colours in spectrum. The order of missing colours is.
- 1) blue, green, and yellow
 - 2) yellow, blue and green
 - 3) green, blue and yellow
 - 4) yellow, green and blue
32. The colour that deviates most during dispersion of white light is.
- 1) orange
 - 2) yellow
 - 3) indigo
 - 4) violet
33. The optical aid that disperses white light in a laboratory is.
- 1) convex mirror
 - 2) glass prism
 - 3) glass slab
 - 4) concave mirror
34. In nature, rainbow is formed due to the dispersion of sunlight. The optical aid responsible for it is/ are.
- 1) molecules of carbon dioxide
 - 2) tiny droplets of water
 - 3) molecules of nitrogen dioxide
 - 4) none of the above
35. The least distance of distinct vision for a normal eye is.
- 1) > 30 cm
 - 2) > 25 cm
 - 3) 25 cm
 - 4) < 25 cm
36. When a person can see far-off objects clearly, but is unable to see clearly the objects at a distance of distinct vision, the defect in the eye is.
- 1) myopia
 - 2) hypermetropia
 - 3) presbyopia
 - 4) none of the above
37. When a person cannot see clearly, the nearer and distant objects, the defect in the eye is.
- 1) hypermetropia
 - 2) myopia
 - 3) presbyopia
 - 4) astigmatism
38. A person can read clearly at a distance of 25 cm, but cannot see clearly far-off objects. The defect in his eye is.
- 1) myopia
 - 2) presbyopia
 - 3) hypermetropia
 - 4) none of the above

- 39. A short sighted eye can be corrected by using a.**
 1) convex lens 2) convace mirror
 3) concave lens 4) none of above
- 40. A long sighted eye can be corrected by using a.**
 1) concave lens 2) convex lens
 3) convex mirror 4) none of above
- 41. When a person cannot see clearly, near and far-off objects, the defect in the eye can be corrected by using.**
 1) convex lens 2) concave lens
 3) bifocal lens 4) none of above
- 42. The common TV remote uses which of the following waves?**
 1) Radiowaves
 2) Lasers
 3) Infrared waves
 4) Ultrasonic waves
- 43. Several colours on the screen of a colour television are produced by mixing lights of colours.**
 1) Red, Yellow and Green
 2) Red, Yellow and Blue
 3) Red, Green and Blue
 4) Yellow, Green and Blue
- 44. Consider the following statements.**
 A) If a person looks at a coin which is in a bucket of water, the coin will appear to be closer than it really is.
 B) If a person under water looks at a coin above the water surface, the coin will appear to be at a higher level than it really is.
Which of the above statements is / are correct?
 1) Both 'A' and 'B' 2) 'A' alone
 3) 'B' alone 4) Neither 'A' or 'B'
- 45. Endoscopy, a technique used to explore the stomach or other inner parts of the body, is based on the phenomenon of**
 1) Total internal reflection
 2) Interference
 3) Diffraction
 4) Polarisation
- 46. The minimum length of a plane mirror in which a person can see himself full length should be.**
 1) Equal to the person's height
 2) Slightly more than his height
 3) Nearly half of his height
 4) Nearly one-fourth his height
- 47. Choose the correct statements form the codes given below.**
 A) A dentist's mirror is a concave mirror
 B) Convex mirrors are used as rear view mirrors in vehicles
 C) The headlamps of cars use parabolic concave mirror.
 Which of the following are correct.
 1) B only 2) A and B only
 3) C only 4) A, B and C
- 48. The sun remains visible for some time after it actually sets below the horizon. This happens due to.**
 A) Atmospheric refraction
 B) Scattering of light
 C) Dispersion
Which of the above statements is/ are correct?
 1) Only A 2) A and B
 3) A and C 4) A, B and C
- 49. The twinkling of stars can be partly attributed to**
 1) Internal shining
 2) Convection currents in the atmosphere
 3) Atmospheric refraction
 4) The unstable motion of the earth

- 50. After a shower of rain a rainbow is seen.**
1) Towards the sun
2) Opposite the sun
3) Anywhere, irrespective of the position of the sun
4) Even in the absence of the sun
- 51. When white light passes through a glass prism, it gets dispersed into colours because.**
1) Glass imparts colours to the light
2) Different colours have different speeds in glass
3) Interference of light occurs
4) Diffraction of light occurs
- 52. Lights of different colours are arranged below in order of increasing wavelengths.**
a) Violet, Indigo, Blue
b) Green, Orange, Red
c) Red, Green, Indigo
d) Blue, Green, Yellow
Which of the above are in proper sequence?
1) Only a 2) a and b
3) b and c 4) a and d
- 53. What are primary colours.**
1) Colours of the rainbow
2) Colours in the spectrum of white light
3) Colours which cannot be produced by mixing other colours
4) Colours found in nature
- 54. In the following list of colours.**
a) Blue b) Green
c) Red d) Yellow
Which are the three primary colours?
1) a, b and c 2) a, b and d
3) b, c and d 4) a, c and d
- 55. The sky, appears blue because.**
1) It is actually blue
2) The atmosphere scatters blue colour more than the others
3) All colours interfere to produce blue
4) In white light the blue component dominates.
- 56. To an astronaut, outer space appears.**
1) White 2) Black
3) Deep blue 4) Crimson
- 57. Soap bubbles show multiple colours when viewed in white light because of.**
1) The scattering of light by thin films
2) The interference of the reflected light
3) The dispersion of light
4) The characteristic colours of the soap solution
- 58. The technique of recording and reproducing three-dimensional images of objects is known as.**
1) Audiography 2) Lexicography
3) Holography 4) Photography
- 59. The part of the human eye which perform a function similar to that of aperture in a photographic camera is**
1) Iris 2) Lens
3) Pupil 4) Ciliary muscles
- 60. The pupil of the eye adjusts automatically for.**
1) Short distance vision
2) Colour
3) Amount of light
4) Object size

- 61. All the colours on a TV screen are produced from.**
 1) Red, Green and Blue
 2) Red, Yellow, Orange and Blue
 3) Blue, Green and Yellow
 4) Violet, Indigo, Blue, Green, Yellow, Orange and Red
- 62. Total internal Reflection can occur when light passes from.**
 1) Denser to rarer medium
 2) Rarer to denser medium
 3) It occurs irrespective of medium of travel
 4) It occurs when light travels from one medium to another of equal refractive index.
- 63. Which one of the following is used to detect fingerprints on a piece of paper.**
 1) X-rays
 2) Infrared rays
 3) Ultraviolet radiation
 4) Gamma rays
- 64. When an eye fails to see vertical and horizontal lines equally distinctly. It is said to be suffering from which one of the following?**
 1) Myopia
 2) Astigmatism
 3) Hypermetropia
 4) Presbyopia

Answers

1) 3	2) 3	3) 2	4) 4	5) 3	6) 2	7) 1	8) 1	9) 2	10) 2
11) 1	12) 4	13) 2	14) 2	15) 3	16) 4	17) 1	18) 1	19) 2	20) 2
21) 3	22) 2	23) 1	24) 4	25) 4	26) 2	27) 1	28) 4	29) 3	30) 2
31) 4	32) 4	33) 2	34) 2	35) 3	36) 2	37) 3	38) 1	39) 3	40) 2
41) 3	42) 3	43) 3	44) 1	45) 1	46) 3	47) 4	48) 1	49) 3	50) 2
51) 2	52) 4	53) 3	54) 1	55) 2	56) 2	57) 2	58) 3	59) 3	60) 3
61) 1	62) 1	63) 3	64) 2						

4. MECHANICS

- Mechanics is a branch of physics dealing with the behaviour of matter under the action of forces.
- Mechanics is classified into three branches.
 - 1) Kinematics
 - 2) Dynamics
 - 3) Statics
- 1) **Kinematics :**
 - Kinematics is the study of the mechanical motion of bodies, without considering the cause of motion like forces, interacting bodies etc.
- 2) **Dynamics :**
 - Dynamics is the mathematical and physical study of the behaviour of bodies under the action of forces that produce motion in a body.
- 3) **Statics :**
 - Statics is the mathematical and physical study of the bodies in equilibrium (Rest) under the action of forces.

Kinematics

Rest :-

- The position of a body does not change with respect to surroundings and time. Then the body is said to be at rest.

Motion :-

- The position of a body changes with respect to surroundings and time. Then the body is said to be in motion.

Distance:

- The change of position of a body without considering direction is called distance.
- It is a scalar quantity.
- Its S.I unit is : metre (m)
- Its C.G.S unit is : centimetre (cm)

Displacement:

- The change of position of a body in a definite direction is called displacement.
- It is a vector quantity.

UNITS :-

- S. I. Unit :- metre (m)
- C.G.S. Unit :- cm

Speed:

- The distance travelled by a body in unit time is called speed.
- It is a scalar quantity $\text{speed} = \frac{\text{Distance}}{\text{time}}$

UNITS :

- S.I. Unit :- m s^{-1}
- C.G.S. Unit : cm s^{-1}

Velocity:-

- The rate of change of position of a body in a definite direction is called velocity.
(or)
- The displacement of a body in one second is called velocity.
- It is a vector quantity.
- Velocity =
- Its S.I. Unit is : ms^{-1}
- Its C.G.S. Unit is : cms^{-1}

Instantaneous velocity :

- The velocity of a body at a particular instant of its travel is called instantaneous velocity.

Uniform Velocity :-

- The velocity of a body is said to be uniform if it travels equal displacements in equal intervals of time.

Non Uniform Velocity (Variable Velocity) :

- The velocity of a body is said to be variable or non-uniform if the body travels unequal displacements in equal intervals of time or equal displacements in unequal intervals of time.

Acceleration (a) :

- The rate of change of velocity is called acceleration.
- It is a vector quantity.
- acceleration =
- Its S.I. Unit : ms^{-2}
- Its C.G.S. Unit : cm s^{-2}

Types of acceleration :**1) positive acceleration :**

- If the velocity of a body gradually increases with time, its acceleration is positive which is called positive acceleration.

Ex: 1) A train leaving from station
2) Motion of a freely falling body

Deceleration (Retardation) :

- If the velocity of a body gradually decreases with time, its acceleration is negative which is called deceleration.

Ex: 1) A train approaching station
2) A body projected upwards travels with deceleration.

Acceleration due to gravity :

- The acceleration of a freely falling body due to earth's attraction is called acceleration due to gravity.

Properties :

- 1) It is denoted by 'g'
 - 2) Its value is constant for all bodies at a given place.
 - 3) Its value changes from place to place on the surface of the earth.
 - 4) It has maximum value at the poles of the earth.
 - The value is nearly 9.83 ms^{-2}
 - 5) It has minimum value at the equator of the earth.
 - The value is nearly 9.78 ms^{-2}
 - 6) The average value of 'g' on earth's surface is 9.8 ms^{-2}
 - 7) On the moon $g = 1.67 \text{ ms}^{-2}$
 - 8) On the surface of sun $g = 1.67 \text{ ms}^{-2}$
- The acceleration due to gravity on moon is of the acceleration due to gravity on earth.
- 9) 'g' is always directed downwards towards the centre of the earth, whether a body is projected upwards or downwards.
 - 10) When a body is falling towards the earth, its velocity increases. Hence 'g' is taken as positive.
 - 11) When a body is projected upwards, its velocity decreases, Hence 'g' is taken as negative.
 - 12) The acceleration due to gravity at the centre of earth is zero.

Variation of 'g' value :-

- 1) The value of 'g' decreases as we move upwards from the surface of the earth.
 - If ' g_h ' is acceleration due to gravity at a height 'h' from the surface of the earth and 'g' is acceleration due to gravity on the surface is

$$g_h = g$$

where R is radius of the earth

- 2) The value of 'g' decreases as we go depth 'd'.
 - If ' g_d ' is acceleration due to gravity at a depth 'd' inside the mine and 'g' is acceleration due to gravity on the surface is

$$g_d = g$$

where R is radius of the earth.

Factors Effecting on ‘g’ value :

S.No	Factors	Effect on ‘g’ value
1.	Height (attitude) increases	‘g’ decreases
2.	Depth increases	‘g’ decreases
3.	Local conditions	‘g’ is slightly effected by geological deposits, massive concrete buildings and topography.
4.	At the poles	‘g’ is maximum
5.	At the equator	‘g’ is maximum

Gravity meters :-

- Sensitive instruments used to measure such small changes in the value of ‘g’ at a given location are called gravity meters.

Ex: i) Boliden Gravity Meter
ii) Gulf Gravity Meter

Equations of Motion of bodies travelling with Uniform acceleration :-

- Consider a body moving with an initial velocity “u”. Let it be travelling along a straight line with uniform acceleration ‘a’. Let ‘V’ be its final velocity after time “t”. The displacement of the body in this time is “s”.

i) $V = u + at$

ii) $S = ut + \frac{1}{2}at^2$

iii) $v^2 - u^2 = 2as$

iv) $s_n = u + \frac{1}{2}a(2n-1)$

Maximum Height (H) :-

- Let a body be projected vertically upwards with an initial velocity ‘u’. The maximum height reached by the body is ‘H’. The final velocity is zero.

where ‘g’ is acceleration due to gravity.

- Hence, maximum height attained by the body is directly proportional to the square of the initial velocity.

Time of ascent (t_a):

- The time taken by the body to reach maximum height is called time of ascent.

$$t_a = \frac{u}{g}$$

Time of descent (t_d) :

- The time taken by the body to reach the ground is called time of descent.

Time of flight (T) :-

- The total time of ascent and time of descent is the time of flight.

Velocity of the falling body on reaching the ground :

- $V = \sqrt{2gh}$

Where 'g' is acceleration due to gravity.

'h' is maximum height.

NOTE : The velocity of the falling body on reaching the ground is equal to the velocity of projection. But they are in opposite directions.

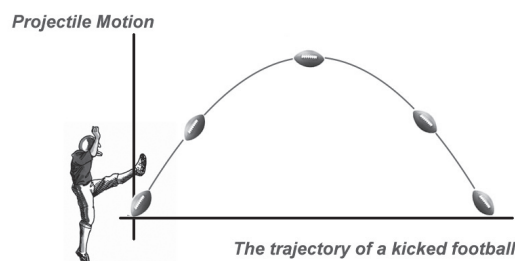
Projectile:

- A body thrown upwards making an angle (other than 90°) with horizontal is called a projectile.
- The path of a projectile is a parabola.
- This science is called Ballistics.

Ex: 1) A cricket ball thrown by a fielder.

2) A bullet fired from a gun.

3) A javelin thrown by an athlete.

**Applications :**

- 1) A ball is thrown outwards vertically upwards by a boy in a moving train. Its trajectory for an observer in the train is straight line. But its trajectory for an observer standing outside the train is parabola.
- 2) A bomb is dropped from an aeroplane then its trajectory for a pilot is straight line. But its trajectory for an observer standing on the ground is parabola.

Range of Projectile (R) :-

- The horizontal distance travelled by the projectile during the time of flight is known as the range of the projectile (R).
- $R = \text{Horizontal velocity} \times \text{time of flight}$.

- $R =$

- The range of projectile is same for angles of projection ' θ ' and $(90 - \theta)$

Ex:- The range will be the same for 30° and 60° angles of projection.

- The value of 'R' is maximum when $\theta = 45^\circ$.

Conceptual Questions :-

- 1) Two bodies of different masses are dropped from the same height in vacuum. Which body reaches the ground earlier?

Ans : Both of them reach the ground at the same time.

Reason :-

- The time of fall for a freely falling body from a height 'h' is given by

$$t =$$

- It does not depend on the mass and shape when moving in vacuum.

2) A stone is thrown horizontally and another is dropped freely from the same height Which of the two will reach the ground early?

Ans :-

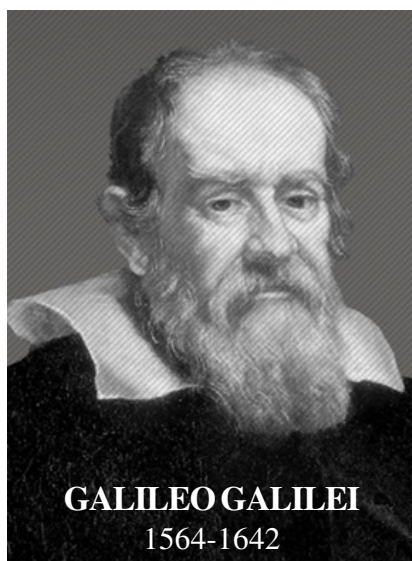
- Both reach the ground in the same time. Both horizontally projected body and the dropped body have same zero initial velocities in vertically downward direction.
- As they have to cover same vertical distance and their accelerations being same, they reach the ground in the same time.

3) Two bullets are fired simultaneously, horizontally and with different speeds from the same place. Which bullet will reach the ground first?

Ans:- The time taken to reach the ground does not depend upon the horizontal velocity and depends on the height only. So both of them reach the ground at the same time.

About Galileo Galilei:-

- Italian mathematician, astronomer and physicist. He demonstrated experimentally that all bodies fall with the same acceleration and he deduced that the trajectory of a projectile is a parabola.
- Galileo performed the experiment of dropping bodies of different weights from the top of the leaning tower of Pisa. He showed that they reached the earth at the same time.



Dynamics

Newton's law of universal Gravitation :-

- Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The force acts along the line joining the two bodies.

$$F = \frac{Gm_1m_2}{r^2}$$

- Where m_1, m_2 = masses of two bodies
 r = distance between two bodies.
 F = gravitational force of attraction
 G = universal gravitational constant
- 'G' value is $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$
- Newton's first explained the earth having gravitational force.

Mass :

- The mass of a body is the quantity of matter contained in it.
- It is the fundamental property of the body.
- It is independent of external factors like position and surroundings.
- The mass of a given body will be the same at the equator as well as at the poles of the earth, or for that matter, any where in the universe.
- The S.I. Unit of mass is "kilogram" (kg)
- The mass of a body is determined by comparing it with known standard mass using "common balance".

Weight :

- Weight of a body is the force with which the body is attracted by the earth. (or)
- It is the product of its mass and the acceleration due to gravity.
- Thus, if 'm' is the mass of a body, and 'g' is acceleration due to gravity, its weight is given by

$$W = mg$$

- Its S.I. Unit is "newton" (N).
- There is another S.I. Unit of Weight is kilogram weight (kgwt) (or) Kilogram force
- $1 \text{ kg wt} = 9.8 \text{ N}$
- Weight is measured by using a "spring balance".
- The principle underlying the spring balance was first investigated by "Robert Hooke".
- Hooke's law helps in the determination of the weight of a body.
- Hooke's law gives the relation between extension in length of a spring and force acting on it.
- The weight of the body is maximum at poles and minimum at equator.

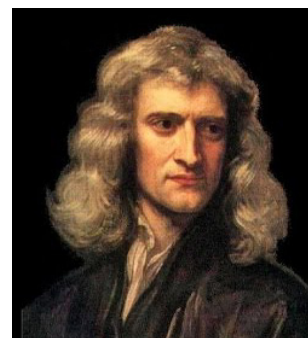
- The weight of the body is zero at centre of the earth due to acceleration due to gravity is zero at centre of the earth.

Newton's Laws of Motion

- Newton proposed 3 laws of motion
 - I) Newton's first law of motion
 - II) Newton's second law of motion
 - III) Newton's third law of motion

I) Newton's first law of motion :

- This law is also called the law of Inertia.

**Statement :**

- Anybody at rest or in uniform motion will remain at rest or in uniform motion unless an unbalanced force is applied to change that state
- Newton's first law defines 'force'.
- Newton's first law gives concept of 'inertia'.

Inertia :

- It is the inability of a body to change by itself its state of rest or of uniform motion in a straight..
- "Mass" is a measurement of inertia.

Types of Inertia :

- i) Inertia of rest
- ii) Inertia of motion
- iii) Inertia of direction

i) Inertia of rest : It is the tendency of a body to remain in its state of rest.

Applications:

1. A person standing in a bus falls backward when the bus suddenly starts moving forward.
 - When the bus moves, the lower part of his body begins to move along with the bus while the upper part of his body continues to remain at rest due to inertia.
2. Dust is removed from hanging a carpet by beating it with a stick. As the carpet is beaten, it suddenly moves forward while the dust particles tend to remain at rest due to inertia and so fall off.
3. When we shake a tree, the fruits or dry leaves fall down from the tree due to inertia of rest.
4. Imagine a pile of books placed on a sheet of paper. If the paper is suddenly pulled with a jerk, the books are left behind, because of the inertia of rest.

ii) Inertia of motion :

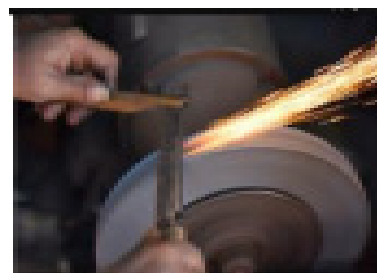
- The tendency of a body to remain in its state of uniform motion in a straight line, is called inertia of motion.

Applications:

- 1) An athlete runs from certain distance before taking a jump so that his inertia of motion may help him to take a longer jump.
- 2) Suppose we are standing in a moving bus, and the driver stops the bus suddenly. We fall forward.
- 3) A rider falls forwards, when a galloping horse stops suddenly. When the horse stops, the rider, on account of the inertia of motion, continues moving, and hence, falls in the forward direction.
- 4) A running body falls in the forward direction if he is tripped by a stone, etc. because the stone stops his foot, whereas rest of the body continues moving forwards, and hence, the boy falls in the forward direction.
- 5) It is very important to run along with moving the bus, and in the same direction in which the bus is moving, while jumping out of it. Otherwise, the feet will suddenly come to rest, while the rest of the body will be in the state of motion, and hence, one can fall down and get seriously injured.

iii) Inertia of direction :

- The inability of a body to change by itself its direction of motion is called inertia of direction.

**Applications :**

- 1) When a car rounds a curve suddenly, the person sitting inside is thrown outwards. This is because the person tries to maintain his direction of motion due to directional inertia while the car turns.
- 2) The rotating wheels of any vehicle throw out mud, if any, tangentially, due to directional inertia. The mudguards over the wheels stop this mud, protecting the clothes etc. of the driver of the bike or motor bike.
- 3) When a knife is sharpened by pressing it against a grinding stone, the sparks fly off along the tangent to the grinding stone, on account of directional inertia.
- 4) When a bus takes a sharp turn towards right, a passenger sitting inside the bus, falls towards left.

II) Newton's Second law of Motion:-

- The rate of change of momentum of a body is directly proportional to the applied force and the change takes place in the direction of the force.
- This law gives measure of force.
- We can write Newton's second law as

$$\mathbf{F = ma}$$

- Where m - mass of the body
a - acceleration of the body
F - force

Definition of force :-

- The external agency which changes or tends to change the state of rest or uniform motion or the direction of motion of a body is called force.

- S.I. unit of force is newton (N).
- C.G.S. unit of force is dyne.

$$1 \text{ newton} = 10^5 \text{ dyne}$$

Definition of one newton :-

- One newton force is that force which produces an acceleration of 1 ms^{-2} in a body of mass 1 kg

Applications of Newton's second law of motion.

Impulse :-

- The forces which act on bodies for short time are called impulsive forces.

Ex: 1) In hitting a ball with a bat

2) In driving nail into a wooden block with a hammer.

Definition :

- It defined as the product of the force acting on a body and the time interval during which the force is exerted.
- Impulse = average force x time
- $I = F \times t$ (or)
- $I = F \times t = \Delta P$
- Where F - force
 t - time interval
 ΔP = change in momentum
- Impulse is a vector quantity.
- Its direction is the same as that of force.
- The S.I. unit of impulse is "newton-second" (N-S)
- $1 \text{ N-S} = \text{kgms}^{-1}$
- The C.G.S. unit of impulse is dyne - sec or gm cms^{-1} .

Applications of the concept of Impulse :

1) A cricket player lowers his hands while catching a ball :

- By lowering the hands, he increases the time of catching the ball. As a result, the rate of change of momentum decreases and by Newton's second law, the force exerted on his hands is less. So he is less likely to hurt.

2) China wares are wrapped in straw paper before packing:

- The straw paper between the china wares increases the time of experiencing the jerk during transportation. Hence, they strike against each other with a less force and are likely to be less damaged.

3) We hurt less when we jump on a muddy floor than on a hard floor:

- When we jump on a muddy floor, the floor is carried in the direction of the jump and the time interval for which force acts is increased. This decreases the rate of change of momentum and hence the force of reaction. Thus, we hurt less.

4) Bogies of a train are provided with the buffers :

- They avoid severe jerks during shunting of the train. Due to presence of buffers, time of impact increases. Therefore force during jerks decreases. Hence the chances of damage decrease.

**5) The vehicles like scooter, car, bus, truck etc. are provided with shockers.**

- When they move over an uneven road, impulsive forces are exerted by the road. The function of shockers is to increase the time of impact. This would reduce the force or Jerk experienced by the rider of the vehicle.

**III) Newton's third law of motion :****Statement :**

- Action and reaction are equal and opposite and they act on different bodies.
- Third law of motion shows that a single force can never exist. The forces always exist in pairs. The two opposing forces are known as action and reaction forces. But the forces of action and reaction always act on two different bodies.

Applications :-**1) Walking :**

- While walking, a person presses the ground in the backward direction (action) by his feet. The ground pushes the person in forward direction with an equal force (reaction). The component of reaction in the horizontal direction makes the person move forward.

2) Swimming :

- A swimmer pushes the water back wards (action). The water pushes the swimmer forward (reaction) with the same force. Hence the swimmer swims.

**3) Firing from a gun :**

- When a gun is fired, the bullet moves forward (action). The gun recoils backwards (reaction).

4) Flight of jet planes and rockets:

- The burnt fuel which appears in the form of hot and highly compressed gases escapes through the nozzle (action) in the backward direction. The escaping gases push the jet plane or rocket forward (reaction) with the same force. Hence jet or rocket moves.

**5) It is difficult to walk on sand or ice :**

- This is because on pushing, sand gets displaced and reaction from sandy ground is very little. In case of ice, force of reaction is again small, because friction between our feet and ice is very little.

6) Rebounding of a rubber ball :

- When a rubber ball is strike against a wall or floor it exerts a force on wall (action). The ball rebounds with an equal force (reaction) exerted by wall or floor on the ball.
- 7) When a sailor jumps out of a rowing boat, the boat moves backwards, As the sailor jumps forward, he applies a backward force (action) on boat and the boat moves backwards due to the force of reaction.

Linear Momentum (Momentum) :-

- Momentum of a body is the quantity of motion possessed by the body. It is equal to the product of the mass and velocity of the body and is denoted by 'p'.
- Momentum = mass x velocity
- where, m - mass of the body
v - velocity of the body
- Momentum is a vector quantity and its direction is the same as the direction of velocity of the body.
- S.I. Unit : kgms^{-1}
- C.G.S. Unit : gmcms^{-1}

Law of conservation of linear momentum (or) Law of conservation of momentum :

- This law states that if a number of bodies are interacting with each other, their total momentum remains conserved before and after the interaction, provided there is no external force acting on them.
- According to Newtons' second law of motion.

$$F = \frac{P_2 - P_1}{t}$$

- If there is no external force, that is $F = 0$

then $\frac{P_2 - P_1}{t} = 0$

Applications of law of conservation of momentum :**1) Recoil of the gun :**

- Before firing, both the bullet and the gun are at rest and initial momentum of the system is zero. As soon as the bullet is fired, it moves forward with a large velocity. In order to conserve momentum, the gun moves backward with such a velocity that the final momentum of the system is zero.

2) Flight of Rockets and Jet Engines :

- When a rocket is fired, it throws a stream of burnt gases in the vertically downward direction with a large velocity. So the outgoing gases have large momentum in the downward direction. According to the law of conservation of momentum, the rocket gets an equal and opposite momentum which causes it to fly up.

Work :

- Work is said to be done whenever a force acts on a body and the body moves in the direction of the force.
- **S.I. Unit of work is joule (J).**

Special Cases :-

- 1) When force acts along the direction of motion of the body.

$$W = FS$$

Where 'F' is force and 'S' is displacement.

- 2) When force 'F' makes an angle 'θ' with displacement 's' of the body

$$W = FS \cos \theta$$

- i) When $\theta = 0^\circ$, $\cos \theta = 1$, and $W = FS$

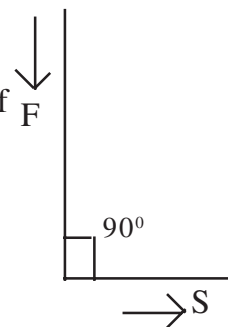
Thus work done is maximum when the displacement of the body is along the direction of the force.

- ii) When $\theta = 90^\circ$, $\cos \theta = 0$ and $W = 0$

Thus, workdone is minimum or zero when the displacement of the body is perpendicular to the direction of force

- iii) When $s = 0$, $\therefore w = 0$

Thus, workdone on a stationary body is zero.

**Applications :**

- 1) The workdone by a person walking on a level road with a load on his head is zero.

Reason :-

- The angle between force 'F' and displacement 's' is 90° .

$$W = FS \cos \theta$$

$$W = FS \cos 90^\circ$$

$$W = 0$$

$$\cos 90^\circ = 0$$

- 2) A body is moving along a circular path, the work is done by the centripetal force is zero.

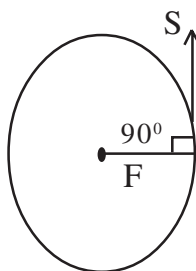
Reason : The centripetal force 'F' acts along the radius towards the centre. The displacement 's' acts along the tangent. Thus the angle between force 'F' and displacement (s) is 90° .

- Hence

$$W = FS \cos \theta$$

$$W = FS \cos 90^\circ$$

$$W = 0$$



Energy

- Energy of a body is defined as its capacity or ability to do work.
 - **S.I. Unit of energy is joule (J).**
 - Energy has many forms.
- Ex:** Mechanical energy, sound energy, heat energy, light energy, chemical energy, Nuclear energy, Electric energy, Magnetic energy, Solar energy etc.

Mechanical Energy :

- The energy produced by mechanical means is called mechanical energy.
- It has two forms.
 - 1) Kinetic Energy (K.E)
 - 2) Potential Energy (P.E)
- Energy is always conserved.

Law of conservation of energy :-

- This law states that energy can neither be created nor destroyed . It can only be converted from one form to another.

Examples:

- 1) **Electric heater :** Electrical energy into heat.
- 2) **Electric Motor :** Electrical energy into mechanical energy
- 3) **Dynamo :** Mechanical into electrical energy
- 4) **Microphone :** Sound energy into electrical energy
- 5) **Loud speaker (Head Phone) :** Electrical Energy into sound energy
- 6) **Hydroelectric power station :** Mechanical energy into electrical energy.
- 7) **Steam engine :** Heat energy into mechanical energy
- 8) **Solar Cell :** Solar energy into electrical energy
- 9) **Photoelectric cell :** Light energy into electrical energy.
- 10) **Electric cell :** Chemical energy into electrical energy.
- 11) **Photosynthesis :** solar energy into chemical energy
- 12) **Thermal power station :** Heat energy converted into electrical energy.

Kinetic energy (K.E) :-

- The energy possessed by a body by virtue of its motion is called its kinetic energy

$$K.E = \frac{1}{2}mv^2 \quad \text{Where, } m - \text{mass of body}$$

v - velocity of a body

- i) K.E of a body is directly proportional to its mass K. E $\propto$ m.
- ii) K.E. of a body is directly proportional to the square of its velocity. K. E $\propto$ V^2

- Ex:-**
- 1) A moving bullet
 - 2) A moving stone
 - 3) Running water (Flowing Water)
 - 4) A speeding car.

Problem based on kinetic energy :-**1) The speed of a car increases four times. Then what is its kinetic energy?**Ans: $K.E \propto V^2$

$$K.E_1 = K.E_1$$

$$K.E_2 = ?$$

$$\frac{K.E_1}{K.E_2} = \frac{V_1^2}{(4V_1)^2}$$

$$V_1 = V_1$$

$$\frac{K.E_1}{K.E_2} = \frac{V_1^2}{16V_1^2}$$

$$V_2 = 4V_1$$

$$\boxed{K.E_2 = 16K.E_1}$$

➤ Its K.E. increases by 16 times.

2) The velocity of a car is doubled then what is its K.E.?**Solution :**

$$K.E = \frac{1}{2}mv^2$$

$$K_1 = K_1$$

Let K.E is K

$$K_2 = ?$$

$$K \propto V^2$$

$$V_1 = V_1$$

$$V_2 = 2V_1$$

$$\frac{K_1}{K_2} = \frac{V_1^2}{(2V_1)^2} = \frac{V_1^2}{4V_1^2}$$

$$\boxed{K_2 = 4K_1}$$

➤ Its K. E becomes 4 times.

Relation between K.E. and momentum :-

$$\boxed{K.E = \frac{p^2}{2m}}$$

➤ Where m - mass of a body
p - momentum of a body.**POTENTIAL ENERGY (P.E) :**

- The energy possessed by a body by virtue of its position or shape (state) is called its potential energy.
- One of the commonest forms of P.E. is that possessed by an object when it is above the level of the earth's surface.
- This is called gravitational potential energy and is described by the expression.

$$\boxed{P.E = mgh}$$

- Where m - mass of the object
 g - acceleration due to gravity
 h - height of the object above the earth's surface.

Examples :

- 1) A stone held at some height above the ground
- 2) Water stored in a dam (Reservoir)
- 3) A stretched or compressed spring (or) wound spring
- 4) A stretched bow
- 5) A stretched rubber band



NOTE: In the following cases the body possesses both kinetic and potential energies.

- i) A flying bird
- ii) A man climbing a hill
- iii) A flying aeroplane

POWER :-

- The rate of doing work is called power.

- $$\text{Power} = \frac{\text{work}}{\text{time}}$$

- $P =$

- Where, w - work done
 t - time

- S.I. Unit of power is "Watt"

- $1 \text{ watt} =$

- Bigger units of power are kilo watt, mega watt and horse power.

- $1 \text{ kilo watt (1 KW)} = 10^3 \text{ watt}$

- $1 \text{ megawatt (1MW)} = 10^6 \text{ watt}$

- $1 \text{ horse power (1 H.P)} = 746 \text{ watt.}$

- 1) A man does 50 j of work in 5 seconds. What is his power?

Solution:-

- Given $W = 50 \text{ J}$

- $t = 5 \text{ seconds}$

- $p =$ $= 10 \text{ watt.}$

Orbital Velocity (V_o) :-

- The speed of a satellite in its orbit is called orbital velocity.

- $V_o = \sqrt{gR}$

- Where ' g ' is the acceleration due to gravity.
at earth's surface
 $g = 9.8 \text{ ms}^{-2}$

'R' is the radius of the earth

$$R = 6.4 \times 10^6 \text{ m}$$

- In the case of earth its value is

$$V_0 = 7.9 \text{ km s}^{-1}$$

Escape Velocity (V_e) :

- The minimum velocity with which a body should be projected to overcome the earth's gravitation field is called the escape velocity.

where 'g' is the acceleration due to gravity at the earth's surface.

- In the case of the earth its value is 11.2 km s^{-1} .
- 'R' is the radius of the earth.
- Its value is constant for all bodies irrespective of their mass on a particular planet.
- It varies from planet to planet
- It is independent of the angle of projection

Satellite :

- A body which revolves round another bigger body (planet) is called a satellite.

Ex:- The moon is a satellite to the earth.

- There are natural and artificial satellites.
- The moon is a natural satellite to the earth.

Artificial satellite :

- If a body is made to revolve round the earth in the same way as the moon, it is an artificial satellite.

Ex: Sputnik

Principle of artificial satellite :

- The principle of launching an artificial satellite into a proper space orbit is to impart sufficient initial horizontal speed such that it revolves round the earth at the chosen height.
- To give the desired speed to a satellite and overcome the force of gravity, the launching of a satellite requires a tremendous force. This is achieved with the help of rockets.
- Since the force of gravity is minimum at the equator, it is easier to launch satellite from equatorial regions.
- Since the earth rotates from west to East satellites are launched in the eastward direction to give them additional push. It is still easier to launch satellites from space shuttles orbiting the earth.

Geostationary or Synchronous Satellites :

- The geostationary satellite is an artificial satellite which revolves round the earth with period of revolution equal to the period of rotation of earth.
- The time period of revolution of geostationary satellite is 24-hours.
- It rotates from west to east.

- A geostationary satellite is one which appears stationary with respect to the earth.
- Height of Geostationary Satellite from earth is 36,000 km (Approximately)
- Geostationary satellites are used for communication and weather forecasting.

Centre of mass : (C.M) :

- The centre of mass of a system of particles is that point lying within the boundary of system where the entire mass appears to be concentrated.
- It refers to mass of the body.
- In the case of ordinary small and regular bodies the centre of mass and the centre of gravity coincide.
- The position of C.M. depends on the shape of the body and the distribution of mass. So the centre of mass may lie within or outside the body.

Centre of gravity (C.G) :

- The centre of gravity of a body is the point where the whole weight of the body can be considered to act.
- In the case of very large bodies the C.M. and C.G. do not coincide. This is because 'g' changes from one part to another part of the very large body.
- For all regular shape bodies the geometric centre is the centre of gravity.

STABILITY :

- The stability as the state of a body depends upon two factors
 - 1) base area
 - 2) height of C.G. stability of the state of a body increases with base area and decreases with the height of C.G.

Applications :-

- 1) Racing cars are built low and with wide wheel bases to reduce the risk of over turning at sharp bends.
- 2) While crossing a river in a boat, passengers are not allowed to stand this keeps the C.G. of the system (boat and passengers) low and ensures stability.
- 3) In a double - decker bus, more passengers are allowed in the lower deck than in the upper deck. Moreover, standing is not allowed in the upper deck. This is done to keep the C.G. fairly low.
- 4) Tall lamp stands have heavy bases so that the C.G. is low.
- 5) A porter carrying a load on his back leans forward to keep the vertical line passing through the C.G. of the System (load and Person) between his feet.
- 6) A person has to bend forward while going up hill and backward while coming down. He bends in order to keep the vertical line passing through his C.G. always between his feet thereby increasing his stability.
- 7) A person walking on a rope, holds a long pole in his hand. He changes the orientation of the pole such that the line of action of the total weight always passes through the rope so that he does not fall down.



FRICTION

- The force which always opposes the motion of one body over another body in contact with it is called friction.

Types of friction :

- 1) Static friction (f_s)
- 2) Kinetic friction (f_k) (or) sliding friction.
- 3) Rolling friction (f_r)

1) Static friction (f_s) :-

- The frictional force between two surfaces in contact before relative motion between them is called static friction.

Ex:- Many labourers try to move a heavy stone on the floor. They are not able to move it even by 1mm. The floor applies frictional force on the stone. This is example of static friction.

2) Kinetic friction (f_k) (or) sliding friction :

- The frictional force present when one body slides over another body is called kinetic friction.
- Kinetic friction is less than static frictions.
- Ex:- The frictional force acts between when book is sliding on inclined plane.

3) Rolling friction (f_r) :

- When a body like a wheel, cylinder or drum rolls over the surface of another body the frictional force acting between them is called rolling friction.
- Rolling friction is less than kinetic friction.

NOTE : Static friction (f_s) > Kinetic friction (f_k) > Rolling friction (f_r)

Normal reaction :-

- When a body is resting on a surface or sliding over it the force exerted by the surface against the body is known as the normal reaction.

Coefficient of friction(μ):

- It is the ratio of the frictional force to the normal reaction.
- It has no units.
- We have three coefficients of friction
- They are
- 1) Coefficient of Static friction (μ_s)
- 2) Coefficient of Kinetic friction (μ_k)
- 3) Coefficient of Rolling friction (μ_r)

NOTE :- $\mu_s > \mu_k > \mu_r$

Advantages of Friction :-

- 1) We can walk safely without slipping because of friction between ground and feet (or shoes). We can also run safely for the same reason. We cannot walk on surface of ice because it is very smooth.
- 2) Brakes can be applied due to friction

- 3) Nails can be fixed in walls due to friction.
- 4) A match stick is lighted because of friction.
- 5) One can hold any object between fingers due to friction.

Disadvantages of Friction :-

- 1) Due to friction there is a loss of energy and hence more work is necessary to operate a machine.
- 2) It causes wear and tear
- 3) It generates heat which damages the machines.

Methods of Reducing Friction :-

Polishing: Friction between two surfaces can be reduced by polishing them.

Bearings: The rolling friction is less than the sliding friction. The free wheels of a cycle, motor car, shafts of motors, dynamos etc are provided with ball bearings to reduce friction considerably. Bearings convert sliding motion into rolling motion.



Lubricants : A lubricant is a substance which form a thin layer between two surfaces in contact and reduces friction

Examples :- grease, oil etc.

- In heavy machines “Graphite” is used as a lubricant.

Streamlining : Auto mobilies and aeroplanes are streamlined to reduce the friction due to air.

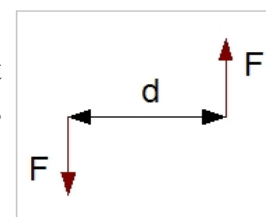
Rotatory Motion : A body is said to have rotatory motion when it moves in a circular path about a fixed point or about an axis.

- Ex:**
- 1) The motion of a grinding stone,
 - 2) Motion of a fly wheel
 - 3) Spin of the earth about its own axis

Couple : Two equal, unlike, parallel forces acting at different points on a rigid body form a couple.

Moment of couple : The moment of a couple is equal to the product of one of forces and the perpendicular distance between the lines of action of the forces.

- The moment of couple is ‘torque’
- It is a vector.
- Its S.I. Unit is newton-metre (N-m).



Angular Momentum : The moment of momentum of a particle about an axis is called angular momentum.

- It is a vector quantity
- S.I. unit: $\text{kg.m}^2/\text{sec}$

Conservation of angular momentum :-

- Angular momentum of a body remains constant when the external torque is zero.

Examples :-

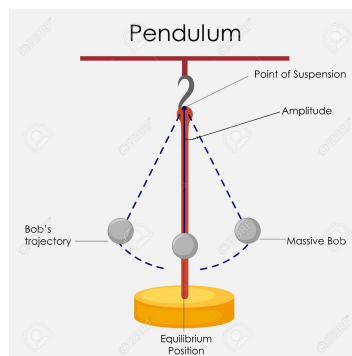
- 1) A diver leaves the spring board with some angular velocity and jumps upwards. He brings his hands and legs closer to his body. His moment of inertia decreases then his angular velocity increases. This helps him to make more number of rotations quickly before touching the water surface.
- 2) The ballet dancers, ice skaters and acrobats use the conservation of angular momentum during their flight and exhibit different spins and orientations of their body.

**SIMPLE HARMONIC MOTION (SHM) :**

- A body is said to have simple harmonic motion, if its acceleration is always directed towards a fixed point on its path and is proportional to its displacement from the fixed point.
- $a \propto -y$
- $a = -ky$
- $a + ky = 0$
- (or)

$$\frac{d^2y}{dt^2} + Ky = 0$$

- Where, a - acceleration and
y - displacement
k - proportionality constant

**Examples :-**

- 1) Motion of cradle
- 2) Motion of sphere (bob) of a simple pendulum.
- 3) Motion of needle of a sewing machine
- 4) Motion of atoms in solids.
- 5) Motion of pendulum in pendulum clock.

**Time Period (T) :**

- The time taken to complete one oscillation of the particle is called time period.
- The time period of simple pendulum is given by
- Where l - is length of the pendulum
 g - is acceleration due to gravity.
- The period of simple pendulum is independent of mass, material, size and shape of the bob of the pendulum.

Seconds Pendulum :

- A pendulum whose time period is 2 seconds is called seconds pendulum.
- Its length on the surface of the earth 100 cm. (approximately)

Pendulum Clock :

- For a clock showing correct time, the period of the pendulum is exactly two seconds.
- 1) In summer the pendulum expands. Its time period (T) increases. It makes less number of oscillations in a day. The clock loses time and goes slow.
- 2) In winter, the pendulum contracts. Its time period (T) decreases. It makes more number of oscillations in a day. The clock gains time and goes fast.

NOTE: To avoid losing or gaining of time pendulums are made of Invar.

- Because its expansion is almost zero.

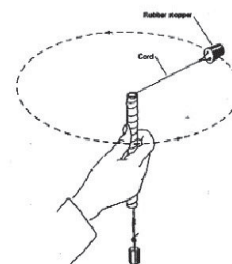
**Centripetal Force :**

- The force which acts continuously on a particle initially moving with a linear velocity and makes it travel along a circular path is called centripetal force.

- where m - mass of the body
v - Velocity
r - radius of the circle
- This force is directed towards the centre.

Examples of Centripetal Force :-

- 1) Centripetal force is a must for the uniform circular motion of a body. Earth goes round the sun because of gravitational force of attraction between them. The gravitational force here acts as the centripetal force.
- 2) The same gravitational force of attraction provides the necessary centripetal force for a satellite revolving round a planet. In this case the centripetal force on the satellite arises due to “force of gravity” between the satellite and the planet.
- 3) Similarly, the finger (the pull in the string) provides the necessary centripetal force for a circular motion of a stone or bob tied at one end of the string.
- 4) In the case of electron going round the nucleus in an atom, the centripetal force is provided by the electrostatic force of attraction between them.

**Centrifugal Force :-**

- The radially outward force on a body in a uniform circular motion, observable only in a rotating frame of reference, is called centrifugal force.
- These forces are pseudo- forces (not real)

- Centrifugal Force = $-\frac{mv^2}{r}$
- Where 'm' is mass of the body 'v' is the velocity of the body, r is the circular path of radius.

Note: Both centripetal force and centrifugal force act on the same rotating body. Therefore, they do not form action-reaction pair in the sense of Newton's third law.

Applications :

Centrifuge :

- A centrifuge is a machine used to separate particles of higher mass from those of lower mass in a given mixture.



Uses :-

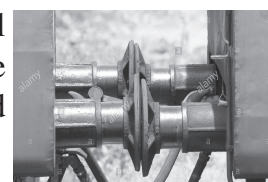
- 1) When milk is poured into the cylindrical vessel of the centrifuge and rotated with high speed, the particles of higher mass (Skimmed Milk) are thrown away from the centre due to greater centrifugal force and lower - mass cream particles collect at the centre.
- 2) Churner for separating butter and butter-milk from curds.
- 3) Sugar crystals are separated from molasses.
- 4) Honey is separated from bee's wax.
- 5) In a non- homogenous mixture of solutions, precipitates and sediments are separated by this method.
- 6) Washing machine (laundry drier)



Banking of Roads and Rail Tracks :

Banking of roads :

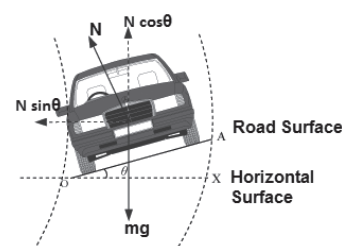
- When a road curved the outer edge of the road is slightly raised above the level of the inner edge to provide “**Centripetal Force**” to the vehicles passing through that curved path. This is called Banking of roads.
- If there is no banking, the car has to take the help of frictional force between its tyres and the road. Since, the frictional force is limited and is not always dependable, banking of curved roads is necessary. Otherwise the car or train would skid.



Angle of Banking (θ) :

- The angle made by the line joining the outer, raised edge of the road to the inner edge with the horizontal line is called angle of banking.

- Thus, the value of the banking angle (θ) depends on expected speed (v) of vehicle and the radius of curvature of road 'r'.



- If the car maintains a speed 'v' less than that required speed (V_{req}) on the curved road.
- i, e $V < V_{required}$ - It slips towards centre
 $V > V_{required}$ - It skids outside
- The same principle is applied for banking of rail tracks.

Banking of rail tracks :

- The centripetal force necessary to make the train travel on the curved track is provided by raising the outer rail slightly above the level of the inner rail towards the centre of curvature of radius. This is called banking of rail tracks.

Geo-Centric Theory :

- This was proposed by 'ptolemy'.
- According to this theory all heavenly bodies like sun, stars, planets, moon etc. revolve round the earth which is stationary - earth is taken as centre of the universe.

Helio - Centric theory :

- This was proposed by "copernicus". According to this theory, sun is stationary and all planets are revolving around it.

Kepler's Laws of planetary Motion :-

i) 1st law (law of orbits) :-

- Every planet revolve round the sun in an elliptical orbit with the sun at one of its foci.

ii) 2nd law (law of areas) :

- The line joining the sun and the planets sweeps equal areas in equal intervals of time.

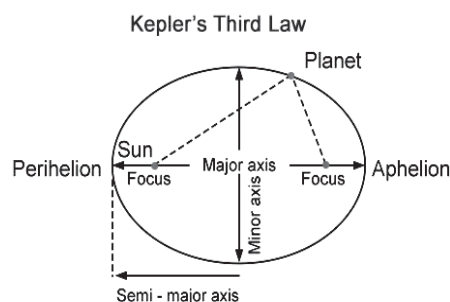
iii) 3rd law (law of periods) :

- The square of period of revolution of a planet around the sun is directly proportional to cube of the semi-major axis length.

$$T^2 \propto R^3$$

- Where, T - Time Period

R - Length of semi major axis



Weight of a body in a lift :-

- Suppose a person is standing on a weighing machine in a lift.

Case (i) : When the lift is stationary the scale shows his actual weight.

Case (ii) : When the lift accelerates upwards, the scale show higher weight because the scale pushes harder against his feet.

Case (iii) : When the lift accelerate down wards the scale shows less weight.

Case (iv) : If the cable of the lift breaks and starts falling freely, the reading on the scale becomes zero and the person experiences weightlessness.

Note: Consider an astronaut orbiting the earth in a spaceship. He feels weightless because he is not pushing against anything. The situation's similar to that in a freely falling lift.

Elasticity

Definition :

- The property by which body regains its original condition, after the deforming forces are removed is called elasticity.

Perfectly elastic body:

- A body which completely regains its original condition after the deforming forces are removed is called perfectly elastic body.
- There is no perfectly elastic body in nature.
- But quartz and phosphor bronze are examples of near perfect elastic bodies.

Perfect plastic (inelastic) body :

- A body which does not regain its original condition after the deforming forces are removed is called perfectly plastic body.
- There is no perfect plastic bodies in nature

Ex:- putty, wax are near plastic bodies.

Stress :

- Restoring force per unit area is called stress.

Stress =

- S.I. Unit is - pascal

Strain :

- Change in one dimension per unit dimension is called strain.

$$\text{strain} = \frac{\text{change in dimension}}{\text{original dimension}}$$

- It has no units.

Elastic limit :

- The maximum value of stress within which the body completely regains its original condition when deformation force is removed is called elastic limit.

Hooke's law :

- Within the elastic limit, stress is proportional to the strain produced in a body.

$$\text{stress} \propto \text{strain}$$

$$\text{stress} = E \quad \text{strain}$$

- where 'E' is called modulus of elasticity.
- Steel is more elastic than copper, brass and Aluminium. For this reason, steel is preferred in heavy duty machines and in structural designs.



Elastic Fatigue :

- The state of temporary loss of elastic nature of a body due to repeated stress is called elastic fatigue.

MECHANICS - Practice Bits

- | | |
|---|---|
| <p>1. The force which pulls all bodies towards the surface of the earth is called.</p> <ol style="list-style-type: none"> 1) mechanical force 2) attractive force 3) nuclear force 4) gravitational force <p>2. Force of friction can.</p> <ol style="list-style-type: none"> 1) oppose the motion of a body 2) produce heat 3) produce wear and tear 4) all the above <p>3. Friction can be reduced by.</p> <ol style="list-style-type: none"> 1) making the surface as smooth as possible 2) by applying lubricant between the surface 3) changing sliding friction to rolling friction 4) all the above <p>4. The bodies of fast moving vehicles are streamlined so that.</p> <ol style="list-style-type: none"> 1) they look nice 2) friction due to air or water decrease 3) they can cut through air or water 4) none of the above <p>5. No work is done when the displacement caused by force is.</p> <ol style="list-style-type: none"> 1) zero 2) at right angles to the direction of force 3) both (1) and (2) 4) none of the above <p>6. In which case work is not done?</p> <ol style="list-style-type: none"> 1) A labourer pulling a cart on the road 2) A girl climbing down the stairs 3) A car waiting at the red signal 4) A fish swimming in water | <p>7. SI unit of work is.</p> <ol style="list-style-type: none"> 1) newton 2) joule 3) pascal 4) newton second <p>8. When a force of one___causes a displacement of 1 m in its own direction, the work done is said to be one joule.</p> <ol style="list-style-type: none"> 1) newton 2) kilogram force 3) gram force 4) dyne <p>9. The force required to do a work of 250 J, when the displacement in the direction of force is 10 m is.</p> <ol style="list-style-type: none"> 1) 25 N 2) 250 N 3) 2.5 N 4) None of the above <p>10. The ability to do work is called.</p> <ol style="list-style-type: none"> 1) force 2) energy 3) pressure 4) density <p>11. The energy possessed by a body by virtue of its position or configuration is called.</p> <ol style="list-style-type: none"> 1) mechanical energy 2) kinetic energy 3) potential energy 4) none of the above <p>12. The energy possessed by a body by virtue of its motion is called.</p> <ol style="list-style-type: none"> 1) kinetic energy 2) potential energy 3) gravitational energy 4) mechanical energy <p>13. A body of mass 'm' is at a height 'h' from the ground level. If the acceleration due to gravity is 'g' its potential energy is.</p> <ol style="list-style-type: none"> 1) mg/h 2) mh/g 3) hg/m 4) mgh |
|---|---|

14. Which one of the following does not possess potential energy.

- 1) A compressed string
- 2) A shooting arrow
- 3) A stretched bow
- 4) A wound up spring of a watch

15. Which one of the following possess potential energy.

- 1) A stone placed at the top of a roof
- 2) A speeding train
- 3) Wind blowing
- 4) A boy on roller skates

16. If 'm' is the mass of a body and 'v' is its velocity, then kinetic energy of the body is.

- 1) mv^2
- 2) mv
- 3) $\frac{1}{2}mv^2$
- 4) $\frac{1}{2}m^2v$

17. A body of mass 2 kg is moving with a constant velocity of 1 ms^{-1} . The kinetic energy possessed by the body is.

- 1) 2 J
- 2) 1 J
- 3) 4 J
- 4) J

Hint: K.E. = $\frac{1}{2}mv^2$

$$\begin{aligned} \text{K.E.} &= \frac{1}{2} \times 2 \times (1)^2 \\ &= 1 \text{ J} \end{aligned}$$

18. A body of mass 4 kg is moving with a velocity v. If the kinetic energy of the body is 32 J, then magnitude of v is.

- 1) 8 ms^{-1}
- 2) 2 ms^{-1}
- 3) 16 ms^{-1}
- 4) 4 ms^{-1}

Hint: K.E. = $\frac{1}{2}mv^2$

$$\begin{aligned} 32 &= \frac{1}{2} \times 4 \times v^2 \\ v^2 &= 16 \\ v &= 4 \text{ ms}^{-1} \end{aligned}$$

19. During photosynthesis the light energy of the sun changes to.

- 1) chemical energy
- 2) mechanical energy
- 3) potential energy
- 4) none of the above

20. When a person speaks in front of a microphone, the sound energy changes to.

- 1) a louder sound energy
- 2) magnetic energy
- 3) electric energy
- 4) none of the above

21. In a loudspeaker connected to a public address system.

- 1) sound energy changes to electric energy
- 2) electric energy changes to sound energy
- 3) sound energy changes to kinetic energy
- 4) none of the above

22. During the charging of a cellphone, the electric energy changes to.

- 1) chemical energy
- 2) magnetic energy
- 3) nuclear energy
- 4) potential energy

23. A body can have potential energy account of.

- 1) its position only
- 2) its configuration only
- 3) either (1) or (2)
- 4) neither (1) or (2)

24. Potential energy due to gravitational force possessed by a body is given by the expression.

- 1) mgh
- 2) mg/h
- 3) mgh^2
- 4) gh

- 25. An aeroplane flying high in the sky possesses.**
1) potential energy only
2) kinetic energy only
3) both (1) and (2)
4) none of the above
- 26. Magnetic energy can be classified as.**
1) kinetic energy
2) potential energy
3) both (1) and (2)
4) none of the above
- 27. In an electric bulb the kinetic energy of a moving electron changes first into.**
1) potential energy and then heat energy
2) light energy and then heat energy
3) heat energy and then light energy
4) none of the above
- 28. According to Newton's first law of motion, unless some external force acts, a body will continue in its.**
1) state of rest in a straight line
2) state of motion in a straight line
3) both (1) and (2)
4) none of the above
- 29. The tendency of a body to continue in its state of rest or uniform motion in a straight line, even when some external force acts on it is called.**
1) pressure 2) momentum
3) inertia 4) force
- 30. The measure of inertia of a body is its.**
1) density 2) mass
3) volume 4) acceleration
- 31. There are two similar spheres, such that one is made from lead and other from wood. Which will have more inertia.**
1) Lead sphere
2) Wooden sphere
3) Both (1) and (2)
4) None of the above
- 32. The passengers sitting in a stationary bus tend to fall backward, when the bus accelerates suddenly on account.**
1) sitting carelessly
2) inertia of rest
3) inertia of motion
4) none of the above
- 33. When a carpet is hit with a stick, the dust flies off on account of.**
1) the force of stick
2) inertia of motion of stick
3) inertia of rest of dust
4) both (2) and (3)
- 34. A rider falls in the forward direction when a galloping horse stops suddenly, on account of.**
1) carelessness of rider
2) inertia of motion
3) inertia of rest
4) none of the above
- 35. The product of mass and velocity of a body is called.**
1) force 2) momentum
3) moment of force 4) pressure
- 36. The SI unit of momentum is.**
1) newton
2) newton/second
3) newton-second
4) none of the above

37. A body of mass m is moving with velocity v . If the mass is increased 3 times and velocity is decreased 3 times, the change in its momentum will be.

- 1) $9mv$ 2) mv
3) $6mv$ 4) zero

Hint: $p = mv$

$$p = (3m)v$$

$$p = mv$$

38. The force acting on a body is directly proportional to.

- 1) momentum of body
2) rate change of momentum of a body
3) rate of change of inertia of a body
4) none of the above

39. The SI unit of force is.

- 1) newton 2) pascal
3) dyne 4) kilogram force

40. One newton is equal to.

- 1) $1 \text{ kg} \times 1 \text{ ms}^{-1}$ 2) $1 \text{ kg} \times 1 \text{ ms}^{-2}$
3) $9.8 \text{ kg} \times 1 \text{ ms}^{-1}$ 4) $10 \text{ kg} \times 1 \text{ ms}^{-2}$

41. One kilogram force (kgf) is equal to.

- 1) 9.8 N 2) 1.6 N
3) 10.0 N 4) 5 N

42. In CGS system, 1 N is equal to.

- 1) 10^5 dynes 2) 10^7 dynes
3) 10^4 dynes 4) 10^3 dynes

43. Action and reaction forces act on.

- 1) the same body
2) two different bodies
3) both (1) and (2)
4) none of the above

44. The rising of the rocket can be explained by.

- 1) Law of conservation of momentum
2) Newton's first law of motion
3) Newton's second law of motion
4) Pascal's law

45. Centre of gravity of a body is.

- 1) imaginary point within the body
2) a point where the whole weight of body acts
3) both (1) and (2)
4) none of the above

46. Some pairs of devices and associated energy changes are given below.

- a) Electric Motor - Electrical to Mechanical
b) Electrical Generator - Mechanical to Electrical
c) Diesel Engine - Light to Electrical
d) Solar Cell - Chemical to Mechanical

Which of the above pairs is/are correctly matched?

- 1) c and d 2) a and c
3) a and b 4) b and d

47. The mass of a body is different from its weight because.

- a) Mass is a variable quantity whereas weight is constant
b) Mass varies very little at different places whereas weight varies a lot
c) Mass is constant but weight decreases as the body moves from the poles to the equator
d) Mass is a measure of the quantity of matter whereas weight is a force
Which of the above statements is/are correct.

- 1) only 'a' 2) 'b' and 'd'
3) only 'c' 4) 'c' and 'd'

48. The weight of a body is.

- a) Same everywhere on the surface of the earth
- b) Maximum at the poles
- c) Maximum at the equator
- d) Less on the hills than in the plains

Which of the above statements is/are correct.

- 1) Only a
- 2) b and d
- 3) c and d
- 4) None

49. If a body is taken from the earth to the moon.

- a) Its mass will be different but weight will remain the same
- b) Both mass and weight will be different
- c) Its mass will remain the same but weight will be different
- d) Its density will remain unchanged

Which of the above statements is/are correct.

- 1) Only a
- 2) Only b
- 3) c and d
- 4) b and d

50. One finds it more difficult to walk on Ice than on a concrete road because.

- 1) Ice is soft spongy whereas concrete is hard
- 2) The friction between the ice and the feet is less than that between the concrete and the feet
- 3) There is more friction on ice than on concrete
- 4) None of these

51. It is easier to roll a barrel than to pull it because.

- 1) The full weight of the barrel comes into play when it is pulled
- 2) Rolling friction is much less than sliding friction

- 3) The surface area of the barrel in contact with the road is more in the case of pulling
- 4) There is a reason other than above mentioned

52. A sheet of paper can be pulled out quickly from under a glass of water without spilling the water. This phenomenon illustrates.

- 1) Lack of friction between paper and glass
- 2) Newton's third law of motion
- 3) Inertia
- 4) Acceleration

53. When the velocity of a body is doubled, its

- a) Acceleration is doubled
- b) Momentum is doubled
- c) Kinetic energy becomes four times
- d) Potential energy is doubled

Which of the above is/are correct?

- 1) b only
- 2) b and c
- 3) c only
- 4) b and d

Hint: $p = mv$

$$p = m(2v)$$

$$p = 2mv$$

$$K.E. = mv^2$$

$$= m(2v)^2$$

$$= 4 \times mv^2$$

54. Conservation of energy means that.

- 1) Energy can be created as well as destroyed
- 2) Energy can be created but not destroyed
- 3) Energy cannot be created but can be destroyed
- 4) Energy can neither be created nor destroyed

- | | |
|---|---|
| <p>55. Treads on the shoe soles are made to increase.
 1) Strength 2) Durability
 3) Friction 4) Elasticity</p> <p>56. Broad wooden sleepers are placed below the rails to.
 1) Act as shock absorber
 2) Fix the fish plates properly
 3) Hold the rails parallel
 4) Reduce pressure exerted by the train.</p> <p>57. Which of the following is used as a lubricant in heavy machines.
 1) Bauxite 2) Sulphur
 3) Phosphorus 4) Graphite</p> <p>58. An iron ball and a wooden ball of the same radius are released from the same height in a vacuum. The time taken by both of these to reach the ground is.
 1) Roughly equal 2) Zero
 3) Exactly equal 4) Unequal</p> <p>59. A rocket works on which of the following principle.
 1) Conservation of mass
 2) Conservation of energy
 3) Conservation of linear momentum
 4) Conservation of angular momentum</p> <p>60. Weight of a body is maximum on the surface of earth at
 1) equator 2) anywhere
 3) poles 4) none of the above</p> <p>61. A spring balance shows a weight of 40 gf at equator for a given body. Its weight at the poles will be
 1) 40 gf 2) less than 40 gf
 3) more than 40 gf 4) both 1 & 2</p> | <p>62. A person on the top of Mount Everest weighs 60 kgf. His weight at sea level will be
 1) 60 kgf
 2) less than 60 kgf
 3) more than 60 kgf
 4) none of the above</p> <p>63. The minimum speed required by a rocket to put a satellite in space is
 1) 11.2 km/s 2) 11.2 m/s
 3) 11.2 miles/h 4) 11.2 km/min</p> <p>64. The man made objects orbiting around the earth are called
 1) meteors
 2) satellites
 3) launch vehicles
 4) space vehicles</p> <p>65. T.V. broadcasting satellites are placed in
 1) polar orbit
 2) equatorial orbit
 3) geostationary orbit
 4) any of the above</p> <p>66. The distance of a satellite from the surface of the earth in geostationary orbit is about
 1) 36000 km 2) 36000 miles
 3) 35000 km 4) 63000 km</p> <p>67. Speed of an object is given by
 1)
 2)
 3) Time distance
 4) None of the above</p> <p>68. The unit of speed is
 1) m/s 2) s/m
 3) m s 4) none of the above</p> |
|---|---|

69. A body moves with a variable velocity when

- 1) magnitude of velocity changes
- 2) direction of the body changes
- 3) any of the 1 or 2
- 4) none of 1 or 2

70. A stone is projected in the vertically upward direction. The acceleration due to gravity on its upward journey is

- 1) negative
- 2) positive
- 3) zero
- 4) none of the above

71. The acceleration due to gravity is maximum at

- 1) equator
- 2) inside the earth
- 3) up in the mountains
- 4) poles

72. The ratio of displacement and time is equal to

- 1) speed
- 2) velocity
- 3) acceleration
- 4) none of the above

Answers

1) 4	2) 4	3) 4	4) 2	5) 3	6) 3	7) 2	8) 1	9) 1	10) 2
11) 3	12) 1	13) 4	14) 2	15) 1	16) 3	17) 2	18) 4	19) 1	20) 3
21) 2	22) 1	23) 3	24) 1	25) 3	26) 4	27) 3	28) 3	29) 3	30) 2
31) 1	32) 2	33) 3	34) 2	35) 2	36) 3	37) 2	38) 2	39) 1	40) 2
41) 1	42) 1	43) 2	44) 1	45) 3	46) 3	47) 4	48) 2	49) 3	50) 2
51) 2	52) 3	53) 2	54) 4	55) 3	56) 4	57) 4	58) 3	59) 3	60) 3
61) 3	62) 3	63) 1	64) 2	65) 3	66) 1	67) 2	68) 1	69) 3	70) 1
71) 4	72) 2								

5. FLUID MECHANICS

- Fluid mechanics is a branch of physics concerned with the mechanics of fluids and the forces on them.

Fluids :

- The substances which can flow from one place to another is called fluids.

Ex:- Liquids and gases.

Surface Tension :-

- The free surface of a liquid behaves like a stretched membrane under tension always trying to have minimum surface area, due to molecular forces is called surfacetension.
- It is the liquid phenomenon.
- Surfacetension of a liquid is defined as the force acting perpendicularly per unit length on an imaginary line drawn at the surface of the liquid.

surface tension (s) =

- Its S.I. Unit is: newton / metre (or) N/m.

Factors affecting surface tension :

Temperature:

- The surface tension of a liquid decreases with increase in temperature.
- Surface tension of a liquid becomes zero at critical temperature.
- When soap, detergent, dettol, phenyl etc are mixed in water then its surface tension decreases.
- When salt is added in water, its surface tension increases.
- When oil spreads over the surface of water its surface tension decreases.

Applications of surface tension :

- 1) The minimum surface area of a given amount of liquid is for spherical shape. Therefore rain drops are spherical.
- 2) Small drops of mercury, water sprinkled on smooth surface assume spherical shape.
- 3) Insects (Mosquitoes) walking on surface of water because the surface of water acts like a stretched elastic membrane.
- When kerosene oil is poured over water its surface of a mosquito sitting on water surface is not balanced by surface tension force and dipped in water.
- 4) When a shaving brush is dipped in water, its hair spread out on taking out the brush from water, the water film formed between the hairs while tending to make its surface area minimum due to surface tension will bring the hairs closer to each other.



Cohesive Forces :

- The intermolecular force of attraction acting between the molecules of same substance is called cohesive force.

Ex:- 1) Intermolecular force of attraction acting between the molecules of water.
2) Intermolecular force of attraction acting between the molecules of mercury.

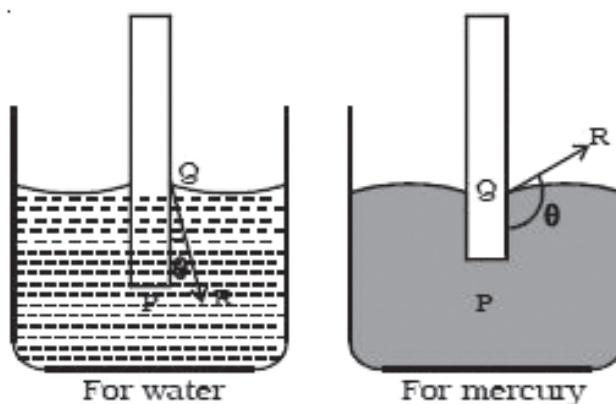
Adhesive Forces :

- Intermolecular force of attraction acting between the molecules of different substance is called adhesive force.

Ex:- Intermolecular force of attraction acting between the molecules of “paper and gum”. (or) paper and ink etc.

Angle of Contact :-

- The angle between the tangent to the liquid surface of the point of contact and the solid surface, inside the liquid is called angle of contact.
- Angle of contact depends upon the nature of liquid and the solid. It is independent of angle of inclination of solid surface.
- Angle of contact increases with increase in temperature of liquid.
- For pure water and clean glass angle of contact, $\theta = 0^\circ$
- For ordinary water and glass $\theta = 18^\circ$
- For mercury and glass, the angle of contact $\theta = 135^\circ - 140^\circ$

**Case (i) :****If angle of contact $< 90^\circ$:-**

- a) The liquid wets the glass
- b) The liquid rises up in a capillary tube
- c) The surface is concave upwards
- d) **Ex:-** water and glass

If angle of contact $> 90^\circ$:-

- a) The liquid does not wet glass
- b) The liquid depressed in a capillary tube.
- c) The surface is convex upwards
- d) **Ex:-** mercury - glass.

Capillarity :

- The property of rise or depression of a liquid due to surface tension in a capillary tube is called capillarity.

Note:- Mercury does not rise in capillary tube because it contains strong cohesive forces and angle of contact is above 90°

Applications of Capillarity :

- 1) Blotting paper absorbs ink because it contains fine pores (hole) between its fibres which act as capillaries.
 - 2) Flow of ink in a pen nib
 - 3) Sucking of water by plants.
 - 4) Oil rises up in the wick of a lamp by capillarity between fibres and burns continuously.
- similarly molten wax in a candle rises up through the wick.
- 5) Piece of sponge, absorbs water because of capillary action of its pores.
 - 6) Towel is used to dry the wet body as it absorbs water by capillary action of its fibres.
 - 7) Bricks contain pores and absorb water in rainy season due to capillary action. To prevent this, cement plastering is done.
 - 8) Ploughing of fields is essential for preserving moisture in the soil. By ploughing, the fine capillaries in the soil are broken. Water from within the soil shall not rise and evaporate off.
 - 9) Sand is drier soil than clay. This is because holes between the sand particles are not so fine as compared to that of clay as to draw up water by capillary action.

**VISCOSITY :**

- Viscosity is the property of a fluid by virtue of which an internal frictional force acts between its layers when it is motion. This force is called viscous force.
- Viscous force acting between two layers of a liquid is given by

$$F =$$

- Where, η = coefficient of viscosity
 A = contact area of layers
 $\frac{dv}{dy}$ = velocity gradient

- If $A = 1\text{m}^2$ and $\frac{dv}{dy} = 1\text{s}^{-1}$ then $F = \eta$.

Coefficient of viscosity :

- It is the viscous force acting between two layers of unit area of cross-section having unit velocity gradient.
- Its S.I. Unit is - NSm^{-2} (or) pascal - second. (or) Poiseuille.
- Its C.G.S. Unit is - dyne. sec/cm^2 (or) poise
- Water has a viscosity of 0.0089 poise at 25°C .
- “Viscometer” is used to measure viscosity of fluid.

Effect of Temperature on Viscosity :

- 1) The Viscosity of liquids decreases with increase in temperature (due to cohesive forces decreases between the molecules of liquid)
- 2) The viscosity of gases increases with increase in temperature (due to exchange of momentum increases)



Examples of Viscosity in Daily Life :

- 1) When a parachutist opens up his parachute, due to buoyancy and viscous drag of air, his velocity reduced and he descends down slowly to the ground.
- 2) The selection of lubricating oil used to reduce friction between various machine parts, depends upon the viscosity of oil.
- 3) When water flows over a mixture of sand and gold particles, the lighter sand particles are carried away by viscosity of water and heavier gold particles are sedimented.
- 4) The speed of rain drops decreases due to viscosity of air.

Bernoulli's Theorem :

- If a non-viscous and incompressible liquid is flowing in stream - lined flow then total energy (sum of pressure energy, kinetic energy and potential energy) per unit volume at every point remains constant.



$$\rho \left[P + \frac{1}{2} \rho v^2 + gh \right] = \text{constant}$$

- For horizontal flow of fluid.

- where, P - pressure energy per unit volume

$$\frac{1}{2} \rho v^2 - \text{K.E per unit volume.}$$

$$gh - \text{P.E. per unit volume}$$

- According to Bernoulli's principle, at the point where the velocity is more, the pressure is less at that points.

Applications :-

- 1) Calender fluttering away from the wall due to fan.
- 2) When wind blows with very high speed very low pressure region is created above the roof. So pressure inside the roof is greater than outside. Therefore it is blown up.
- 3) For the same reason, wind blowing horizontally creates a low pressure above the kite. So it is pushed up by the high pressure existing below it.
- 4) Aeroplanes are able to fly upward in air against their own weight



basing on Bernoulli's principle.

- 5) Motion of spinning (Magnus) can be explained on the basis of Bernoulli's principle.
- 6) Venturi metre and aspirator pump works on Bernoulli's principle.
- 7) Atomiser or sprayer, scent and nasal sprayer are based on this principle.

Thrust :

- The total force exerted by a body on any surface in contact with it is called thrust.
- Its S.I. Unit is - newton (N).

Pressure :-

- The thrust acting per unit area of a surface is called pressure.
- Pressure =
- $P =$
- where, F-is force, A-area of surface.
- Its S.I. Unit - pascal (pa) (or) N/m^2

Some everyday application of the concept of pressure :

- 1) Broad wooden sleepers are placed below the rails to reduce the pressure exerted by the weight of a train.
- 2) **A sharp knife cuts better than a blunt one :**
 - The area of sharp edge is much less than the area of blunt edge. For the same total force, the pressure for the sharp edge is more than the bludge. Hence, a sharp edge cuts better.
- 3) **Pins and nails are made to have pointed ends :**
 - Their pointed ends have very small area. When a force is applied over the head of a pin or nail, it transmits a large pressure on the surface and hence easily penetrates the surface.

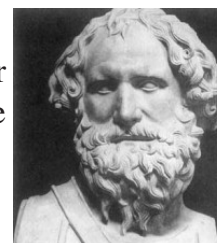


Buoyancy (Upthrust) :

- The upward force acting on a body immersed in a fluid is called upthrust or force of buoyancy and the phenomenon is called buoyancy.

ARCHIMEDE'S PRINCIPLE :

- This principle states that when a body is immersed fully or partially in a fluid, it experiences an upward thrust equal to the weight of the fluid displaced by it.



Applications :

- 1) This principle is used in designing ships and submarines.
- 2) Lactometers based on this principle used to measure purity of milk.
- 3) Hydrometers used to measure density of liquids are based on this principle.
- 4) It is because of the higher density of sea water that it is easier to swim in the sea than in river.

- 5) Life saving belt are based on this principle.
- 6) When a ship passes from river water (less density) to sea water (more density) ship rises a little.
- 7) A solid chunk of iron will sink in the water but float in mercury because the density of iron is more than that of water but less than that of mercury.

Laws of floatation :

- The law of floatation was proposed by Archimede's.
- 1) A body will float in a liquid if its density is less than the density of the liquid.
- 2) A body will be in equilibrium at any submerged depth if the density of the body is equal to the density of the liquid.
- 3) A body will sink in a liquid if its density is greater than the density of the liquid.

Density :

- The density of a substance is defined as its mass per unit volume.
- Density =
- Its S.I. Unit is - kg/m^3

Relative Density :

- It is the ratio of the density of the substance to the density of water.

$$\text{Relative density} = \frac{\text{Density of substance}}{\text{Density of water}}$$

- It has no units

- Density of water is 1000 kgm^{-3}
- Density of mercury is 13600 kgm^{-3}
- Density of gold is 19300 kgm^{-3}

Substance	Density(kg/m^3)
Alcohol	790
Aluminium	2,700
Copper	8,900
Gold	19,300
Lead	11,300
Mercury	13,600
Platinum	21,500
Steel	7,800
Tuopentine	870
Water	1000
Wood	700

Pressure in Liquids :

- A diver experiences pressure in the water due to the weight of water above him. The pressure at any point in a liquid acts in all directions. The pressure 'p' at a depth 'h' in a liquid of density 'ρ' is given by the relation
- $p = h \rho g$
- where 'g' is acceleration due to gravity.

Applications :

- 1) Since the pressure of water increases with depth, the bottom of a dam is made much thicker than the top.
- 2) The pressure and hence speed of water obtained from the ground floor tap is much higher than that from the top floor tap.

Pascal's law :

- It states that in increase of pressure at any point in a liquid at rest is transmitted equally in all directions without any change.

Applications :

- 1) Hydraulic press (Bramah press) is used to compress soft materials like cotton, cloth and paper into compact bales. Its working is based on pascal's law.
- 2) The working of "hydraulic brakes" and "air brakes" are based on pascal's law.
- 3) Hydraulic lift works on this principle only.
- 4) It is used to compress oil seeds for getting oil.

**Atmospheric Pressure :-**

- Air exerts pressure on each and every body located on earth's surface which is called atmospheric pressure.
- It is due to the pressure of the atmosphere that ink rises in the tube of a fountain pen, or liquid rises in a syringe when the piston pulled.
- At high altitudes where atmospheric pressure is less, nose bleeding may occur due to the greater pressure of blood.

BAROMETER:

- Atmospheric pressure is measured with an instrument called the "Barometer"
- Barometers are also used for weather forecasting.
- If the barometric height falls suddenly it indicates the coming of a "storm"
- A gradual fall in the barometric height indicates the possibility of rain.
- A gradual increase in the barometric height indicates fair weather.
- At sea level, atmospheric pressure is 76 cm of Hg. (or) 1.01×10^5 pascal.
- The unit of atmospheric pressure used for metreological purposes is called bar. Where
1 bar = 10^5 pa



FLUID MECHANICS - Practice Bits

- | | |
|--|--|
| <ol style="list-style-type: none"> 1. Pressure exerted by a substance is the force acting. <ol style="list-style-type: none"> 1) per unit length 2) per unit area 3) per unit volume 4) any of the above 2. SI unit of pressure is. <ol style="list-style-type: none"> 1) pascal 2) newton 3) watt 4) joule 3. The device used to measure atmospheric pressure is called. <ol style="list-style-type: none"> 1) calorimeter 2) ammeter 3) voltmeter 4) barometer 4. The liquid used in cistern barometer is <ol style="list-style-type: none"> 1) glycerine 2) water 3) mercury 4) alcohol 5. Which is not a correct statement, when mercury is used as a barometric liquid. <ol style="list-style-type: none"> 1) It does not stick to the walls of glass tube 2) It exerts maximum pressure on account of its high density. 3) It can be easily obtained in pure state 4) It vaporises under vacuum conditions 6. On a hill 1000 m high, the atmospheric pressure will be. <ol style="list-style-type: none"> 1) 76 cm of mercury 2) more than 76 cm of mercury 3) less than 76 cm of mercury 4) none of the above | <ol style="list-style-type: none"> 7. At a given place the barometric height falls suddenly and the air is dry. The weather forecast can be. <ol style="list-style-type: none"> 1) dust storm 2) rain storm 3) gentle breeze 4) mild drizzle 8. At a given place the barometric pressure rises suddenly. The weather forecast is. <ol style="list-style-type: none"> 1) dry weather 2) rain 3) anticyclonic weather with strong winds 4) fair weather 9. The force of attraction between the similar kind of molecules is called. <ol style="list-style-type: none"> 1) force of cohesion 2) force of adhesion 3) gravitational force 4) molecular force 10. When small amount of mercury is poured over a clean china plate, it forms tiny droplets of mercury. It is because. <ol style="list-style-type: none"> 1) there is large force of cohesion between the molecules of mercury 2) there is no force of adhesion between the molecules of mercury and the china plate 3) there is large force of adhesion between the molecules and the china plate 4) both (1) and (2) 11. The force of attraction between the different kinds of molecules is called force of. <ol style="list-style-type: none"> 1) gravitation 2) cohesion 3) adhesion 4) none of the above |
|--|--|

- 12. The molecules of ink stick to the molecules of paper, due to the force of.**
1) gravitation 2) cohesion
3) adhesion 4) none of the above
- 13. Water wets the surface of glass, due to the force of.**
1) adhesion 2) gravitation
3) cohesion 4) none of the above
- 14. The pressure exerted by a liquid is independent of its.**
1) depth
2) density
3) acceleration due to gravity
4) area of cross-section
- 15. When pressure is applied to any part of a liquid enclosed in a vessel, it is transmitted equally in all directions. The above statement is.**
1) Archimedes principles
2) Law of floatation
3) Pascal's law
4) none of the above
- 16. The approximate pressure exerted by the atmosphere at sea level is.**
1) 10^5 Pa 2) 10^6 Pa
3) 10^4 Pa 4) 4×10^4 Pa
- 17. The simple instrument used for reading atmospheric pressure is.**
1) ammeter 2) anemometer
3) barometer 4) galvanometer
- 18. The atmospheric pressure can support a maximum of.**
1) 76 cm vertical column of mercury
2) 80 cm vertical column of mercury
3) 72 cm vertical column of mercury
4) 40 cm vertical column of mercury
- 19. The vacuum caused in the barometer tube, above the surface of mercury is called.**
1) Pascal's vacuum
2) Galileo vacuum
3) Torricelli vacuum
4) Archimedes' vacuum
- 20. The upward force experienced by a solid, when immersed in a liquid is called.**
1) upthrust
2) buoyant force
3) apparent weight
4) both (1) and (2)
- 21. According to Bernoulli's principle, with increase in the speed of wind the atmospheric pressure.**
1) increase
2) decreases
3) remains same
4) none of the above
- 22. The buoyant force is directly proportional to the.**
1) volume of liquid displaced
2) weight of liquid displaced
3) density of the solid
4) none of the above
- 23. The pressure exerted at any point in an enclosed vessel is transmitted.**
1) only towards the sides of the vessel
2) only towards the base of the vessel
3) in all direction of the vessel
4) none of the above

24. When an ink fountain pen leaks in high altitude.

- 1) the atmospheric pressure is more than the pressure inside the pen
- 2) the atmospheric pressure is less than the pressure inside the pen
- 3) the density of ink on high altitudes decreases
- 4) the density of ink on high altitudes increases

25. A tumbler, filled to the brim with water, has an ice cube floating in it, A part of the floating ice cube is above the surface of the water. What happens when the cube starts melting.

- 1) Water starts spilling over
- 2) Water level remains the same
- 3) Water level starts falling
- 4) Water level first falls and then rises

26. Oil is sprinkled on stagnant water as it.

- a) Lowers the surface tension of water
- b) Drowns and kills the mosquito larvae
- c) Repels mosquitoes due to its pungent smell

Choose the correct statement/statements from the above.

- 1) Only a
- 2) Only b
- 3) a and b
- 4) a, b and c

27. When a ship enters a sea from a river.

- 1) It rises a little
- 2) It sinks a little
- 3) It remains at the same level
- 4) It rises or sinks depending on the material it is made of

28. A steel ball floats on mercury because.

- 1) Mercury does not allow any metallic ball to sink in it
- 2) Mercury is also a metal in the liquid form
- 3) The density of mercury is higher than that of steel
- 4) A steel ball can be made to float on any liquid by suitable adjustment

29. In a barmeter, mercury is preferred over water because.

- 1) Mercury is a good conductor of heat
- 2) Mercury is shining and therefore its level can be read easily
- 3) Mercury is available in pure form
- 4) Mercury has high density and low vapour pressure.

30. On a clean galss plate a drop of water spreads to form a thin layer whereas a drop of mercury remains almost spherical because.

- 1) Mercury is a metal
- 2) Density of mercury is greater than that of water
- 3) Cohesion of mercury is greater than its adhesion with glass
- 4) Cohesion of water is greater than its adhesion with glass

31. The swing of a spinning cricket ball in the air can be explained on the basis of.

- 1) Sudden change in wind direction
- 2) Buoyancy of air
- 3) Turbulence caused by wind
- 4) Bernoulli's theorem

32. Hydraulic brakes in automobiles work on.

- 1) Bernoulli's principle
- 2) Posieulle's principle
- 3) Pasccal's principle
- 4) Archimede's principle

33. Soap helps in cleaning clothes because of which of the following reasons.

- 1) It absorbs all the dirt present in the clothes
- 2) Chemicals of soap diminish the dirt
- 3) It increases the surface tension of the solution
- 4) It lowers the surface tension of the solution

34. Ball pen functions on whcih of the following principle.

- 1) Viscosity
- 2) Boyle's law
- 3) Gravitational force
- 4) Surface tension

Answers

1) 2	2) 1	3) 4	4) 3	5) 4	6) 3	7) 1	8) 3	9) 1	10) 1
11) 3	12) 3	13) 1	14) 4	15) 3	16) 1	17) 3	18) 1	19) 3	20) 4
21) 2	22) 2	23) 3	24) 2	25) 2	26) 3	27) 1	28) 3	29) 4	30) 3
31) 4	32) 3	33) 4	34) 4						

6. ELECTRICITY

- Electricity is a form of energy.
- It is classified into two types
 - 1) Static Electricity
 - 2) Current Electricity

1) Static Electricity:-

- The electric charge generated by (friction) rubbing is called static electricity because these charges do not transmit.
- Depending on the nature of the objects, one acquires a positive charge and the other an equal negative charge.

Examples:-

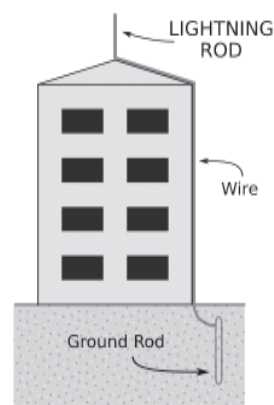
- 1) If a glass rod is rubbed with silk, the glass rod acquires positive charge and the silk an equal negative charge.
 - 2) When an ebonite rod is rubbed with wool (flannel), the rod acquires negative charge and the wool an equal positive charge.
- It is found that the like charges repel and unlike charges attract.

Uses of Static Electricity:-

- 1) It is used in photocopier (Xerox)
- 2) Spray painting car parts
- 3) When a hollow metallic conductor is charged with static electricity it is found experimentally that the charge resides entirely on the outside of the conductor, the inner surface remains uncharged.
- 4) If a car is struck by lightning persons sitting inside are shielded from the electricity and not harmed at all since the charge remains on the outer surface and may arc to the ground through the lowest metallic part of the car.
- 5) If a pear-shaped conductor is charged, it is found that concentration of charge on and near the pointed end is much greater. If the charge on the conductor is increased, the pointed end starts losing charge. It can be shown that a pointed end not only enables a conductor to lose charge, it can also act as a collector of charge. The lightning conductor is based on this principle.

Lightning conductor:-

- A lightning conductor is used to save big buildings from the damage by the lightning flashes.
- Lightning is a gigantic electric discharge occurring between two charged clouds or between a charged cloud and the earth.
- A good lightning conductor is made of “Copper rod”, on the one end of which are provided sharp copper points.
- It is installed on the highest point of the building. Its lower end is connected to a thick copper strip that runs along the



height of the building. The lower end of this copper strip is attached to a flat copper plate and buried deep inside the earth.

- When lightning strikes on the sharp points on the lightning conductor, as copper is a very good conductor of electricity, therefore all the electric discharge from the cloud flows into the earth without damaging the building.
- Lightning is a dazzling bluish white light produced in the clouds. It is followed by a loud noise called thunder.
- **Benjamin Franklin** proved that it is caused by the static electric charges in the clouds.



Current Electricity:-

- The flowing of charge through a conductor is called current electricity.

Electric Current:-

- The rate of flow of charge from a body at a higher potential to a body at a lower potential is called electric current.
- If 'Q' units charge flows through a conductor in time interval 't' such that 'I' is electric current then

$$I = \frac{Q}{t}$$

- S.I. Unit of current is 'ampere'

- 1 ampere =

Smaller units of current:

1) Milliampere (mA)

-

2) Microampere (μA)

- $1\mu\text{A} = 10^{-6} \text{ A}$

Electric Charge:-

- Electric charge is an intrinsic property of protons and electrons and only two types of charge have been discovered they are
 - i) Positive charge
 - ii) Negative charge
- It is a scalar quantity.
- **It's S.I. unit is:** Coulomb
- A body is said to have one coulomb charge, if it has 6.25×10^{18} electrons on it.
- Thus, 6.25×10^{18} electrons have a charge = 1Coulomb
- $\therefore$ 1 electron has a charge =

$$= 1.6 \times 10^{-19} \text{ C}$$

Laws of electrostatic attraction and repulsion:-

- 1) Like charges (similar charges) repel each other.
- 2) Unlike charges (dissimilar charges) attract each other.
- 3) The force of attraction or repulsion operates even when the charges are placed in vacuum.

Note: Repulsion is the sure test for electrification.

Law of conservation of electric charges.

- Electric charge can neither be created nor destroyed, they can only be transferred from one body to another.

Electric Field:-

- Electric field is the region around a charged body within which its influence can be experienced i.e. within which it can attract or repel another charged body.

Electric field intensity:

- It is defined as the force experienced per unit positive charge at a placed in the electric field.

Mathematically

$$E = \frac{F}{q}$$

Where, F - Force

q - charge

S.I. unit is: newton/coulomb.

It is a vector quantity.

$\frac{W}{Q} \text{ Volt} = \frac{1 \text{ Joule}}{1 \text{ Coulomb}}$
--

Electric potential:-

- The electric potential at a point in an electric field is defined as the work done in moving a unit positive charge from infinity to that point.
- Potential is denoted by symbol 'v'.
- Its S.I. unit is Volt.

Potential Difference:

- The potential difference between two points in an electric circuit is defined as the amount of work done in moving a unit charge from one point to other point.
- S.I. unit of potential difference is 'Volt'.

- Potential difference (p.d) =

- Where W = work done

Q = Quantity of charge

Definition of 1 Volt:-

- The potential difference between two points is said to be 1 volt if 1 joule of work is done in moving 1 coulomb of electric charge from one point to the other.

- The potential difference is measured by means of an instrument called “Voltmeter”.
- The voltmeter is always “connected in parallel” across the two points where the potential difference is to be measured.
- A voltmeter has a high resistance so that it takes a negligible current from the circuit.
- Resistance of an ideal voltmeter is infinity.

Electric Circuit:-

- A continuous conducting path consisting of wires and other electric bulb and a switch, between the two terminals of a cell or a battery along which an electric current flow is called a circuit.

Resistance:-

- It is the property of a conductor to resist the flow of charges through it. (or)
- It is the property of a conductor by virtue of which it opposes the flow of current through it.
- It is equal to the ratio of the potential difference applied across its ends and the current flowing through it.
- Resistance = $\frac{\text{Potential difference}}{\text{Current}}$

$$R =$$

S.I. unit of Resistance is: Ohm (Ω)

“Ohm” definition:-

- A conductor has a resistance of one ohm if a current of one ampere flows through it on applying a potential difference of 1 volt across its ends.
- 1 Ohm = (or) $1\Omega =$

Resistor:- A conductor which has some appreciable resistance is called resistor.

Factors on which resistance of a conductor depends:-

- 1) The resistance of a conductor is directly proportional to its length ()
---- (1)
- 2) The resistance of a conductor is inversely proportional to its area (A) of cross - section
---- (2)
- 3) The electrical resistance of a conductor depends on the nature of the material of which it is made.

Ex:- If we take two similar wires, having equal lengths and diameters of copper metal and Nichrome wire, we will find that the resistance of Nichrome wire is about 60 times more than that of the copper wire.

- This shows that the resistance of a conductor depends on the nature of the material of the conductor.

- 4) The resistance of all pure metals increases on raising the temperature and decreasing on lowering the temperature.
- But the resistance of alloys like Manganin, constantan and Nichrome is almost unaffected by temperature.
- From equations (1) and (2)

$$R \propto \frac{l}{A}$$

$$\rho = \frac{RA}{l}$$

- Where (rho) is a constant known as resistivity of the conductor.
- Resistivity is also known as specific resistance.
- Let us take a conductor having a unit area of cross section of 1m^2 and a unit length of 1m .
- So putting $A = 1\text{m}^2$ and $l = 1\text{m}$ in above equation.

we get

$$\rho = R$$

- Thus, the resistivity of a substance is numerically equal to the resistance of a rod of that substance which is 1 metre long and 1m^2 in cross section.

S.I. unit of Resistivity is : ohm-metre

- Resistivity of a substance does not depend on its length or thickness.
 - It depends on the nature of the substance and temperature.
 - The resistivity of a substance is its characteristic property.
 - The resistivities of alloys are much more higher than those of pure metals.
- 1) The resistivity of conductor (like metals) is very low. The resistivity of most of the metal increases with temperature.
 - 2) The resistivity of insulators like ebonite glass and diamond is very high and does not change with temperature.
 - 3) The resistivity of semi-conductors like silicon and germanium is in between those of conductors and insulators and decreases on increasing the temperature.

Conductance:-

- The reciprocal of a resistance of a substance is called conductance.
- Conductance =
- Its S.I. unit is “mho”.

CONDUCTIVITY:-

- The reciprocal of resistivity of a substance is called conductivity.
- Its S.I. unit is: mho/metre

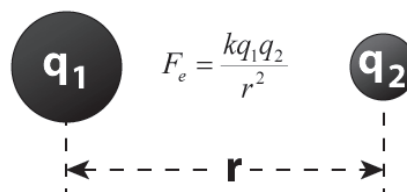
Conductivity =

Coulomb's inverse square law:

- The electrostatic force of attraction or repulsion between two point charges is directly proportional to the product of magnitude of charges and inversely proportional to the square of the distance between them.

$$F = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{q_1 q_2}{r^2}$$

$$F = 9 \times 10^9 \times \frac{q_1 q_2}{r^2}$$



- Where ϵ_0 - Permittivity of free space.
 $q_1 q_2$ - electric charges
 r - distance between two charges.
 F - Force of attraction or repulsion.

Ohms Law:-

- At constant temperature, the electric current (i) in a conductor is directly proportional to the potential difference between its ends.

- Where 'R' is the proportionality constant.
- This 'R' is called electrical resistance.

Ohmic conductors:-

- The conductors which obey ohm's law are called ohmic conductors.
- All metallic conductors are ohmic conductors.

Non-ohmic conductors:-

- The conductors which do not obey ohm's law are called non-ohmic conductors.

Ex:- Semiconductors and electrolytes.

Problems on Ohm's law:-

- 1) The potential difference across a bulb is 240V and a current of 3A flows through it. Find the resistance of the bulb.

Solution: Given $V = 240\text{v}$
 $i = 3\text{ A}$

- According to ohm's law $v = iR$

$$R = \frac{V}{i}$$

$$R = 80\Omega$$

- 2) An immersion heater of resistance 23Ω is connected to a mains of 230V supply. How much current flows through it?

Solution: Given $R = 23\Omega$
 $V = 230\text{V}$

According to ohm's law

$$V = iR$$

$$i = \frac{V}{R}$$

$$i = 10\text{ A}$$

Electromotive force (e.m.f.):-

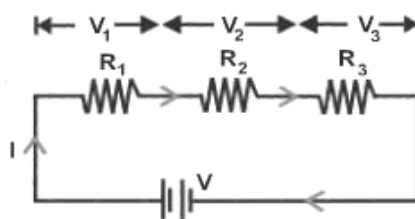
- The e.m.f. is defined as the amount of work done by the cell on charge carriers to force them to go to the point of higher potential.

Combination of Resistances

- i) Resistance in series.
- ii) Resistances in parallel.

Resistances in series:-

- If a number of resistances are joined end to end so that the same current flows through each of them in succession, then the resistances are said to be connected in series (or) when two or more resistances are connected end to end consecutively, they are said to be connected in series.



- Let us consider two resistances R_1 , R_2 connected in series then its resultant resistance is $R = R_1 + R_2$

Laws of resistances in series:-

- i) Current through each resistance is same.
- ii) Total voltage across the combination = sum of voltage drops.
- iii) Voltage drop across any resistor is proportional to its resistance.
- iv) Equivalent (Resultant) resistance = sum of the individual resistances
- v) Resultant resistance is larger than the largest individual resistance.

Note:-

- 1) If $R_1, R_2, R_3, \dots, R_n$ resistors connected in series then its effective resistance is $R_{\text{eff}} = nR$

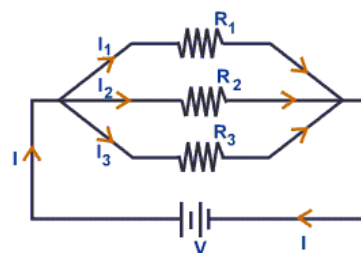
Resistances in parallel:-

- If a number of resistances are connected in between two common points so that each of them provides a separate path for current, then they are said to be connected in parallel. (or)
- The reciprocal of the combined resistance of a number of resistances connected in parallel is equal to the sum of the reciprocals of all the individual resistances.

- Let R_1, R_2 resistances are connected in parallel, then

their resultant resistance is given by $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

(or)
$$R = \frac{R_1 R_2}{R_1 + R_2}$$



Laws of resistances in parallel:-

- Voltage across each resistance is same and is equal to the applied voltage.
- Total current = sum of the currents through the individual resistances.
- Currents through various resistances are inversely proportional to the individual resistances.
- Reciprocal of equivalent resistance = sum of reciprocals of individual resistances.
- Equivalent resistance (Resultant resistance) is less than the smallest individual resistance.

Note:- If $R_1, R_2, R_3, \dots, R_n$ resistors connected in parallel then its effective resistance is

$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

$$R_{eff} = \frac{R}{n}$$

SUPER CONDUCTORS:-

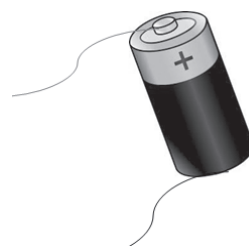
- Prof. H.K. Onnes in 1911 discovered that certain metals and alloys at very low temperature lose their resistance considerably. This phenomenon is known as super-conductivity.
- As the temperature decreases the resistance of the material also decreases, but when the temperature reaches a certain value, its resistance becomes zero. This particular temperature is called critical temperature.
- Then the material behaves as if it is a super-conductor.
- (or)
- The particular temperature at which the resistance of a conductor becomes zero, that is it does not offer any resistance to the passage of electric current is called critical temperature.
- The phenomenon is called super conductivity and the conductor itself is called a superconductor.
- This is invented by prof. H.K. Onnes in 1911.
- 1) The critical temperature of mercury is 4.2K
 - 2) The critical temperature of Lead is 7.25K
 - 3) The critical temperature of Niobium 9.2K

Applications of Superconductors:-

- 1) These are used for making very strong electromagnets
- 2) Superconductivity is used to produce very high speed computers.
- 3) Superconductors are used for the transmission of electric power.

ELECTRIC CELL:-

- The device used for maintaining constant potential difference is called cell.
- The current flows in the cell from negative electrode to positive electrode.
- In a electric cell chemical energy is converted into electric energy.
- Electric cell was first invented by Alessandro volta.
- In voltaic cell positive electrode is copper, negative electrode is zinc and electrolyte is dilute sulphuric acid.

**Types of cells**

- 1) Primary cells
- 2) Secondary cells

1) Primary Cells:-

- Dry cells used in torches, radios are primary cells.
- A dry cell consists of a negative electrode, a positive electrode and an electrolyte.
- The negative electrode is made of zinc.
- The positive electrode is a carbon rod surrounded by a mixture of carbon and Manganese dioxide.
- The electrolyte consists of a mixture of Ammonium Chloride and Zinc Chloride made into a paste.
- It produces about 1.5 volt.

2) Secondary Cells:-

- Lead cells used for ignition and lighting on motor cars are secondary cells.
- Secondary cells are known as storage cells or accumulators. Because of their low internal resistance, secondary cells are capable of giving large currents. Moreover, they can be recharged after they have run down by passing a current through them.

Electromotive Force of Cell (emf):-

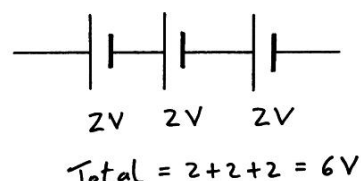
- When no current is drawn from a cell is said to be in an open circuit.
- The potential difference at the terminals of a cell in an open circuit is called emf.
- The emf of cell depends on the following factors.
 - 1) It depends upon material of plates used in the cell.
 - 2) It depends upon nature of the electrolyte used.
 - 3) The emf of a given kind of cell does not change, with the change in shape or size of the plates of cell.

Closed Circuit:-

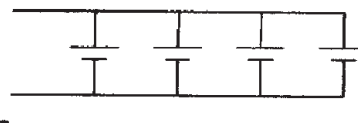
- When the path of electric current which starts from one terminal of the cell, ends at the other terminal of the cell, without any break, then such a circuit is called closed circuit.
- Electric current will flow only if circuit is closed.

Cells are connected in series:-

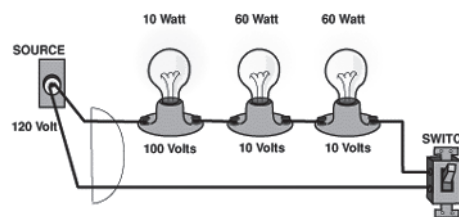
- When negative terminal of a cell is connected to the positive terminal of the next cell, then the cells are said to be in series.
- When cells are connected in series, the total emf of the combination is the sum of the emf of individual cells.
- Let the e.m.f. of three cells connected in series be E_1 , E_2 and E_3 . Then the total emf in the circuit is $E = E_1 + E_2 + E_3$.

**Cells are connected in Parallel:-**

- When all positive terminals of two or more cells are connected to a single point and similarly all the negative terminals are connected to another single point, then the cells are said to be connected in parallel.
- When cells of different emf are connected in parallel the effective emf is equal to the emf of that cell which has the greatest emf.
- Let three cells of emf E_1 , E_2 and E_3 such that $E_1 > E_2 > E_3$ be connected in parallel.
- The effective e.m.f. E is given by $E = E_1$.

**Bulbs in Series:-**

- In an electric circuit, bulbs are said to be connected in series, if the second terminal of one bulb is connected to the first terminal of the next bulb and so on.
- When a number of bulbs are connected in series in a circuit, if one of them is removed or if it fails to work, the rest of the bulbs will not glow. This is because removal of the bulb or its failure causes a break in the circuit.

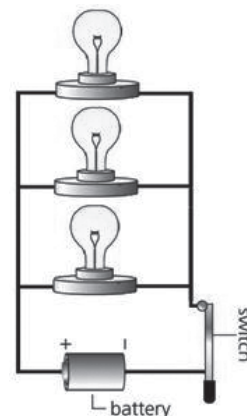


Ex:- The electric bulbs decoration are usually connected in series circuit with the 220volt a.c. power supply.

Bulbs in Parallel:-

- In an electric circuit, bulbs are said to be connected in parallel, if the first terminals of all the bulbs are connected to a common point and similarly all the second terminals to another common point.
- When a number of bulbs are connected in parallel in a circuit, if one of them is removed or if it fails to work, the rest of the bulbs will continue to glow. This is because only this particular part of the circuit breaks and all other parts of the circuit remain unaffected even after removal of the bulb or failure of the bulb.

Ex:- The electric bulbs in a house are connected in parallel circuit with the 220V a.c. power supply.



- All other electric appliances (Fan, T.V) in a house connected in parallel.

Conductors (Electric conductors):-

- Those substances through which electricity can flow are called conductors.
- The electrons of the outer shells of metallic atoms are loosely bound to the nucleus, these electrons are quite free to move. So metals have high electrical conductivity.

Ex:- Silver, copper, Aluminium, Iron, Graphite, Human body, Acids, Alkalies, Water solution of salts.

- Silver is the best conductor of electricity.
- All metals have large no. of free electrons.

Insulators:-

- Those substances through which electricity cannot flow are called insulators.
- The electrons present in insulators are strongly held by the nuclei of their atoms. Since there are “no free electrons” in an insulator.

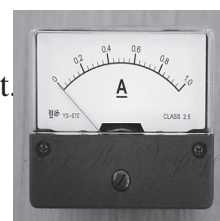
Ex:- Diamond, Rubber, Glass-wood, Mica, Plastic, ebonite, Distilled water.

Uses:-

- Electricians wear rubber handgloves while working with electricity because rubber is an insulator and protects them from electric shocks.

Electrical Measuring Instruments:-**Ammeter:-**

- Current is measured by instrument called ammeter.
- The ammeter is always connected in series with the given circuit.
- Resistance of an ammeter should be low.
- Resistance of an ideal ammeter is zero.

**Galvanometer:-**

- It is used for measuring smaller current in the circuit.
- It is always connected in series with the given circuit.

Voltmeter:-

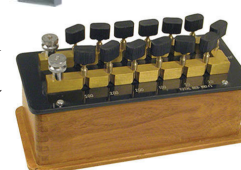
- The potential difference is measured by means of an instrument called voltmeter.
- The voltmeter is always connected in parallel to the component across which the p.d. is to be measured.
- The Resistance of a voltmeter should be very high.
- Resistance of an ideal voltmeter is infinity.

Rheostat:-

- A resistance whose magnitude can be changed, but the exact change in magnitude is not known is called a rheostat.

**Resistance box:-**

- An arrangement of a number of known resistances, which can be added or removed from an electric circuit is called a resistance box.

**Inductor:-**

- It consists of an insulated copper coil of a large no. of turns. It behaves like a magnet on the passage of electric current.

Battery:-

- A combination of two or more cells is called a battery.

Plug key:- A plug key is an electric switch.

- When plug key is open, current does not flow in the electric circuit.
- When plug key is closed, current flows in the electric circuit.

Electroscope:-

- It is used for the presence of electric charge.
- These are two types
 - 1) pith ball electroscope
 - 2) Gold leaf electroscope

Potentiometer:-

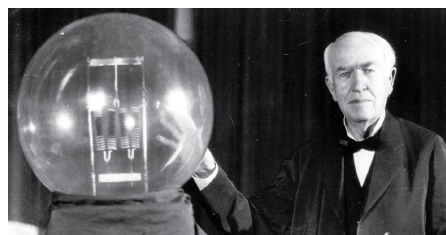
- 1) It measures the e.m.f. of a cell without drawing any current.
- 2) Comparison of e.m.f's of two cells.
- 3) Determining the internal resistance of a cell.

Metre bridge:-

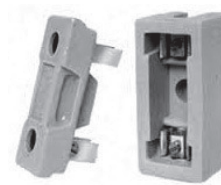
- It is used for determining the resistance of a conductor.

Effects of Electric Current:-**1) Heating effect of Electric current:-****Electric Bulb (Incandescent electric bulb):**

- It is invented by “**Thomas Alva Edison**”.
- It consists of a filament of a fine metallic wire enclosed in a glass bulb filled with chemically inactive gases like “Nitrogen” and “Argon”.
- The filament material should have “high resistivity” and “high melting point”.
- Therefore “tungsten” is used for bulb filament.
- Melting point of tungsten is 3380°C when a current is passed through the filament, it gets heated to a high temperature and emits light.

**Electric Fuse:-**

- It is a safety device used to protect electrical appliances from strong currents.
- A fuse wire must have “high resistivity” and “low melting point”.
- It is usually made from an alloy of “Tin”(Sn) and “Lead” (pb)
- It is put in “series with the live wire” of the circuit.
- When the current exceeds the safety limit, the fuse wire melts and breaks the circuit. The electric installations are thus saved from getting damaged.

**Household Heating appliances:-**

- Most of the electrical appliances used in daily life are based on heating effect of current such as room heater, electric toaster, electric iron, electric oven, electric kettle, geyser.



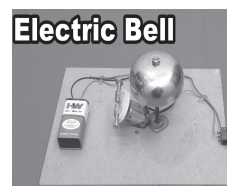
- The heating element used in all those heating appliances contain coils of high resistance wire made of “Nichrome alloy”

Reasons for Nichrome element is used in household heating appliances:

- 1) Its melting point is high
- 2) Its resistivity is high
- 3) It can be easily drawn into thin wires.
- 4) It is not easily oxidised (burn) by the oxygen of the air when heated.

2) Magnetic Effect of Electric Current:-

- The term magnetic effect of electric current means that an electric current flowing in a wire produces a magnetic field around it. (or) Electric current can produce magnetism.
- The magnetic effect of current was discovered by oersted in 1820.



Applications:-

- Electro-magnets, electric motors, telephones, telegraph, electric bell, electric generator and radio all utilize the magnetic effect of current.

3) Chemical Effect of Electric Current:-

- An electric current passed through a solution results in the decomposition of the solution into negative and positive ions.
- Negative ions collect at the positive electrode (Anode) and the positive ions collect at the negative electrode (cathode). This phenomenon is known as electrolysis.
- It is a chemical effect of current.

Applications:-

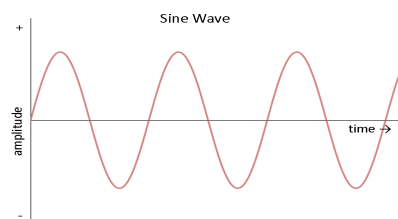
- 1) Working of electric cells is based on this effect.
- 2) Electrolysis is widely used in electroplating i.e. coating of a base metal with a layer of more expensive metal.
- 3) **Electroplating:** To avoid rusting of iron, it is coated with Nickel or Chromium by electrolysis.
- 4) Electronic components used in computers etc are gold plated to avoid atmospheric corrosion.
- 5) This effect is used in extraction of metals from the ores.
- 6) Ornaments made of cheap metal like brass can be coated with gold using electrolysis.
- 7) For the purification of metals
- 8) In electroplating D.C. current is used.
- 9) Electrotyping and electroplating is done by chemical effect of electric current.

Electric current is of two types

- i) Alternating Current (AC)
- ii) Direct Current (DC)

i) Alternating Current:-

- When a current changes its magnitude and direction after fixed interval of time in a particular circuit, it is said to be alternating current or AC.



Ex:- The current produced by emergency generators or in hydroelectric power houses is alternating current.

- It has a frequency of 50Hz. i.e. it changes its direction 50 times in a second in an electric circuit.

Note:- The alternating current (A.C.) produced in India has a frequency of 50 Hz and (voltage) potential difference is 220V.

ii) Direct Current:-

- When a current flows in the circuit in one particular direction and does not change its magnitude and direction for any duration of time, it is said to be direct current (D.C.)



Ex:- The current produced by the cells and storage battery is direct current.

Rectifier:-

- It is an electrical device which converts an alternating current (A.C.) into a direct current (D.C.)

Rectifiers are classified into two types.

i) Half wave rectifier:-

- It has only one diode.
- It converts upto 40.6% A.C. into D.C.

ii) Full wave rectifier:-

- It contains two diodes.
- It converts upto 81.2% A.C into D.C.

Inverter:-

- It is an electrical device which converts Direct current (D.C.) into Alternating current (A.C.)
- The inverters in homes and offices are specially designed to convert DC from a battery to AC and charge the battery.

Electric power:-

- The amount of electric work done in one second is called the electric power.

$$P = \frac{W}{t} = I^2 R = \frac{V^2}{R}$$

(or) $P = IV$

- Where, W - electric work
t - time
p - electric power
I - electric current
v - potential difference
- S.I. unit of power is “watt”.

Definition of one watt:-

- When a current of one ampere flows through a conductor at a potential difference of 1 volt, then the electric power is said to be one watt.

Commercial units of electric energy:- (Cost of electric energy)

- We know that

$$\text{Electric power} = \frac{\text{Electric work (or electric energy)}}{\text{Time}}$$

- Electric Energy = Electric power × time
- The bigger unit of commercial energy at which electric energy is sold to consumers is called kilowatt-Hour (KWh)

$$1 \text{ KWh} = 1000 \text{ watt} \times 1 \text{ hour} \\ = 1000 \text{ watt} \times 60 \times 60 \text{ sec}$$

$$1 \text{ KWh} = 36 \times 10^5 \text{ watt.sec}$$

(or)

Definition of kilowatt-hour:

- It is defined as the electric energy consumed by an appliance of 1000 watt in one hour.

$$\text{No. of units (kwh)} = \frac{[\text{wattage} \times \text{Hours of use per day} \times 30]}{1000}$$

Electromagnetic Induction:-

- Whenever the magnetic lines of force passing through a closed circuit change, a voltage and hence a current is induced in it. This phenomenon is called electromagnetic induction.

It was discovered by **Michael Faraday** in 1831.

“Electric generator” or “Dynamo”:-

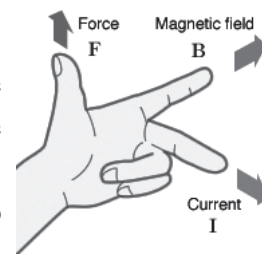
- It is a device to convert mechanical energy into electrical energy.
- It is based on the principle of “Electromagnetic induction”.

Electric Motor:-

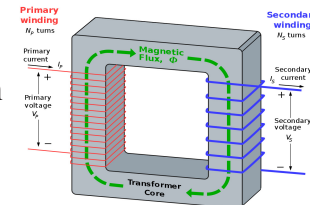
- An electric motor works on the principle that current carrying conductor placed in a magnetic field experiences a force, the direction of force is given by Fleming's left hand rule.

Fleming's left hand rule:-

- Stretch the forefinger, the central finger and the thumb of the left hand mutually perpendicular to each other. If the forefinger points in the direction of the magnetic field (B), the central finger in the direction of current (i), then the thumb points in the direction of force on the conductor (F).

**Transformer:-**

- It is a device which can change a low voltage of high current into a high voltage of low current and vice-versa.
- Transformer works on the principle of “mutual induction”.



There are two types of transformers**i) Step-up Transformer:-**

➤ It converts a low voltage of high current into high voltage of low current.

➤ In this transformer

$$N_s > N_p, \quad E_s > E_p \quad \text{and} \quad I_p < I_s$$

Where N_s - no. of turns in the secondary coil

N_p - no. of turns in the primary coil

E_s - e.m.f. induced in the primary coil

E_p - e.m.f. induced in the secondary coil

I_p - current in the primary coil

I_s - current in the secondary coil

Step-down transformer:-

➤ It converts a High voltage of low current into a low voltage of high current.

➤ In this transformer

$$N_s < N_p, \quad E_s < E_p \quad \text{and} \quad I_p > I_s$$

➤ A transformer uses iron core to minimise power losses.

Uses of Transformers:-

- 1) They are used in electric grids to save electric energy during transmission through over head wires.
- 2) Step up transformers are used in Television, X-ray's, tubes, discharge tubes etc.
- 3) Step down transformer are used in electric bells, Radio-sets, Transistors, battery chargers and in bed lamps.

Self Induction:-

➤ The production of induced e.m.f. in an isolated coil due to a change in the current in the same coil is called self induction.

$$e = -L \frac{di}{dt}$$

➤ Where, e - induced e.m.f.

di - current

dt - time

L - proportionality constant called self inductance.

➤ Units of self inductance is "Henry".

Mutual Induction:-

➤ The production an induced e.m.f. in one coil due to change in current in another close by coil is known as mutual induction.

$$e = -M$$

➤ Where M is proportionality constant is called "Mutual Inductance".

➤ Units of Mutual inductance is "Henry".

Capacitor (or) Condenser:-

- It is a device over which a large amount of charge can be stored without changing its dimensions.

**Capacitance:-**

- The capacitance of conductor is equal to the ratio of the charge (q) given to the conductor to change in its potential (V) is given by

- S.I. unit of capacitance is coulomb/volt (or) farad.
- “Farad” is the large unit of “capacitance”.
- Its practical unit is microfarad (μf) (or) nano farad (nF)
 $1\mu\text{F} = 10^{-6}\text{F}$
 $1\text{nF} = 10^{-9}\text{F}$

Uses of capacitors:-

- 1) Television
- 2) Transistors
- 3) In wireless sets
- 4) In Tubelights

Problem on the Household consumption of Electrical Energy:

- 1) A house is fitted with 10 lamps of each 60 watt. If each lamp burns for 5 hours a day on average, find the cost of consumption in a month of 30 days at 80 paise per unit

Solution:

- No. of units (kwh) =

No. of units =

No. of units (kwh) = 90

cost of 1 unit = 80 paise

cost of 90 units = $90 \times 80 = 7200$ paise

= Rs.72.00

Electrolyte:-

- When an electric current passes through a liquid conductor called ‘electrolyte’.

Electrolysis:-

- The process of decomposition of a chemical compound in a solution when an electric current passes through it is called electrolysis.

Electrolytic Cell (Voltmeter):-

- The vessel which contains an electrolyte and allows electrolysis to take place is called a voltmeter or Electrolytic cell.

Faraday's laws of Electrolysis:-**Faraday's First law of Electrolysis:**

- The Faraday's first law of electrolysis states that the mass(m) of ions liberated from an electrolyte is directly proportional to the strength of the current (i) and the time (t) for which the current passes.

$$m = zit$$
- Where ' z ' is called electro chemical equivalent (e.c.e) of the substance.
- units of ' z ' is: gm/coulomb.

Faraday's second law of Electrolysis:-

- It states that when the same quantity of electricity passes through different electrolytes, the masses of ions liberated at the respective electrodes are proportional to their chemical equivalents.

$$m_1 : m_2 : m_3 = E_1 : E_2 : E_3$$

(or)

$$Z_1 : Z_2 : Z_3 = E_1 : E_2 : E_3$$

Where

E_1, E_2, E_3 are chemical equivalents

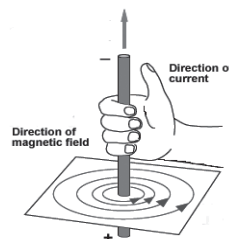
Z_1, Z_2, Z_3 are electrochemical equivalents

Maxwell Cork-Screw Rule:-

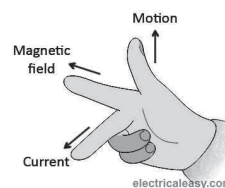
- When the head of a cork-screw is rotated such that the tip of the screw advances in the direction of the current, then the direction of rotation of the head represents the direction of the magnetic field around the conductor.

Ampere's Right Hand Rule:-

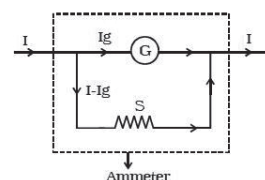
- When a current carrying conductor is held in right and such that the thumb points along the direction of the current, then the remaining fingers indicate the direction of the magnetic field around the conductor.

**Fleming's Right Hand Rule:-**

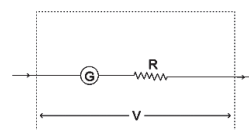
- When the thumb, the fore finger and the central finger of the right hand stretched mutually perpendicular to each other and are held such that the forefinger is along the direction of magnetic field (B) and the thumb is along the direction of motion (M) of the linear current, then the central finger points along the direction of the induced current (C).

**Conversion of Galvanometer to Ammeter:**

- In order to convert Galvanometer into an Ammeter, a very low resistance (shunt) is connected in parallel to Galvanometer.

**Conversion of Galvanometer to Voltmeter:**

- In order to convert a Galvanometer into Voltmeter a very high resistance connected in series with the Galvanometer.



Some Important Conceptual Questions:**1) Why is an ammeter connected in series in a circuit?**

A: An ammeter is low resistance device. When it is connected in series in a circuit, the total resistance of the circuit does not increase appreciably and hence the current in the circuit remains unaffected.

2) Why is a voltmeter connected in parallel across a circuit element?

A: A voltmeter is a high resistance device when it is connected in parallel across any component, it draws very small current from the main circuit and most of the current passes through that component. Hence potential difference ($V=iR$) across that element is not affected materially.

3) What happens when ammeter is placed in parallel with a circuit element?

A: An ammeter is a low resistance device. When it is connected in parallel, the resistance of the circuit reduces considerably. A large current flows through the circuit which may damage the ammeter.

4) What happens when a voltmeter is connected in series in a circuit?

A: A voltmeter is a high resistance device. When it is connected in series, the resistance of the circuit becomes high. The current decreases considerably. Voltmeter will not read the actual potential difference.

5) Why should a voltmeter have a high resistance?

A: The resistance of a voltmeter is kept very high so that it takes a negligibly small current from the main circuit.

6) What is the resistance of air gap?

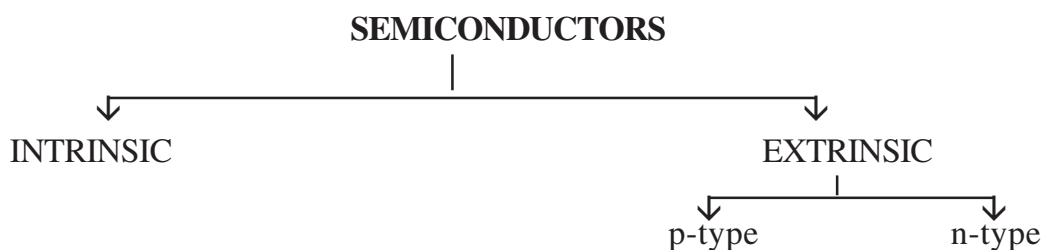
A: Infinity

7) What is the resistance of a closed plug-key?

A: Zero

Semiconductors

- It allows the current partially at room temperature.
(or)
- These are the substances whose resistivities lie in between those of conductors and insulators i.e. between 10^{-6} to $10^4 \Omega \text{ m}$.
- Germanium (Ge) and Silicon (Si) are typical semiconductors which are of great importance in chips fabrication.
- At 0 K (Zero Kelvin) semiconductors behave like insulators of electricity.
- The conductivity of a semiconductor depends on its temperature.
- If temperature of semiconductors increases then its resistance decreases.
- Silicon (Si) is used in computer IC chips.
- Silicon (Si) is used in photo voltaic cells.



Intrinsic Semiconductors:

- Pure semiconductors, i.e. without any impurities in them are called intrinsic semiconductors.

Ex: Pure Germanium and pure Silicon.

Extrinsic Semiconductor:

- When select impurities in small quantities are added to an intrinsic semiconductor an extrinsic semiconductor is formed. Then the semiconductor is said to be doped.

Doping:

- The process of introducing impurities in small quantities into a material is called doping.
- The extrinsic semiconductors can be of two types depending on the type of impurity. They are
 - i) p-type semiconductors
 - ii) n-type semiconductors

i) p-type semiconductor:

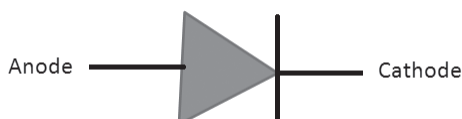
- A p-type semiconductor is formed when a small quantity of acceptor impurity is added to the pure semiconductor.
- Examples of acceptor impurities or trivalent impurities are Gallium (Ga), Indium (In), Aluminium (Al), Boron (B).
- In p-type semiconductor 'holes' are majority charge carriers.
- Whereas 'electrons' are minority charge carriers in a p-type semiconductor.

ii) n-type semiconductor:

- A n-type semiconductor is formed when a small quantity of donor impurities or pentavalent impurities is added to a pure semiconductor.
- Examples of pentavalent impurities are Arsenic (As), Antimony (b), Phosphorus (P).
- In n-type semiconductor electrons are majority charge carriers whereas Holes are minority charge carriers in n-type semiconductor.

p-n Junction Diode:

- The p-n junction consisting p-type and n-type semiconductor provided with two terminals is called a p-n junction diode.

Symbol of p-n junction diode:**Uses of p-n junction Diode:**

- 1) A p-n junction diode can be used as electronic switch.
- 2) In rectifier circuit, a p-n junction diode is used to convert alternating current (AC) into direct current (DC). This conversion is utilized in power supply devices found in electronic appliances such as TV, Radio, Tape-recorders, computers etc.
- 3) Certain p-n diodes glow on supply of electricity. Such diodes are called Light Emitting Diodes (LED). These are used in digital clocks and digital (electronic) calculators.
- 4) Zener diode is used as voltage regulator.

Transistor:-

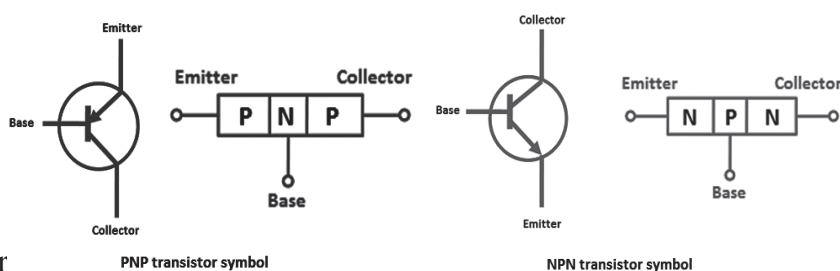
- The transistor was invented in 1948 by J.Bardeen, W.H.Brattain and William Shockley who shared the 1956 Nobel Prize in Physics.
- The transistor is a small semiconductor device which is at the heart of modern-day electronic appliances.
- A transistor consists of two p-n junctions formed by sandwiching either a p-type semiconductor or a n-type semiconductor between a pair of opposite type semiconductors.

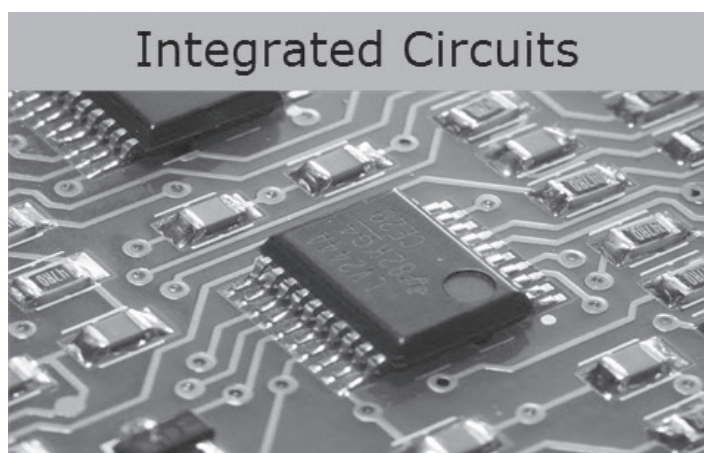
Two types of Junction Transistors:

- 1) p-n-p Transistor
- 2) n-p-n Transistor

p-n-p Trnasistor:**n-p-n Transistor:**

Where, E - emitter
B - Base
C - Colleter



**Uses of Transistors:-**

- 1) Transistor acts as an amplifier. An electronic device which can raise the magnitude of current or voltage input signals is called an amplifier. The junction transistor is the main component in an amplifier circuit.
- 2) Transistors are used in electronic circuits called oscillators to produce signals of different frequencies.
- 3) Transistors are also in stabilised power supplies.
- 4) Transistors form important components of micro-electronic systems called IC's (Integrated Circuits) or Chips.
 - These chips are used in all electronic devices including computers.

ELECTRICITY - Practice Bits

- | | |
|---|--|
| <p>1. A device in an electric circuit which can turn on/off the electric circuit is called.
 1) an electric regulator
 2) an electric meter
 3) an electric switch
 4) all the above</p> <p>2. The overhead electric cables used for the transmission of electric current are made of.
 1) aluminium 2) zinc
 3) steel 4) lead</p> <p>3. In a simple voltaic cell, the negative plate consists of.
 1) copper 2) zinc
 3) lead 4) tin</p> <p>4. In a simple voltaic cell, the positive plate consists of.
 1) zinc 2) lead
 3) tin 4) copper</p> <p>5. The liquid used in the voltaic cell is.
 1) dilute sulphuric acid
 2) dilute nitric acid
 3) dilute sodium hydroxide
 4) none of the above</p> <p>6. The emf of a simple voltaic cell about.
 1) 0.5 V 2) 0.75 V
 3) 1.08 V 4) 1.25 V</p> <p>7. The positive terminal of a simple le-clanche cell consists of.
 1) zinc 2) carbon
 3) copper 4) none of the above</p> <p>8. The negative terminal of a simple leclanche cell consists of.
 1) zinc 2) aluminium
 3) iron 4) copper</p> | <p>9. The liquid used in le-clanche cell is.
 1) dilute HCl 2) dilute NaOH
 3) saturated ammonium chloride solution
 4) saturated copper sulphate solution</p> <p>10. The emf of le-clanche cell is about.
 1) 1.5 V 2) 1.1 V
 3) 1.75 V 4) 2.0 V</p> <p>11. The liquid used in bichromate cell is.
 1) potassium bichromate solution acidified with sulphuric acid
 2) potassium chromate solution acidified with sulphuric acid
 3) potassium sulphate solution acidified with sulphuric acid
 4) none of the above</p> <p>12. The positive terminal of a bichromate cell is.
 1) carbon rod 2) zinc rod
 3) copper rod 4) none of the above</p> <p>13. The emf of a bichromate cell is about
 1) 1.75 V 2) 1.50 V
 3) 1.25 V 4) 2.00 V</p> <p>14. Bichromate cell is not commonly used because it contains.
 1) a liquid which is corrosive in nature
 2) a liquid which is poisonous in nature
 3) a liquid which is explosive in nature
 4) both (1) and (2)</p> <p>15. When a number of cells are connected end to end such that the positive terminal of one cell is connected to the negative terminal of the next cell, then the combination of cells are in.
 1) series 2) parallel
 3) mixed circuit 4) none of the above</p> |
|---|--|

- 16. When the cells are connected in series, the electromotive force of the cells is.**
1) more than one cell
2) less than one cell
3) same as one cell
4) none of the above
- 17. When the cells are connected in series, the current output of the combination.**
1) increases 2) decreases
3) remain same 4) none of the above
- 18. When the number of cells are connective is such a way that they have common positive terminal and common negative terminal, then the cells are connected in.**
1) series 2) parallel
3) mixed circuit 4) none of the above
- 19. When the cells are connected in parallel, the electromotive force of cells is.**
1) more than one cell
2) less than one cell
3) same as one cell
4) none of the above
- 20. When the cells are connected in parallel, the current output of combination.**
1) increases 2) decreases
3) remains same 4) none of the above
- 21. Copper or aluminium wires do not get hot when electric current flows through them because.**
1) they are cool wires
2) they offer no resistance to the electric current
3) they offer resistance to the electric current
4) none of the above
- 22. Which device is not based on heating effects of electric current?**
1) Hot plate 2) Toaster
3) Transformer 4) Geyser
- 23. The material used to make an element of electric heater is.**
1) brass 2) nichrome
3) steel 4) magnesium
- 24. The magnetic effect of an electromagnet can be increased.**
1) by increasing the number of turns in its coil
2) by inserting soft iron core in the coil
3) by increasing the magnitude of current in the coil
4) all the above
- 25. Electromagnets are used in.**
1) telephones 2) radios
3) both (1) & (2) 4) none of the above
- 26. When electric current is passed through acidulated water, it splits into hydrogen and oxygen. This effect of electric current is called.**
1) chemical effect
2) magnetic effect
3) electromagnetic effect
4) all the above
- 27. The process of bringing about chemical reaction by passing an electric current through some conducting solution is called.**
1) chemical decomposition
2) electric decomposition
3) electrolysis
4) none of the above
- 28. The cations are.**
1) positively charged
2) negatively charged
3) neutral
4) none of the above

- | | |
|---|---|
| <p>29. During electroplating the article to be electroplated is connected to the.
 1) cathode 2) anode
 3) eiether 1 or 2 4) none of above</p> <p>30. Which is not an application of electrolysis.
 1) Electroplating
 2) Electrotyping
 3) Purification of metals
 4) Radio-telegraphy</p> <p>31. During purification of a metal, the impure metal is connected to.
 1) cathode 2) anode
 3) either 1 or 2 4) none of above</p> <p>32. The surest test for the electrification of a body is.
 1) attraction 2) repulsion
 3) both 1 and 2 4) none of above</p> <p>33. The protons and neutrons are held firmly in the nucleus by strong.
 1) electrostatic forces
 2) gravitational forces
 3) magnetic forces
 4) nuclear forces</p> <p>34. When a body has excess of electrons as compard to the protons, the body is.
 1) negatively charged
 2) positively charged
 3) neutral
 4) none of the above</p> <p>35. Electrification by friction is due to the migration of.
 1) protons 2) free electrons
 3) bound electrons 4) both 2 and 3</p> <p>36. Which is not a good conductor of electricity?
 1) Copper 2) Lemon juice
 3) Wool 4) Dilute sulphuric acid</p> | <p>37. The device used to detect static electric charges on a body is called.
 1) gold leaf inductor
 2) gold leaf conductor
 3) gold leaf electroscope
 4) none of the above</p> <p>38. A gold leaf can.
 1) detect static electric charge
 2) find the nature of static electric charges
 3) both (1) and (2)
 4) none of the above</p> <p>39. The phenomenon due to which an uncharged conductor gets electrically charged, when held near a charged conductor is called.
 1) electric conductor
 2) electric induction
 3) electric transmission
 4) none of the above</p> <p>40. The brilliant spark produced in the clouds is commonly called.
 1) electric discharge
 2) earthing
 3) lightning
 4) none of the above</p> <p>41. The tall buildings can be protected from lightning.
 1) by fixing reflectors on the roof top
 2) painting the roof black
 3) fixing a lightning conductor on the top of the building
 4) praying to God</p> <p>42. According to Coulomb's law, force of attraction or repulsion between the two point charges is proportional to.
 1) sum of the charges
 2) difference of the charges
 3) product of the charges
 4) none of the above</p> |
|---|---|

- 43. The space around a charged body or a system of charged bodies, where its influence can be felt is called.**
 1) electric strength
 2) electric area
 3) electric field
 4) none of the above
- 44. The standard unit of electric charge is.**
 1) ampere 2) volt
 3) ohm 4) coulomb
- 45. A body is said to have 1 coulomb charge, if it has**
 1) 6.25×10^6 electrons
 2) 6.25×10^{16} electrons
 3) 6.25×10^{18} electrons
 4) 6.25×10^9 electrons
- 46. 500 microampere current is equal to.**
 1) 5×10^{-4} A 2) 5×10^{-3} A
 3) 5×10^{-6} A 4) 5×10^{-5} A
- 47. The SI unit of current is.**
 1) ohm 2) volt
 3) coulomb 4) ampere
- 48. When one coulomb charge flows through a conductor in one second, the current flowing through the conductor is.**
 1) 1 A 2) 10 A
 3) 2 A 4) 4 A
- 49. It 6.25×10^{18} electrons flow through a conductor in 0.25s, the magnitude of the current in the electric circuit is.**
 1) 2 A 2) 1 A
 3) 0.4 A 4) 4 A
- Hint:**
 $i = \frac{q}{t} = \frac{1}{0.25} = 4\text{A}$
- 50. The condition that determines the flow of electric charge is called.**
 1) electric current
 2) electric conductance
 3) electric potential
 4) none of the above
- 51. If “V” is the potential difference between the two points. When W is the work done to move a charge Q, then.**
 1) $V = \frac{W}{Q}$ 2)
 3) 4) both (1) and (2)
- 52. One coulomb charge is moved between two points by doing a work of 1 Joule, the potential difference between the two points is.**
 1) 1 V 2) 10 V
 3) 4 V 4) 5 V
- 53. A device in which potential difference is maintained between the two terminals by converting chemical energy into electric energy is called.**
 1) electric generator 2) dynamo
 3) electric cell 4) none of the above
- 54. Dry cell was invented by.**
 1) Leclanche George 2) Alessandro Volta
 3) R. Daniell 4) Edison
- 55. The negative terminal and the electrolyte in case of dry cell are.**
 1) carbon rod and manganese dioxide
 2) zinc cylinder and manganese dioxide
 3) zinc cylinder and ammonium chloride
 4) carbon rod and ammonium chloride

- 56. The obstruction or friction offered by a conductor to the passage of electricity is called.**
 1) resistance 2) potential difference
 3) inductance 4) conductance
- 57. The SI unit of resistance is.**
 1) mho 2) ohm
 3) ampere 4) volt
- 58. The electric resistance of a conductor is.**
 1) directly proportional to its length
 2) directly proportional to its area of crosssection
 3) inversely proportional to its area of crosssection
 4) both (1) and (3)
- 59. The instrument used for measuring current is.**
 1) ammeter 2) voltmeter
 3) galvanometer 4) voltameter
- 60. The instrument used for measuring potential difference is.**
 1) ammeter 2) voltmeter
 3) voltameter 4) none of the above
- 61. In a series circuit.**
 1) current in all resistors is same, but not the potential difference
 2) current in all resistors is different, but potential difference is same
 3) both current and potential difference is the same
 4) both current and potential difference is different
- 62. In a parallel circuit all resistors have.**
 1) The same potential difference but different magnitude of current
 2) The same potential difference and same magnitude of current
 3) different potential difference and different magnitude of current
 4) different potential difference and same magnitude of current
- 63. According to Ohm's law.**
 1) 2)
 3) 4)
- 64. Three resistors of 2Ω , 6Ω , and 12Ω are connected in series. The combined resistance will.**
 1) less than 18Ω 2) less than 14Ω
 3) less than 20Ω 4) 20Ω
- 65. Three resistors P, Q and R are connected in such a way that the combined resistance is less than any of the individual resistors. The resistors are connected in.**
 1) series circuit 2) parallel circuit
 3) mixed circuit 4) none of the above
- 66. If W is the electric work, Q the charge and V the potential difference, then.**
 1) 2)
 3) $W = Q \times V$ 4) $W = V + Q$
- 67. 5 kWh is equal.**
 1) $1.8 \times 10^7 \text{ J}$ 2) $1.8 \times 10^6 \text{ J}$
 3) $1.8 \times 10^9 \text{ J}$ 4) $1.8 \times 10^4 \text{ J}$
- 68. Commercial unit of electric energy is.**
 1) Wh 2) kWh
 3) W 4) kW
- 69. In an electric circuit, the fuse wire should be.**
 a) Always connected in the live (phase) wire
 b) Made of a material with low melting point
 c) Made of tungsten
Which of the above statements is/are correct
 1) a 2) a and b
 3) b and c 4) a, b and c

70. Henry is the unit of which one of the following.

- 1) Inductance
- 2) Intensity of sound
- 3) Quantity of electricity
- 4) Thermal capacity

71. Which among the following is the best conductor of electricity.

- 1) Zinc
- 2) Copper
- 3) Gold
- 4) Silver

72. A capacitor (condenser) is used in an electrical circuit to.

- 1) Step down voltage
- 2) Step up voltage
- 3) Store electric charge
- 4) Produce electric charge

73. Which of the following statements is/are correct.

- a) Silicon can be used as a semiconductor material.
 - b) IC chips for computers are made of silicon
- 1) a only
 - 2) b only
 - 3) Both a and b
 - 4) Neither a nor b

74. Choose the correct statements from the codes given below.

- a) An electric motor converts electric energy into mechanical energy
- b) Dynamo converts mechanical energy into electric energy

- 1) a only
- 2) b only
- 3) Both a and b
- 4) Neither a nor b

75. Which of the following statements is/are correct.

- a) The function of the fuse in an electric circuit is to reduce the flow of current
 - b) The fuse is connected to the live wire
 - c) The material used in the fuse wire should have a high melting point.
- 1) b only
 - 2) a and b only
 - 3) c only
 - 4) None of these

76. An electrical appliance is earthed to

- a) Protect the appliance against any damage
- b) Prevent electric shocks
- c) Avoid short-circuiting

Which of the above is/are correct?

- 1) a only
- 2) b only
- 3) Both a and b
- 4) Neither b nor c

77. A common fluorescent tube contains.

- 1) Sodium vapour
- 2) Argon at low pressure
- 3) Mercury vapour at low pressure
- 4) Mercuric oxide and neon

78. What is the source of electric energy in an artificial satellite?

- 1) A mini nuclear reactor
- 2) A dynamo
- 3) A thermopile
- 4) Solar cells

79. Which of the following are semiconductors.

- a) Silicon b) Ceramic
- c) Germanium d) Quartz

Which of the above statement is/are correct.

- 1) b and d 2) a and c
- 3) a, c and d 4) All the above

Answers

1) 3	2) 1	3) 2	4) 4	5) 1	6) 3	7) 2	8) 1	9) 3	10) 1
11) 1	12) 1	13) 4	14) 4	15) 1	16) 1	17) 3	18) 2	19) 3	20) 1
21) 2	22) 3	23) 2	24) 4	25) 3	26) 1	27) 3	28) 1	29) 1	30) 4
31) 2	32) 2	33) 4	34) 1	35) 2	36) 3	37) 3	38) 3	39) 2	40) 3
41) 3	42) 3	43) 3	44) 4	45) 3	46) 1	47) 4	48) 1	49) 4	50) 3
51) 4	52) 1	53) 3	54) 1	55) 3	56) 1	57) 2	58) 4	59) 1	60) 2
61) 1	62) 1	63) 1	64) 4	65) 2	66) 3	67) 1	68) 2	69) 2	70) 1
71) 4	72) 3	73) 3	74) 3	75) 1	76) 2	77) 3	78) 4	79) 2	

7. MAGNETISM

- It is very much interesting to note that magnetism was first discovered by a shepherd when his shoes containing iron nails, got stuck on a rock due to attraction.
- In early days small ships and boats in seas were being attracted by hills and used to be destroyed due to collision.
- It was found that some stones in Magnesia in Asia minor, possess this magic property of attraction. These stones are nothing but ores of iron.
- This ore of iron is called “**Magnetite**” and the property is named as magnetism.
- The materials containing this property of attraction are called lodestones which are natural magnets.

Natural Magnets:

- The magnets found in nature like magnetite ore called natural magnets.
- Earth is a huge natural magnet.

Artificial Magnets:

- If Iron and Steel bars are suitably rubbed with natural magnets, they acquire magnetic property and their magnetic strength is greater than that of natural magnets. These magnets are called Artificial magnets.
- Artificial magnets can be prepared in suitable shapes.

Ex:- Bar magnet, Cylindrical magnet, Horse-shoe magnet etc.



Properties of Magnets

1) Attractive Property:-

- Magnets possess the property of attracting small iron pieces.
- The points inside the magnet where this attraction property is maximum are called poles.

Magnetic substances:-

- The substances which are strongly attracted by a magnet are called magnetic substances.

Ex:- Iron, Steel, Nickel, Cobalt.

Non-magnetic substances:-

- The substances which are not attracted by a magnet are called non-magnetic substances.

Ex:- Paper, Wood, Brass, Aluminium.

Directive Property:-

- If a magnet is suspended freely and horizontally with silk thread, it oscillates to and fro for a short time and then comes to rest in an approximate Earth's North (N) and South (S) direction.
- The end which points towards North is called North pole.
- The end which points towards south is called South pole.

Law of Magnetic poles:-

- Like poles repel each other and unlike poles attract each other. This is called law of magnetic poles.

Note: Repulsion sure test for magnetism.

Pair Property:-

- Magnetic poles in pairs and they can not be separated.
- Mono poles or single poles cannot exist.

Magnetic Induction (or) Magnetic Flux Density:-

- A magnet induces magnetic properties in some substances like Iron, Cobalt, Nickel, Steel. This property is called magnetic induction.

Definition of magnetic fluxdensity (B):-

- It is defined as the magnetic flux passing through a unit normal area.

where Φ - magnetic flux

A - area

(or)

- The magnetic field induction (B) due to a magnet at a point in a medium is defined as the force acting on a unit north pole placed at that point.

$$1 \text{ Tesla} = \frac{1 \text{ Weber}}{1 \text{ m}^2}$$

Where 'F' is force and 'm' is pole strength.

S.I. unit magnetic Flux Density:-

- Tesla
-
- C.G.S. unit:- "Gauss"
-

Pole Strength:-

- The ability of a pole of a magnet to attract or repel another magnetic pole is called its pole strength.
- S.I. units of pole strength is "ampere-metre".

Magnetic Moment (M):-

- Magnetic moment (M) of a bar-magnet is measured by the product of its pole strength (m) and its magnet length (2l).
- Magnetic moment (M) = 2ml
- S.I. unit of M is ampere - metre² (A-m²)

Axial Line:-

- The imaginary line joining the north and south poles and passing through the centre of mass of a magnet is called Axial line.

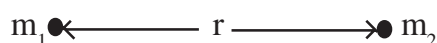
- Magnetic Field Induction on the axial line is given by

Equatorial line:-

- The imaginary line passing through the centre of mass and perpendicular to the line joining the poles of a bar magnet is called equatorial line.
- Magnetic field Induction on the equatorial line is given by $B = \frac{\mu_0 M}{4\pi d^3}$

Inverse square law of Magnetism:-

- It states that the force of attraction or repulsion between two magnetic poles is directly proportional to the product of their pole strengths and inversely proportional to the square of the distance between them.



- According to this law

$$F \propto m_1 m_2 \quad - (1)$$

$$F \propto \frac{1}{r^2} \quad - (2)$$

- from equations (1) and (2)

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = \frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^2}$$

- Where μ_0 is called magnetic permeability of free space.
- The value of μ_0 is $4\pi \times 10^{-7}$ Henry/metre.

Making of Artificial Magnets:

- 1) Single Touch Method
- 2) Double Touch Method
- 3) Electrical Method
- 4) Induction Method

Electrical Method:

- An Electro-magnet is one which gets magnetised when current is passed through the surrounding coil and loses magnetism when current is switched off.
- Soft iron is used to make electro-magnet because its susceptibility is more.
- Steel is used to make permanent magnet due to its retentivity is high.

Induction Method:

- When an iron body is placed in the magnetic field of a magnet, induction takes place and so gets magnetised. Due to induction, a magnet attracts an iron body.

Magnetic Field:

- The space around a magnet in which a force is experienced by a magnetic pole is called magnetic field.

Magnetic line of force:

- It is a line joining any two points in the direction of magnetic field.
- The number of magnetic lines of force present per unit area is proportional to the intensity of magnetic field in the area.

Magnetic Flux:-

- The no. of magnetic lines of force present per unit volume and perpendicular to its cross-section is called as magnetic flux.
- **Its S.I. unit:** weber (wb)
- **C.G.S. unit:** maxwell.

Magnetic properties of Materials:-**Intensity of Magnetisation (E):**

- The magnetic moment acquired by substance per unit volume is defined as intensity of magnetisation.
- Its S.I. unit: ampere/metre

Intensity of Magnetic field (H):-

- The strength of a magnetic field at a point in C.G.S units is defined by the term intensity of magnetic field or magnetic field strength.
- Its unit is A/m.

Uses of Magnets:

- 1) A magnetic needle when freely suspended always shows North and South direction. This directive property is used in constructing “**marine compass**”.
- It is used to know the position of ship in mid sea and to drive it in required direction for reaching their destinations.
- 2) Magnets are used to separate iron ores from ordinary rocks and the process is called Magnetic separation of ores.
- 3) Powerful electromagnets are used in electric cranes which can lift heavy girders and weights.
- 4) Electro magnets are extensively used in tape recorders, speakers, Transformers, Dynamos, Telephones and motors.

**Magnetic Susceptibility:-**

- It is defined as the ratio of the intensity of magnetisation (I) and the intensity of applied magnetic field (H).

$$\chi = \frac{I}{H}$$

- It has no units.

Retentivity:-

- Retentivity of a magnetic material is a property by virtue of which it retains certain amount of intensity of magnetisation even after the removal of applied magnetic field.

Magnetic Permeability:-

- It is defined as its ability to allow the magnetic lines of force to pass through it or to allow itself to be influenced by magnetic field.

Magnetic classification of materials:-

- These are classified into three types.
 - 1) Dia
 - 2) Para
 - 3) Ferro-magnetic material.

Diamagnetic Materials:-

- Diamagnetic materials are repelled by a magnetic field.

Ex:- Air, water, bismuth (Bi), glass, zinc (zn), gold, alcohol, mercury (Hg), Hydrogen, Copper (cu), Antimony (sb)

- Its relative permeability is less than one ($\mu_r < 1$)
- The magnetic susceptibility of this material is small and negative.

Paramagnetic Materials:-

- These materials are weakly attracted by a magnetic field.

Ex:- Aluminium (Al), platinum (pt), chromium (cr), oxygen (O_2), calcium (ca), manganese (Mn).

- The relative permeability of a paramagnetic substance is slightly greater than one.
- The magnetic susceptibility of this substance is small and positive.

Ferromagnetic Materials:-

- The ferromagnetic substances are strongly attracted by a magnet.
- They align themselves easily in the direction of the applied magnetic field.
- The relative permeability of this substance is very high
- The magnetic susceptibility of this substance is very large and positive.
- Therefore, they can be easily magnetised and made into magnets.

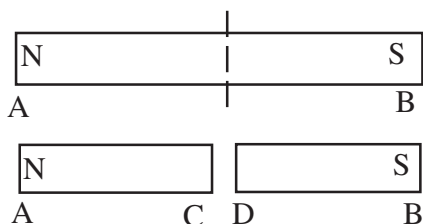
Curie temperature:

- The temperature at which ferromagnetic material is converted to paramagnetic material is called Curie temperature.

MAGNETISM - Practice Bits

- | | |
|---|---|
| <p>1. Magnetic force is.
 1) Only attractive in nature
 2) Repulsive in nature
 3) both attractive and repulsive in nature
 4) None of the above</p> <p>2. Which one of the following is not attracted by magnetic force.
 1) Iron 2) Nickel
 3) Cobalt 4) Copper</p> <p>3. An ore of iron that acts as a natural magnet is.
 1) haematite 2) magnetite
 3) both 1 and 2 4) none of above</p> <p>4. The directive property of magnetic was first used by.
 1) soldiers 2) carpenters
 3) masons 4) sailors</p> <p>5. The ends of a bar magnet are called.
 1) tips of magnet
 2) vertices of magnet
 3) poles of magnet
 4) none of the above</p> <p>6. Most of the permanent magnets are made from.
 1) soft iron 2) chromium
 3) cobalt 4) steel</p> <p>7. The substances that cannot be magnetised/attracted by a powerful magnet are called.
 1) unmagnetic substances
 2) paramagnetic substances
 3) non-magnetic substances
 4) none of the above</p> | <p>8. Which is a non-magnetic substance.
 1) iron 2) nickel
 3) chromium 4) steel</p> <p>9. Which is a magnetic substance.
 1) cobalt 2) glass
 3) copper 4) carbon</p> <p>10. The maximum magnetic strength of a bar magnet is at its.
 1) north pole 2) south pole
 3) in the middle 4) both 1 and 2</p> <p>11. A freely suspended or pivoted magnet will point in.
 1) any geographic direction
 2) east-west direction
 3) north-south direction
 4) none of the above</p> <p>12. Similar poles of the bar magnets.
 1) attract each other
 2) repel each other
 3) neither attract nor repel each other
 4) none of the above</p> <p>13. One end of a bar magnet is brought near the south pole of a freely suspended magnetic needle. It is seen that south pole of the magnetic needle gets repelled. The polarity of the end of the bar magnet is.
 1) north 2) south
 3) either of 1 or 2 4) none of the above</p> <p>14. Which one is used for making temporary magnet.
 1) copper 2) steel
 3) ferrite 4) soft iron</p> |
|---|---|

15. A bar magnet AB is cut into two pieces AC and DB. The ends C and D will have



- 1) north polarity
 - 2) south polarity
 - 3) C as south polarity and D as north polarity
 - 4) C as north polarity and D as south polarity
16. A bar magnet is hammered for some time. Its magnetic strength.
- 1) is likely to increase
 - 2) is likely to decrease
 - 3) will not change
 - 4) none of the above
17. A bar magnet is heated strongly. Its magnetic strength.
- 1) is likely to remain same
 - 2) is likely to increase
 - 3) is likely to decrease
 - 4) none of the above

18. The magnet used in a magnetic compass is.

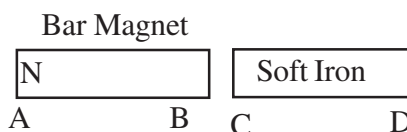
- 1) electromagnet
- 2) permanent magnet
- 3) temporary magnet
- 4) none of the above

19. When a powerful magnet is held near a piece of soft iron, the latter changes to magnet on account of the phenomenon of.

- 1) magnetic conduction
- 2) magnetic induction
- 3) magnetic attraction
- 4) none of the above

20. The polarity induced at the end of D of soft iron is.

- 1) north
- 2) south
- 3) none
- 4) none of the above



21. Which is the sure test for the magnetism.

- 1) attraction
- 2) repulsion
- 3) Both 1 and 2
- 4) None of the above

Answers

1) 3	2) 4	3) 2	4) 4	5) 3	6) 4	7) 3	8) 3	9) 1	10) 4
11) 3	12) 2	13) 2	14) 4	15) 3	16) 2	17) 3	18) 2	19) 2	20) 2
21) 2									

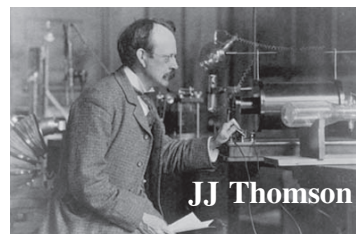
8. MODERN PHYSICS

Atom:

- The smallest unit of any element, which carries all the properties of the element is called an atom.
- Atom mainly consists of three particles electron, proton and the neutron.
- Rutherford's atomic model consists of positively charged massive nucleus at the centre with electrons revolving around it.
- Bohr's atomic model assumes stationary circular orbits of definite energy for electrons with nucleus at the centre.
- Nucleus consists of protons and neutrons
- Collectively these are called nucleons.
- According to nuclear atomic model, protons and neutrons exist inside the nucleus and electrons exists outside the nucleus.

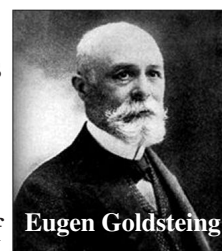
Electron:

- 1) Electron was invented by J.J. Thomson
- 2) The name electron was given by Johnstone stoney.
- 3) It is negatively charged particle.
- 4) Its mass is 9.1×10^{-31} kg.
- 5) Its charge is $- 1.6 \times 10^{-19}$ coulomb.
- 6) The charge-to-mass ratio (e/m) of electron is 1.758×10^{11} coulomb/kg



Proton:

- 1) Proton was discovered by Eugen Gold stein. and this name is given by Rutherford.
- 2) It is positively charged particle.
- 3) Its charge is $+ 1.6 \times 10^{-19}$ coulomb.
- 4) Its mass is 1.672×10^{-27} kg. which is about 1837 times the mass of the electron.



Neutron:

- 1) Neutron was discovered by "James chadwick"
- 2) Its charge is neutral
- 3) Its mass is 1.6750×10^{-27} kg.
- 4) The mass of the neutron is slightly more than that of the proton.



James Chadwick

Atomic number (z) :

- The number of protons in the nucleus is called atomic number.

Ex: Atomic number of Hydrogen is 1.

Atomic mass number (A) :

- The total number of protons and neutrons is called mass number $A = Z + N$.
 where Z is atomic number
 N is number of neutrons.

Atomic mass unit :

- The mass of an atom is measured in units called “atomic mass unit”. This is denoted by “amu”
- The amu is defined as of the mass of neutral carbon atom.
- $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
(or)
- $1 \text{ amu} = 931.5 \text{ Mev}$

Isotopes :

- The nuclei having the same atomic number (z) but different mass numbers (A) are called isotopes.

Ex:- 1) are the isotopes of hydrogen

2) are the isotopes of oxygen.

Isobars :

- The nuclei having the same mass number (A), but different atomic number (Z) are called isobars.

➤ Ex (1)

(2)

Isotones :

- The nuclei having same neutron number (N) but different atomic numbers (Z) and different mass numbers are called isotones.
- Ex:

Cathode Rays :

- Cathode rays are produced when pressure inside the discharge tube is nearly 0.001 mm of Hg.

Properties of cathode rays:

- 1) They travel in straight line.
- 2) They are material particles.
- 3) Cathode rays carry negative charge and are known as stream of electrons.
- 4) The rays produce fluorescence
- 5) They are deflected by electric and magnetic field.
- 6) The rays when strike a heavy solid metal produce x-rays.
- 7) Cathode rays are made up of fast moving electrons. The velocity of cathode ray is of velocity of light.

- 8) Cathode ray particles have a mass of _____ times the mass of Hydrogen atom.
- 9) The rays penetrate through thin foils (metal foils)
- 10) They Ionise gas through which they pass.
- 11) These rays was discovered by J.J.Thomson.

Applications :

- 1) Cathode rays are used in T.V. Picture tubes, computer display tubes, glow signs for advertisements and tube lights in the houses.

Properties of anode (or) Canal Rays :

- 1) These rays travel in straight line
- 2) They carry the positive charge.
- 3) They are deflected by electric and magnetic fields
- 4) They ionise gas through which they pass.
- 5) They are material particles i.e it has K.E.
- 6) These rays was discovered by Gold Stein.

Radio Activity :

- The self disintegrating, spontaneous activity in the form of emission of radiations by some unstable nuclei with atomic number greater than 83 is called natural radio activity.
- Natural radio activity was discovered by “**Henri Becquerel**”.
- Examples of radio active elements are Uranium, Thorium, Polonium, Radium.

**Units of Radio Activity:-**

- S.I. Unit is Becquerel.
- Its other units are curie and Rutherford
- 1 Curie = 3.7×10^{10} decays / sec
- 1 rutherford = 10^6 decays / sec
- 1 becquerel = 1 decays / sec.
- For measuring radio activity “Geiger Muller Counter” is used.
- Classification of Radiations from a Radio active substance
 - (1) α -particles
 - (2) β -particles
 - (3) γ -rays

Properties of α -particles :

- 1) It consists of two protons and two neutrons
- 2) It is identical to a doubly ionised helium atom. Its mass is 4 times the mass of proton and its charge is 2 times the charge of proton.
- 3) They ionises the gases.
- 4) They have small penetrating power compared to β and γ Radiations.
- 5) They are deflected by electric and magnetic fields.

Properties of β - Particles

- 1) β - particles are electrons. The mass and charge of a β - particles are same as those of an electron.

- 2) Speed of this particles is 10^8 ms^{-1}
- 3) The ionising power of β -particles is less than that of α -particles but more than γ -rays.
- 4) Penetrating power of β -particles is more than that of α -particle but less than γ -rays
- 5) They are deflected by electric and magnetic fields.

Properties of γ - rays :

- 1) γ -rays are electro magnetic radiation
- 2) travelling with speed of light
- 3) Ionising power is less than α and β particles
- 4) They are not deflected by electric and magnetic fields
- 5) They are more affect on phtographic plate
- 6) They have no mass and charge but possess high energy.

Artificial Radio Activity :

- A nuclear process in which a stable element is convered into another element, which is radio active in nature is called artificial radio activity.
- Artificial Radio Activity was First Discovered by “**Medam Curie.**”



Uses of Radio Isotopes :

- These are used in Various applications in different fields including medicine, industry, agriculture and meteorology and radioactive dating.
 - 1) cobalt - 60 is used to kill cancer cells.
 - 2) Radio active Isotopes are used in chemical analysis.
 - 3) Uranium - 235 is used production of electricity.
 - 4) Dating of plants, animals / human beings obtained from ancient times after excavation by using carbon-14.

Radio active (radio isotope) dating :-

- The method of determining age of old-objects like dead plants, rocks, animals is called radioactive dating.

Carbon Dating:-

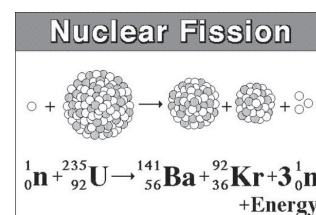
- Carbon dating is used to determining the age of fossils (dead plant or animal).
- In carbon dating c^{14}/c^{12} ratio is used.

Uranium Dating :

- U^{235} is used to determine the age of the earth or rocks by finding the present ratio of U^{235} and the stable -lead isotope.

Nuclear Fission :

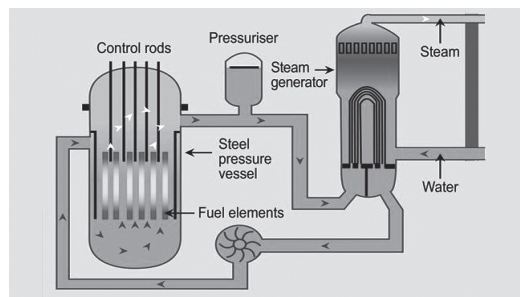
- A nuclear process in which a heavy nucleus when bombarded with neutrons, splits into two nuclei of nearly equal mass with the release of enormous amount of energy is called nuclear fission.



- This was invented by Ottohann and Strassmann.
- The principle of “Atom Bomb” is uncontrolled nuclear fission reactions (or uncontrolled chain reaction)

Nuclear Reactor :

- The working of a nuclear reactor is based on controlled chain reaction.
- The main parts of nuclear reactor are following.



Fuel : , Are used as fuel

Moderator : The material that slows down the neutrons in a nuclear reactor is called moderator.

- Heavy water, graphite are used as moderator

Control rods : “Boron” and “Cadmium” are suitable materials for capturing excess, neutrons are called absorbers. The absorbers are used in the form of rods are commonly called control rods.

Coolant :

- Water, Molten Sodium, CO_2 used as coolant.

Protective shield :

- 10m thickness of concrete wall or lead (pb) used as shield.

Uses of Nuclear reactor :

π

- 1) Production of electric power
- 2) For producing radio isotopes

Breeder Reactor :

- A reactor which produce fissionable material than its consumption.
- Pu^{239} is produced from U^{238}

Nuclear Fusion :

- The process of combining of two lighter nuclei to form one heavy nucleus is called nuclear fusion. In this process a large amount of energy released.
- Nuclear Fusion takes place at very high temperature.
- The source of the sun’s energy is the nuclear fusion process taking place at the sun.
- Hydrogen bomb works on the principle of uncontrolled fusion reaction.

X-Rays :-

- X - Rays were discovered by **Roentgen** in 1895.
- X - Rays can be produced if the cathode rays are made to strike against the metal target.



Properties of X-Rays :

- 1) Travel in straight line.
- 2) They produce fluorescence
- 3) They travel with velocity of light
- 4) They are not deviated by electric and magnetic fields.
- 5) The rays can ionise a gas.
- 6) The rays can penetrate through a matter.
- 7) The rays affect photographic plates
- 8) X-rays are electromagnetic waves.
- 9) Wavelength of X-rays in the range of 0.01 to 10 nm.

Uses :-

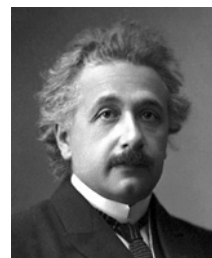
- 1) They are useful in detecting a fracture of bone.
- 2) They are very useful in crime investigation
- 3) In customs department x-rays are used for detecting weapons and the atoms that are likely to be smuggled.
- 4) They are used in studying crystal structure.

Nuclear Forces :

- Nuclear forces are the strongest in nature.
- Nuclear forces are very short range attractive force.
- Gravitational forces are weakest in nature

Einstein mass-energy Relation:

- According to Einstein, when m mass is lost the produced energy is given by mc^2
- Where 'C' is the speed of light

**Photoelectric Effect:**

- The phenomenon of emission of electrons from a metal surface when light of suitable frequency is incident on it is called photoelectric effect

$$h\nu = K.E_{\max} +$$

$$(\text{or}) E = K.E_{\max} +$$

Where $E = h\nu$ is energy of photon.

h - Planck's constant

ν - frequency of incident light

ϕ - work function

- Einstein received Nobel Prize in 1922 for his discovery of this effect.

MODERN PHYSICS - Practice Bits

- | | |
|--|---|
| <p>1. The subatomic particles of an atom are.
 1) electrons 2) protons
 3) neutrons 4) all the above</p> <p>2. For a nuclear reactor enriched uranium fuel is needed. What is enriched uranium.
 1) Pure U-235
 2) Fuel having 50% U-235
 3) Fuel having nearly 5-6% U-235
 4) Pure U-238</p> <p>3. Hydrogen bomb is based on the principle of.
 1) Controlled fission reaction
 2) Uncontrolled fission reaction
 3) Controlled fusion reaction
 4) Uncontrolled fusion reaction</p> <p>4. Radio Activity is a property of.
 1) Atomic nuclei
 2) Excited electrons
 3) X-rays
 4) Ultraviolet light</p> <p>5. Which of the following rays are produced when very high energy electrons are stopped suddenly at a metal target.
 1) α - rays 2) β - rays
 3) γ - rays 4) X - rays</p> <p>6. The nucleus of an atom contains.
 1) Protons and neutrons
 2) Protons and electrons
 3) Neutrons and electrons
 4) Only neutrons</p> | <p>7. Radio isotope are.
 1) Isotope those are radioactive
 2) Some nuclei which emit radio waves
 3) Isotopes used in radio sets
 4) None of these</p> <p>8. Radiocarbon dating technique is used to estimate the age of.
 1) Rocks 2) Soil
 3) Fossils 4) Buildings</p> <p>9. The age of the earth can be estimated by.
 1) Uranium dating
 2) Carbon dating
 3) Atomic clocks
 4) Biological clocks</p> <p>10. The process of the splitting up of a heavy nucleus into roughly two equal fragments with the emission of energy is called.
 1) Transmutation 2) Fission
 3) Fusion 4) Implosion</p> <p>11. The process of fission occurs mainly in
 1) Nuclear reactors
 2) Chemical reactions
 3) The hydrogen bomb
 4) The sun</p> <p>12. The function of a moderator in a nuclear reactor is.
 1) To moderate the number of secondary neutrons
 2) To slow down the secondary neutrons
 3) To increase the number of fissile (or fissionable) nuclei
 4) To control the fission reaction</p> |
|--|---|

13. A breeder reactor is that which

- 1) Produces more fissionable material than it burns
- 2) Does not require fissionable material at all
- 3) Uses only heavy water
- 4) None of these

14. The enormous energy released in an atomic explosion is due to the.

- 1) Conversion of mass into energy
- 2) Conversion of chemical energy into heat energy
- 3) Conversion of mechanical energy into nuclear energy
- 4) Conversion of neutrons into protons

15. The process by which energy is generated in the sun is the.

- 1) Fission of uranium
- 2) Fusion of helium
- 3) Fusion of hydrogen
- 4) Combination of all these processes

16. Which of the following are Einstein's discovered.

- 1) Photoelectric effect and X-rays
- 2) Radioactivity and the theory of relativity
- 3) Photoelectric effect and the theory of relativity
- 4) Radio activity and X-rays.

17. Natural radioactivity was discovered by

- 1) Medam Curie
- 2) Henri Beaquerel
- 3) Wilhelm Roentgen
- 4) Albert Einstein

18. Which of the following is used as control rod in nuclear reactor

- 1) Heavy water
- 2) Boron
- 3) Cadmium
- 4) '2' and '3'

19. Canal rays which one of the following?

- 1) Light rays
- 2) a stream of uncharged particles
- 3) a stream of electrons
- 4) a stream of positively charged particles

20. Curie is the unit for

- 1) radioactivity
- 2) temperature
- 3) heat
- 4) Energy

Answers

1) 4	2) 3	3) 4	4) 1	5) 4	6) 1	7) 1	8) 3	9) 1	10) 2
11) 1	12) 2	13) 1	14) 1	15) 3	16) 3				
17) 2	18) 4	19) 4	20) 1						

9. UNITS AND MEASUREMENTS

- Measurement is the comparison of an unknown quantity with a known standard quantity.

Magnitude of physical quantity :

- The number times a standard quantity is present in a given physical quantity is called magnitude of physical quantity

$$\text{physical quantity} = (\text{magnitude}) \times (\text{unit})$$

Fundamental Quantities :

- The physical quantities which are independent to each other are called fundamental quantities and their units are called fundamental units.

Ex: Length, Mass, Time, Temperature, Luminous intensity, Current.

- There are seven fundamental quantities in SI system.

Fundamental Quantities in S.I. System

S.No.	Fundamental Quantity	Fundamental Unit	Symbol
1.	Length	Metre	m
2.	Mass	Kilogram	kg
3.	Time	Second	s
4.	Electric current	Ampere	A
5.	Temperature	Kelvin	k
6.	Luminous intensity	Candela	cd
7.	Amount of substance	Mole	mol

Derived Quantities:

- The physical quantities which are obtained with the help of fundamental quantities are called derived quantities and their units are called derived units.

Ex:- Velocity, Force, Work, Density, Momentum, etc

S.I. Units And Symbols Derived Quantities

S.No	Physical Quantities	Unit	Symbol
1.	Area	(metre) ²	m ²
2.	Volume	(metre) ³	m ³
3.	Density	Kilogram / metre ³	kgm ⁻³
4.	Velocity	metre/second	ms ⁻¹
5.	Acceleration	metre/second ²	ms ⁻²
6.	Momentum	Kilogram metre/sec	kgms ⁻¹
7.	Force	Newton	N
8.	Impulse	Newton second	NS
9.	Work or energy	Joule	J
10.	Power	Watt	W

11.	Pressure	Pascal	Pa
12.	Coefficient of Viscosity	Poiseuille	Pi
13.	Luminous flux	Lumen	Lm
14.	Frequency	Hertz	Hz
15.	Young's Modulus	Newton/ (metre) ²	Nm ⁻²
16.	Surface tension	Newton / metre	Nm ⁻¹
17.	Coefficient of linear expansion	Per deg. kelvin	K ⁻¹
18.	Specific heat	Joule / kg.kelvin	Jkg ⁻¹ k ⁻¹
19.	Latent heat	J/kg	Jkg ⁻¹
20.	Thermal conductivity (or) Coefficient of thermal conductivity	Watt/metre kelvin	Js ⁻¹ m ⁻¹ k ⁻¹ (or) wm ⁻¹ k ⁻¹

System of Units:

➤ Physical quantities are measured in four system of units.

1) FPS 2) CGS 3) MKS 4) SI

S.No	Fundamental	FPS	CGS	MKS
1.	Length	Foot	Centimeter	Meter
2.	Mass	Pound	Gram	Kilogram
3.	Time	Second	Second	Second

Some other units of length :**1) Astronomical Unit :-**

- It is the average distance between earth and the sun
- One Astronomical unit (Au) = $1.496 \times 10^{11}\text{m}$ $1.5 \times 10^{11}\text{m}$

2) Light Year :

- The distance travelled by light in one year in vaccum
- 1 Light year = $9.46 \times 10^{15}\text{m}$

Parsec :

- It is the largest unit of length.
- 1 parsec = $3.086 \times 10^{16}\text{m}$
- (or) 1 parsec = 3.26 light year

Scalars and Vector Quantities

- On the basis of magnitude and direction, physical quantities are classified as below.

Vectors :

- The physical quantities which have both “Magnitude” as well as “direction” are called vector quantities.

Ex: Velocity, acceleration, momentum, force, torque, displacement, angular momentum, electric field, magnetic field etc.

Scalars :

- The physical quantities which have only magnitude and no direction are called scalars

Ex: Speed, mass, time, volume, density, current, Temperature, work, energy etc.

- Nuclear radii are measured in “fermi”.
1 Fermi = 10^{-15}m
- The speed of a ship is measured in “knot”.
1 knot = 1.852 km / hr.
- “Fathom” is the unit for measuring depth of water
- 1 fathom = 1.82 meters
- Flow of fluids is measured in cusec which is equal to 1 cubic foot.
- 1 cusec = $0.0283 \text{ m}^3/\text{sec}$.
- The lux is the S.I. derived unit of illuminance.
- 1 Lux = 1 Lumen / m^2

Scientific Instruments

Altimeter	:	It measures the height of a flying plane.
Ammeter	:	It measures electric current.
Anemometer	:	It measures the speed of air (wind)
Barometer	:	It is used in measuring the atmospheric pressure.
Calorimeter	:	It is made up of copper and measures the quantities of heat.
Chronometer	:	An instrument used in ships to calculate accurate time.
Colorimeter	:	An instrument for comparing intensities of colour.
Compass Box	:	A device that indicates the north and south directions at any place.
Crescograph	:	An instrument that used in recording growth of plants. It was invented by Dr.J.C. Bose.
Electroscope	:	An instrument for detecting the presence of electric charge.
Fathometer	:	An instrument for measuring the depth of the ocean.
Geiger muller counter	:	A device used to detect the presence of radiation of a radio active source.
Gravimeter	:	An instrument to record oil deposits under water.
Hydrometer	:	An instrument to measure the specific gravity of liquids.
Hydrophone	:	An instrument for recording sound under water.
Hygrometer	:	An instrument to measure the humidity in air
Lactometer	:	An instrument used to test the purity of milk
Manometer	:	An instrument for determining the pressure of gases.
Microphone	:	An instrument used for converting sound waves into electric signals

Microscope	:	An optical instrument the produces enlarged images of minute objects.
Odometer	:	An instrument used to determine the distance covered by wheel vehicles.
Potentiometer	:	It is used to compare the e.m.f of cells to measure the thermal e.m.fs to determine the large potential difference currents and measure the low resistances.
Pyrometer	:	An instrument to record the high temperature of a distant obejct, e.g., the sun.
Radar	:	It is used to detect the direction and range of a flying aircraft by means of radio waves.
Rain Guage	:	An instrument for recording the rainfall at a particular place.
Seismograph	:	An instrument used to record the intensity and distance of an earthquake.
Speedometer	:	An instrument that indicates speed at which a vehicle is moving.
Sphygmomanometer	:	An instrument that measures arterial blood pressure.
Stethoscope	:	An instrument used to hear and analyse the movements of heart and lungs. It is used by doctors.\
Stop Watch	:	An instrument used to record small intervals of time in laboratory, in races etc.

UNITS & MEASUREMENTS - Practice Bits

1. **The SI unit of length is.**
 - 1) metre 2) centimetre
 - 3) yard 4) foot length
2. **Any quantity which we measure in the study of physics is called.**
 - 1) measurable quantity
 - 2) physical quantity
 - 3) material quantity
 - 4) natural quantity
3. **In FPS system the mass is measured in.**
 - 1) grams 2) kilograms
 - 3) pounds 4) ounce
4. **The metric system of measurement are.**
 - 1) CGS and FPS 2) CGS and MKS
 - 3) FPS and MKS
 - 4) none of the above
5. **SI system of measurement includes the measurement of.**
 - 1) temperature
 - 2) luminous intensity
 - 3) electric current
 - 4) all the above
6. **One millimetre is _____ of a metre.**
 - 1) 2) 100 th
 - 3) 4)
7. **The magnitude of light year is.**
 - 1) 9.46×10^{12} km
 - 2) 4.96×10^{12} km
 - 3) 9.64×10^{12} km
 - 4) 6.94×10^{12} km
8. **One angstrom ($^{\circ}\text{A}$) is equal to.**
 - 1) 10^{-8} m 2) 10^{-7} m
 - 3) 10^{-10} m 4) 10^{-9} m
9. **The most appropriate unit to measure the diameter of an atom is.**
 - 1) millimetre 2) angstrom
 - 3) micrometre 4) none of the above
10. **The most appropriate unit to measure distance of stars from the earth is.**
 - 1) light year 2) kilometre
 - 3) hectometre 4) metre
11. **A spring balance measures the.**
 - 1) mass of a body
 - 2) weight of a body
 - 3) both (1) and (2)
 - 4) none of the above
12. **The SI of weight is.**
 - 1) newton 2) kilogram-force
 - 3) dyne 4) gram-force
13. **Which one is a constant quantity?**
 - 1) mass 2) weight
 - 3) both (1) and (2)
 - 4) none of the above
14. **SI unit of time is.**
 - 1) hour 2) minute
 - 3) second 4) none of the above
15. **One second is _____ part of mean solar day.**
 - 1) 2)
 - 3) 4)

16. The SI unit of temperature is.

- 1) degree celsius ($^{\circ}\text{C}$)
- 2) degree fahrenheit ($^{\circ}\text{F}$)
- 3) degree Reaumur
- 4) Kelvin (K)

17. The SI unit of area is.

- 1) metre
- 2) metre square
- 3) metre cube
- 4) none of the above

18. In order to find the area of a rose leaf the best option is.

- 1) centimetre graph paper
- 2) inch graph paper
- 3) millimetre graph paper
- 4) none of the above

19. The SI unit of volume is.

- 1) square metre 2) cubic metre
- 3) cubic millimetre 4) litre

20. One cubic metre is equal to.

- 1) 10^4 cm^3 2) 10^2 cm^3
- 3) 10^3 cm^3 4) 10^6 cm^3

21. 1 ml volume of a liquid is equal to.

- 1) 1 cm^3
- 2) 10 cm^3
- 3) 100 cm^3
- 4) $10,00,000 \text{ cm}^3$

22. Which one is not a physical quantity?

- 1) Force 2) Length
- 3) Pain 4) Volume

23. Which one is not a fundamental quantity

- 1) Mass
- 2) Temperature
- 3) Luminous intensity
- 4) Pressure

24. Which one is not a derived quantity.

- 1) Mole 2) Newton
- 3) cm^3 4) pascal

25. Candela is the fundamental unit of.

- 1) thermodynamic temperature
- 2) amount of substance
- 3) electric current
- 4) luminous intensity

26. The SI unit of density is.

- 1) g cm^{-3}
- 2) kg cm^{-3}
- 3) kgm^{-3}
- 4) none of above

27. Which of the following is/are units of length.

- a) nanometre
 - b) Light year
 - c) Astronomical Unit
 - d) Parsec
- 1) Only a
 - 2) a and b
 - 3) a, b and c
 - 4) All the four

28. Distances of air and sea travel are measured in units of nautical mile. One nautical mile equals.

- a) 1.6 mile
- b) 1.151 mile
- c) 1.852 km

Which of the above is/are correct.

- 1) Only a 2) only b
- 3) a and c 4) b and c

29. A device used for measuring the depth of the sea is called.

- 1) Altimeter
- 2) Fathometer
- 3) Hydrometer
- 4) Manometer

30. Some quantity and unit pairs are given below.

- a) Force - newton
- b) Power-joule
- c) Energy-watt
- d) Pressure-pascal

Which of the above pairs are correctly matched.

- 1) a, b and c
- 2) b and c
- 3) a and d
- 4) a, b, c, and d

Answers

1) 1	2) 2	3) 3	4) 2	5) 4	6) 4	7) 1	8) 3	9) 2	10) 1
11) 2	12) 1	13) 1	14) 3	15) 1	16) 4	17) 2	18) 3	19) 2	20) 4
21) 1	22) 3	23) 4	24) 1	25) 4	26) 3	27) 4	28) 4	29) 2	30) 3

10. OUR UNIVERSE

- The universe is the whole of all matter, energy planets, galaxies and space.
- In Astrophysics we study about the space, its origin and about the heavenly bodies.

STARS:

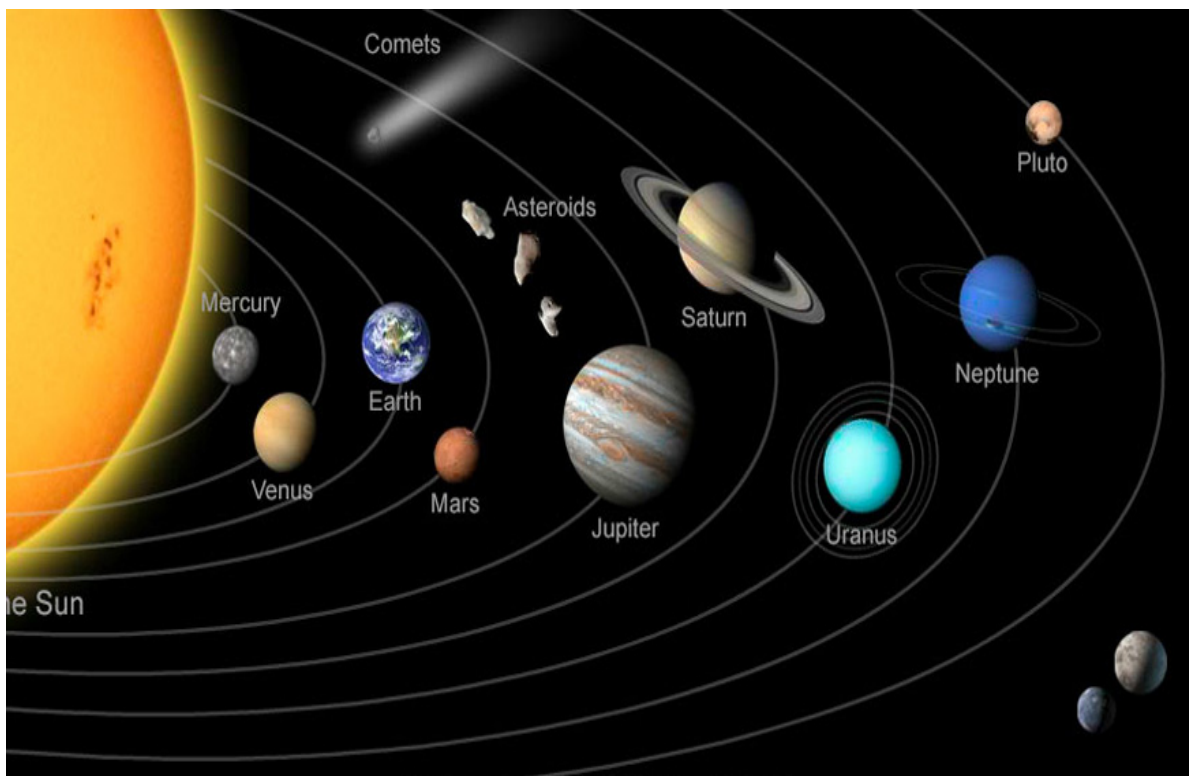
- A star is a luminous heavenly body, which emits light of its own. There are millions of stars. The stars contain 70% hydrogen, 28% Helium, 1.5% Carbon nitrogen and Oxygen and 0.5% other heavy elements colour of a star depends upon its temperature. Sun is also a star.

Solar System:

- The group of various planets, satellites, comets, asteroids and several other heavenly bodies revolving around the sun is called the solar system.
- The solar system is considered to be originated from the sun.
- Its surface temperature is 6000°C.
- It is 150×10^6 km away from the earth.
- Its mass is 330,000 times heavier than the earth.

Planets of solar system:

- The names of the planets of the solar system in order of increasing distance from the sun are:
 - 1) Mercury
 - 2) Venus
 - 3) Earth
 - 4) Mars
 - 5) Jupiter
 - 6) Saturn
 - 7) Uranus
 - 8) Neptune



1) Mercury:-

- Mercury is the closest planet to the sun. It completes its one revolution around the sun in 88 days.
- It is the smallest planet of our solar system.
- Mercury has no satellite of its own.

2) Venus:-

- It is the second planet from the sun.
- It is almost equal to earth in size and weight.
- It completes one revolution around the sun in 225 days.
- It is nearest to earth.
- It is very hot planet with very high temperatures.
- It is the “brightest” planet in the sky.
- Sometimes venus appears in the eastern sky before sunrise.
- Sometimes it appears in the western sky just after sunset. Therefore, it is called a morning star or an evening star.
- Venus has no satellite of its own.
- It rotates from east to west while the earth rotates from west to east.

3) Earth:-

- Earth is the third planet from the sun.
- It is situated between Venus and Mars.
- The earth is the only planet in the solar system on which life is known to exist.
- its diameter is 12,756 km and the average distance from the sun is 14.96×10^6 km.
- It completes one revolution around the sun in 365.25 days.
- Moon is the only natural satellite of the earth.

4) Mars:-

- Mars is the fourth planet from the sun.
- It appears slightly reddish and therefore it is also called the “**Red planet**”.
- Mars has two small natural satellites namely Demos and Phobos.

5) Jupiter:-

- It is the fifth planet from the sun.
- Jupiter is the largest planet of the solar system.
- The mass of Jupiter is about 318 times that of our earth.
- It rotates very rapidly on its axis.
- Jupiter has a large number of satellites.

6) Saturn:-

- It is the sixth planet from the sun.
- Saturn is the second largest planet in the solar system.
- It completes its one revolution around the sun in 29.5 years.
- Saturn is surrounded by rings which are composed of small particles.
- This planet appears as a yellow star in the sky.
- It has a large number of satellites.

7) Uranus:-

- It is the seventh planet from the sun.
- It is the third largest planet in the solar system.
- It appears green in colour when observed by the telescope.
- It takes 84 years to complete its one revolution around the sun.
- Uranus also rotates from east to west.

8) Neptune:-

- It is the eighth planet from the sun.
- It is the fourth largest planet in the solar system.
- It was discovered in 1846 by a German Astrologer Johann Gale.
- It completes its revolution around the sun in 164 years and takes 16 hours to rotate on its axis.

Note:-

- 1) The first four planets, mercury, venus, earth and mars are much nearer the sun than the other four planets. The above four planets are called inner planets. The inner planets have very few moons.
- 2) The planets outside the orbit of Mars namely Jupiter, Saturn, Uranus and Neptune are much farther away from the sun than the inner planets. Therefore the above four planets are called outer planets. The outer planets have a large number of moons.

Pluto:-

- Till 25th August 2006 we used to say there are nine planets in our solar system. The ninth planet at that time was Pluto.
- In the 26th general assembly of International Astronomical Union (IAU) was decided that pluto is no more a planet. The decision was taken since it was observed that pluto does not follow the rule of cleared the neighbourhood. That means some times it is entering into the orbit of Neptune.

Comparison between planets:

Name of the Planet	Comparative Diameter with Diameter of Earth	Distance from the Sun in Crore km.	Period of Revolution	No. of satellites (detected so far)
Mercury	0.38	5.79	88 days	0
Venus	0.95	10.80	225 days	0
Earth	1.00	15.0	365 days	1
Mars	0.53	22.80	687 days	2
Jupiter	11.19	77.80	12 years	50
Saturn	9.40	148.70	29.5 years	53
Uranus	4.04	286.90	84 years	27
Neptune	3.88	449.70	165 years	13

Some other members of the solar system:-**Asteriods:-**

- There is a large gap in between the orbits of Mars and Jupiter.
- This gap is occupied by a large number of small objects (rock pieces) that revolve around the sun. These are called asteroids.
- The largest asteroid “ceres”. It was discovered in 1801 by an astronomer Piyaji.

Comets:-

- Comets are also members of our solar system. They revolve around the sun in highly elliptical orbits. Their period of revolution round the sun is usually very long.
- A comet appears generally as a bright head with a long tail. The length of the tail grows in size as it approaches the sun.
- The tail of a comet is always directed away from the sun.
- Many comets are known to appear periodically. One such comet is “Halley’s comet”.
- Halley’s comet is the most famous and regular comet.
- Its time period is 76.3 years.
- It was seen in 1986 and now expected in 2062 A.D.
- The other comets are Donatis (1858) and Kohoutek’s comet (1973).

Milky way:-

- Milky way is a very large group of stars held together by attractive forces.
- Milky way galaxy looks like a stream of milk across the sky. It was discovered by Galileo that this galaxy consists of billions of stars.
- 68% of stars of the milky way are larger in size than the sun.
- The space between the stars in the Milky way is filled with dust and gases called interstellar matter. About 90% of interstellar matter is in the form of hydrogen.

**Moon:-**

- It is the earth’s satellite. Its average distance from the earth is 3,84,000 km and its diameter is 3476 km. Its weight is 1/81th part of earth’s.
 - It completes its one revolution around the earth in 27 days. There is no atmosphere on moon.
- Land, mountains and deep holes are found on its surface. American space traveller was the first person to land on moon.



OUR UNIVERSE - Practice Bits

1. Which of the following planets do not have satellites revolving around them.
 a) Mars b) Venus
 c) Mercury d) Neptune
 1) 'a' and 'b' 2) 'b' and 'c'
 3) 'a' and 'c' 4) 'b' and 'd'
 2. Which planet has the maximum number of satellites
 1) Jupiter 2) Uranus
 3) Mars 4) Saturn
 3. The tail of a comet always points away from the sun due to
 1) Force of repulsion
 2) Centrifugal force
 3) Solar radiation and solar wind
 4) An unknown reason
 4. Which planet does not have atmosphere around it?
 1) Mercury 2) Mars
 3) Uranus 4) Neptune
 5. What is the total number of planets orbiting the sun?
 1) 7 2) 8 3) 9 4) 10
 6. We get energy directly from
 1) The sun 2) Ocean
 3) Space 4) Atmosphere
 7. Planets are
 1) Luminous bodies that twinkle
 2) Non-luminous bodies that twinkle
 3) Luminous bodies that do not twinkle
 4) Non-luminous bodies that do not twinkle
 8. In upper space astronauts
 1) Cannot see stars during the day
 2) Cannot see stars at all
 3) Can see stars during the day opposite the sun
 4) Can see stars during the day in all directions
 9. The solar system belongs to the galaxy called
 1) Andromeda nebula 2) Milky way
 3) Radiogalaxy 4) Magellanic cloud
 10. Which of the following do not belong to the solar system.
 1) Asteroids 2) Comets
 3) Planets 4) Nebulae
 11. Which planet is known as the red planet
 1) Jupiter 2) Mars
 3) Venus 4) Pluto
 12. The planet closest to the sun is
 1) Mercury 2) Venus
 3) Earth 4) Mars
 13. The heaviest among the inner planets is the
 1) Earth 2) Mars
 3) Mercury 4) Venus
 14. The planet farthest from the sun is
 1) Jupiter 2) Neptune
 3) Saturn 4) Uranus
 15. Which planet is also called as Morning or Evening star
 1) Neptune 2) Jupiter
 3) Venus 4) Uranus

Answers

1) 2	2) 1	3) 3	4) 1	5) 2	6) 1	7) 4	8) 4	9) 2	10) 4
11) 2	12) 1	13) 1	14) 2	15) 3					